8 Ecology MCQ

1. A scientist is evaluating a proposal for raising large numbers of fish in ocean pens for human consumption. As part of the evaluation, the scientist is designing a plan for investigating how the fish in the ocean pens might affect nearby ecosystems.

Which of the following is the most appropriate factor to use as the dependent variable in the experimental investigation?

(A) The amount of metabolic wastes in the water where the fish are being raised ✓

(B) The water temperature in the natural habitat of the fish being studied

(C) The types of fish that are preferentially consumed by humans

(D) The amount of time it takes for fish to digest a meal

2. A biologist is studying territoriality in three related species of birds found in the same section of a forest. For each observed pair of nesting adults, the biologist measured the distance to the next closest nesting pair of the same species (nearest-neighbor distance). An analysis of the results is shown in the table below.

Species	Mean Nearest-Neighbor Distance (km)	Median Nearest-Neighbor Distance (km)	Range of Nearest-Neighbor Distances (km)
I	0.35	0.47	0.25–0.50
II	0.31	0.32	0.30–0.32
III	0.80	1.10	0.10–1.50

Which of the following would be most appropriate for identifying the species with the most even distribution of nests?

(A) Area of the forest in which the nearest-neighbor measurements were made

(B) Total number of nearest-neighbor observations collected during the study period

(C) Standard deviation of the nearest-neighbor distances for each species ✓

(D) Proportion of nearest-neighbor distances that were less than the median

3. A colony of termites was exposed to an atmosphere of 100 percent oxygen for three days. The insects were not immediately harmed by the treatment, but the protozoa that lived in the termites' guts were eliminated. The treated termites continued to behave normally and to eat wood, but they began to starve after a short time.

When the treated termites were instead fed wood contaminated with the feces of untreated termites, the treated termites regained the ability to digest wood and no longer starved. The best analysis of the results of the experiment is that

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(A) infection with protozoa causes the termites to switch to fermentation

(B) cooperative interactions between the termites and the protozoa allow termites to extract energy from wood to survive ✓

(C) the termite immune system eliminates protozoa from the gut

(D) termites digest protozoa as their main source of energy

4. A group of students summarized information on five great extinction events.

Mass Extinction	Time of Extinction	Organisms Greatly Reduced or Made Extinct
End of the Ordovician period	443 million years ago	Trilobites, brachiopods, echinoderms, and corals
End of the Devonian period	354 million years ago	Marine families on tropical reefs, corals, brachiopods, and bivalves
End of the Permian period	248 million years ago	Trilobites, mollusks, brachiopods, and many vertebrates
End of the Triassic period	206 million years ago	Mollusks, sponges, marine vertebrates, and large amphibians
End of the Cretaceous period	65 million years ago	Ammonites, dinosaurs, brachiopods, bivalves, and echinoderms

The students are sampling a site in search of fossils from the Devonian period. Based on the chart, which of the following would be the most reasonable plan for the students to follow?

(A) Searching horizontal rock layers in any class of rock and try to find those that contain the greatest number of fossils

(B) Collecting fossils from rock layers deposited prior to the Permian period that contain some early vertebrate bones ✓

(C) Looking in sedimentary layers next to bodies of water in order to find marine fossils of bivalves and trilobites

(D) Using relative dating techniques to determine the geological ages of the fossils found so they can calculate the rate of speciation of early organisms

8 Ecology MCQ

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6. All of the following statements concerning characteristics of predator-prey relationships are correct EXCEPT:

(A) A rise in the population of prey is often followed by a rise in the population of predators.

(B) A rise in the population of predators is followed by a decrease in the population of prey.

(C) Camouflage is an adaptation that protects prey.

(D) The production of large numbers of offspring within very short periods of time ensures the survival of some prey populations.

(E) The population of predators most often eliminates the population of prey. ✓

7. Some scientists claim that amphibians are currently experiencing a period of mass extinction.

Which of the following should be included in an alternative hypothesis that would best support this claim?

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(A) The current extinction rate of amphibians compared with the background rate of extinction as determined by the fossil record ✓

(B) The extinction rate of amphibians in the 21st century compared with the extinction rate of amphibians during the 20th century

(C) The number of species of amphibians currently alive compared with the number of amphibian species known to be extinct

(D) The number of new amphibian species being discovered per year compared with the number of amphibian species becoming extinct during the same time period

Answer A

Correct. A mass extinction requires that the extinction rate of a certain group of organisms, in this case amphibians, be greater than the background extinction rate. Claiming that this difference exists would be an alternative hypothesis.

8. Scientists have found that the existing populations of a certain species of amphibian are small in number, lacking in genetic diversity, and separated from each other by wide areas of dry land. Which of the following human actions is most likely to improve the long-term survival of the amphibians?

(A) Cloning the largest individuals to counteract the effects of aggressive predation

(B) Reducing the population size by one-fifth to decrease competition for limited resources

(C) Constructing a dam and irrigation system to control flooding

(D) Building ponds in the areas of dry land to promote interbreeding between the separated populations ✓

9. An overlap in the niches of two species will most frequently result in

(A) interspecific cooperation

(B) a hybridization of species

(C) a mutualistic symbiotic relationship

(D) an increase in the biomass

(E) interspecific competition ✓

10. A parasite may be described as an organism that derives nourishment from another living organism and

(A) becomes necessary to the life of the host

(B) is neutral in its effect on the host

(C) benefits the host nutritionally

(D) provides a supportive structure for the host

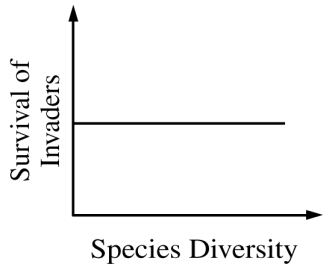
(E) harms the host in the process ✓

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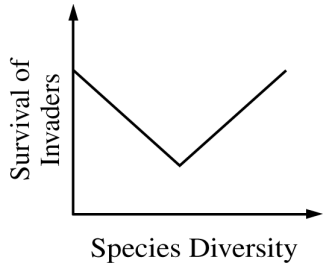
11. A researcher is investigating the relationship between the existing species diversity in a community and the ability of an introduced nonnative species to destabilize the community.

Which of the following graphs is most consistent with the claim that communities with high diversity are more resistant to change than are communities with low diversity?

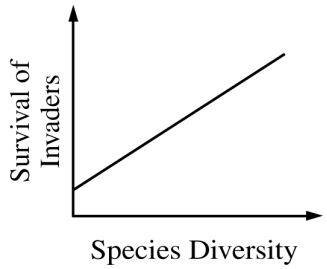
(A)



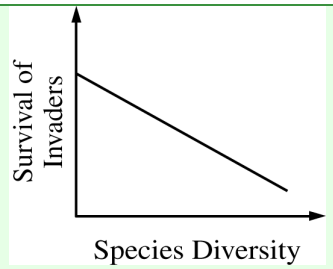
(B)



(C)

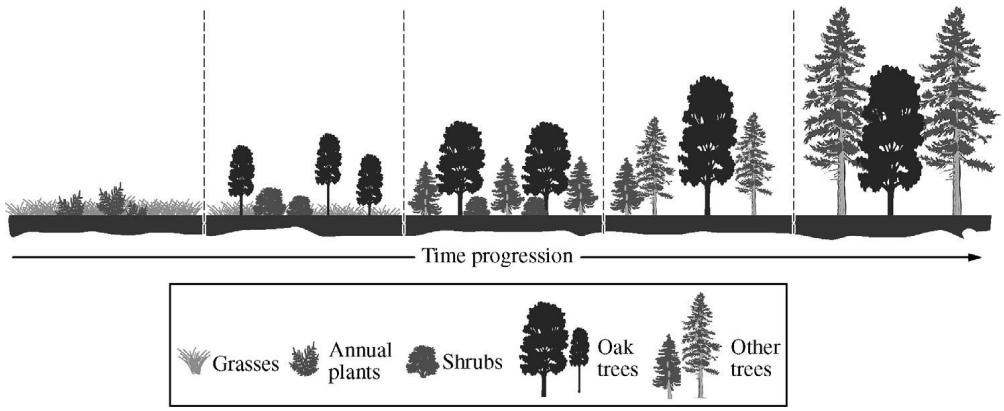


(D) ✓



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12.



The diagram above shows the progression of ecological events after a fire in a particular ecosystem. Based on the diagram, which of the following best explains why the oak trees are later replaced by other trees?

(A) Eventually the other trees grow taller than the oak trees and form a dense canopy that shades the understory. ✓

(B) Oak trees alter the pH of the soil, making the forest better suited for shrubs and other trees.

(C) Roots of shrubs proliferate in the soil of the forest and prevent the oak trees from obtaining water.

(D) Oak trees succumb to environmental pollutants more readily than do either the shrubs or the other trees.

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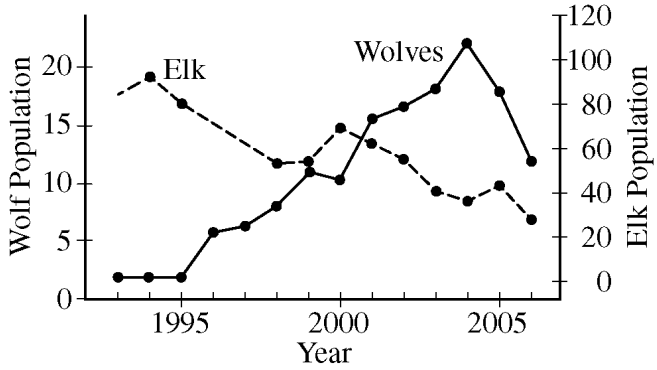


Figure 1. Wolf and Elk Population Sizes in Yellowstone National Park

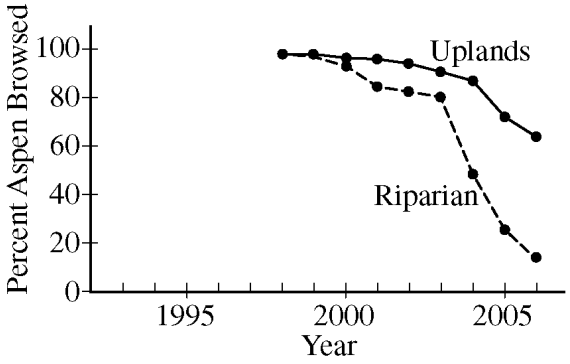


Figure 2. Browsing of Aspen in Yellowstone National Park

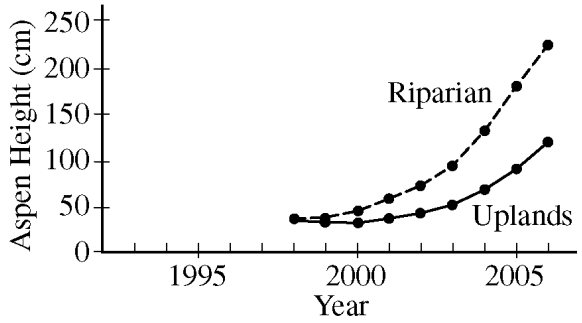


Figure 3. Growth of Aspen in Yellowstone National Park

Wolves, a top predator, were reintroduced to Yellowstone National Park in 1995 after a 50-year absence. In a multiyear study, the numbers of wolves and elk were monitored. The data are shown in Figure 1. In two different environments scientists monitored the percent of aspen trees browsed by herbivores (Figure 2) as well as the growth of the trees (Figure 3). The upland environments consist mostly of flat forested areas. The riparian environments are areas along streams with steep, wooded banks.

13. Based on the data, which of the following behaviors in elk could account for the differences between the percent of aspens browsed by herbivores and the height of aspen trees in riparian and upland environments?

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(A) Elk tend to avoid riparian areas where the steep, wooded riverbanks make it difficult to escape predators. ✓

(B) Elk tend to prefer riparian areas where there is easy access to water.

(C) Elk tend to avoid upland areas where trees are too tall to be easily eaten.

(D) Elk tend to prefer upland areas where there are richer sources of mineral nutrients, such as potassium and iodine.

14. Based on the data, which of the following is the best explanation for the changes in the elk population size between 2000 and 2005 ?

(A) The heights of aspen trees increased during that time period.

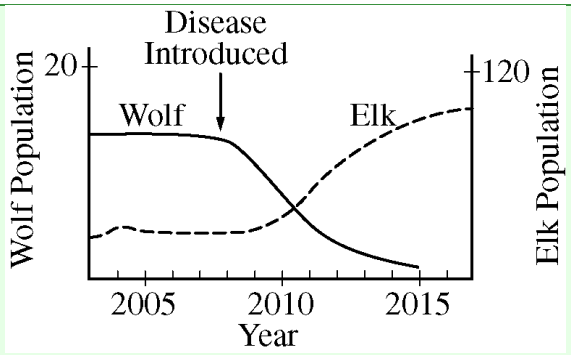
(B) Predation by wolves was higher than before 1995. ✓

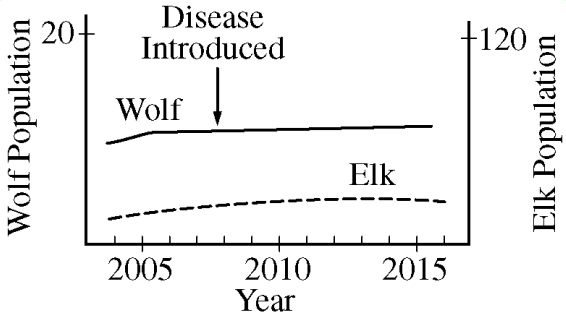
(C) The numbers of aspen trees increased during that time period.

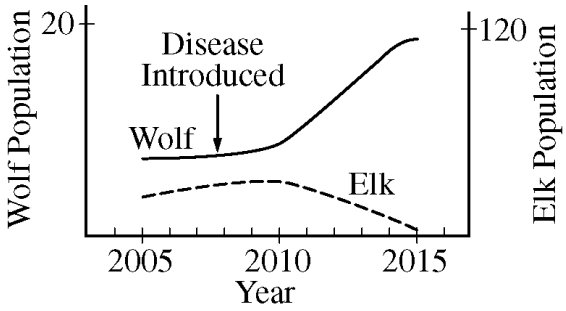
(D) Wolf populations increased more rapidly in the upland areas.

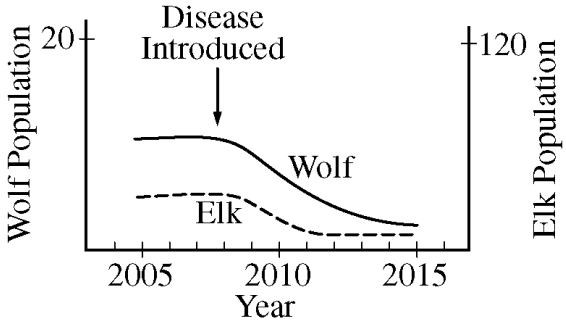
15. Which of the following models best predicts the effect on the ecosystem if a deadly communicable disease is introduced into the wolf population?

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(A)


(B)


(C)


(D)


16. Which of the following predictions about the community is most likely true?

(A) A decrease in the elk population will cause wolves to feed on aspen trees.

(B) An increase in the wolf population will lead to a decrease in aspen growth.

(C) An increase in the growth of aspens will lead to a decrease in the wolf population.

(D) A decrease in the wolf population will lead to a decrease in the mean aspen height.

8 Ecology MCQ

In an investigation of interspecies competition, researchers grew the unicellular protozoan *Paramecium aurelia* in a 5 mL culture and *Paramecium caudatum* in a separate 5 mL culture. *P. aurelia and P. caudatum* were grown together in a third 5 mL culture. Each day a small sample of each culture was removed so the total number of individuals could be estimated, and the remainder of the population was transferred to fresh growth medium. The experimental results are represented in the graphs below.

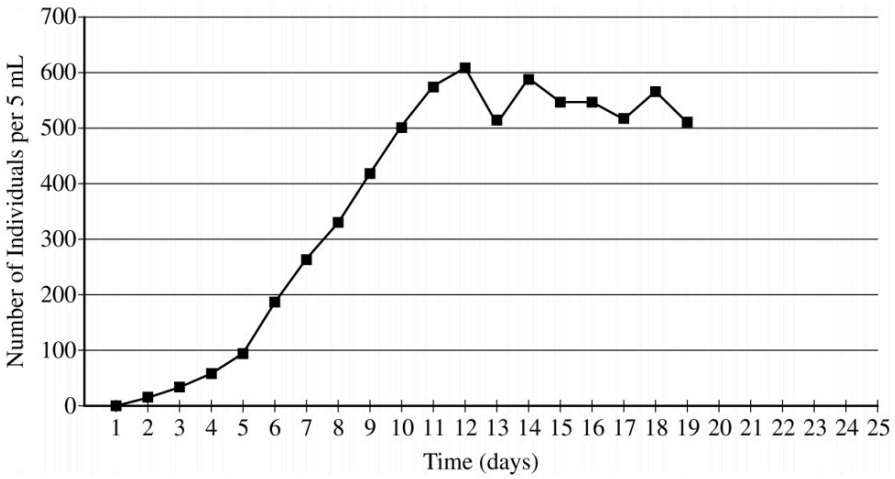


Figure 1. Population of *P. aurelia* cultured alone

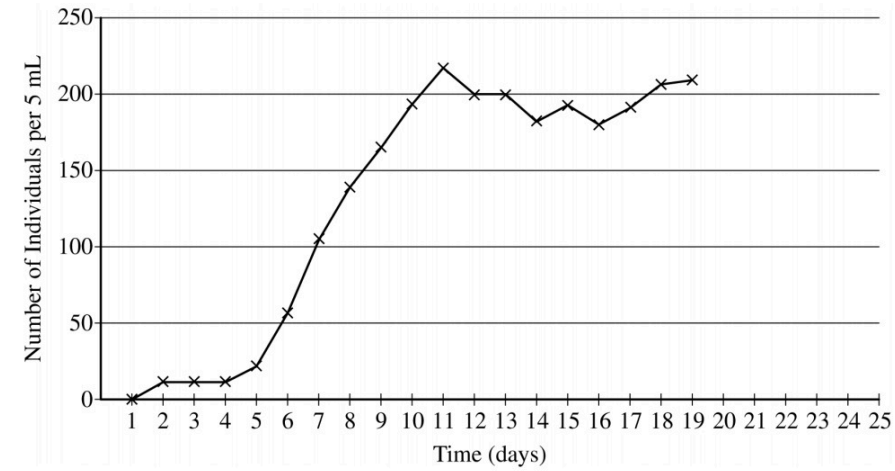


Figure 2. Population of *P. caudatum* cultured alone

8 Ecology MCQ

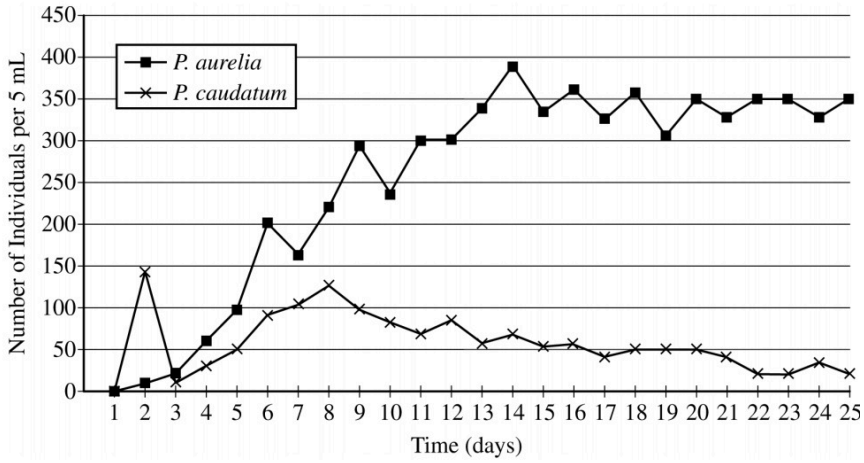


Figure 3. Populations of *P. aurelia* and *P. caudatum* cultured together

17. Based on the experimental results, which of the following statements best describes the relationship of the two populations that were studied in the investigation?

(A) The populations have a parasitic relationship in which one species requires the presence of the other to reproduce.

(B) The populations have a symbiotic relationship in which both species require the presence of the other to reproduce.

(C) The populations have a competitive relationship in which each species limits the ability of the other to reproduce. ✓

(D) The populations have a predator/prey relationship in which one species positively influences the ability of the other to reproduce.

18. If the mixed populations were cultured for an additional 25 days, which of the following predictions would be consistent with the data?

(A) Both populations would stabilize at a small size.

(B) Both populations would stabilize at a large size.

(C) Both populations would be eliminated from the culture.

(D) One population would stabilize at a large size, and the other population would be eliminated from the culture. ✓

19. Which of the following conclusions is best supported by the results of the experiment?

(A) Neither population grows as large when cultured together as each does when cultured separately. ✓

(B) *P. caudatum* prefers different food when cultured with *P. aurelia* than when cultured separately.

(C) Competition for mates increases when the species are cultured together compared with when they are cultured separately.

(D) Individuals of the species *P. caudatum* are smaller when cultured together with *P. aurelia* than when cultured separately.

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20. Which of the following statements best justifies the use of the experimental results in an investigation of interspecies competition?

(A) When two populations use different sources of energy, one of the populations will thrive in a shared environment and the other will become extinct.

(B) When two populations use the same sources of energy, resource availability in a shared environment limits the growth of both populations. ✓

(C) Because the change in entropy of a system containing two different species is always positive, populations typically grow faster in shared environments.

(D) Because the free energy of a system containing two different species is always negative, populations that utilize similar resources cannot coexist in nature.

21. The difference in carrying capacity between the two species when cultured separately is closest to which of the following values?

(A) 0 individuals per 5.0 mL

(B) 60 individuals per 5.0 mL

(C) 150 individuals per 5.0 mL

(D) 350 individuals per 5.0 mL ✓

---

22. Bt corn varieties are genetically modified to produce a toxin that specifically kills corn borers, a type of insect that eats and damages corn crops. Insects that are not closely related to corn borers are not affected by the toxin. Figure 1 shows the increasing production of Bt corn varieties and the change in the use of broad-spectrum pesticides that were the primary means of killing corn borers before the invention of Bt corn.

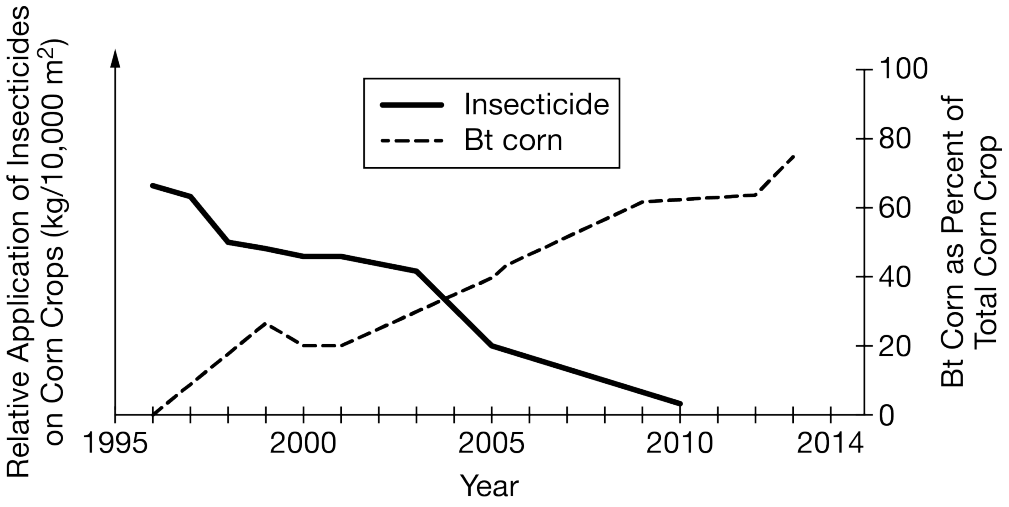


Figure 1. Relative use of pesticides on corn fields and percent of Bt corn in crops

How do the data in Figure 1 support the alternative hypothesis that increased use of Bt corn reduces the impact of corn farming on the natural environment?

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(A) The increased use of Bt corn will result in rapid mutation in corn borers.

(B) The toxin in Bt corn kills only the corn pests, leaving other insects unharmed. ✓

(C) Other insect species will replace corn borers and require additional applications of insecticides.

(D) Since Bt corn is not natural like non-Bt corn, it will not interact with the rest of the environment.

Answer B

Correct. Bt corn is modified to be toxic to certain insects; thus, it doesn't require as much insecticide. The graph shows this trend.

23. Paleoclimatic analysis has generated a claim that there is a possible correlation between an extinction event around 29 million years ago that allowed grasslands to become abundant and the adaptive radiation of several groups of butterflies.

Which of the following proposes the best alternative hypothesis for the claim?

(A) Climate analysis conducted by sampling pollen preserved in amber shows an increase in grass pollen 29 million years ago, but there is no significant difference between speciation rate, as well as species richness with regard to butterflies, before and after 29 million years ago.

(B) Climate analysis conducted by sampling pollen preserved in amber shows an increase in grass pollen 29 million years ago, which indicates there was an increase in grasslands that correlates to an increase in butterfly speciation during the same time period. ✓

(C) There is a correlation between the rapid and wide-ranging decrease in species richness of insects in general 29 million years ago and the increase in species richness and diversity among the butterfly population during the same time period.

(D) Evidence of rapid speciation among butterfly species 29 million years ago is evenly distributed among all environments of that period.

Answer B

Correct. Of the two conditions being studied, grassland distribution seems to correlate to butterfly species radiation. Saying the change in the butterfly speciation rate is due to the change in grassland is an alternative hypothesis.

24. In the year 2000, specimens of *Caulerpa taxifolia*,a green alga used in tropical aquariums, were found off the coast of California. Native to the Indian Ocean, *C. taxifolia* is known for aggressive growth and an ability to compete with sea grasses.It is currently on an international list of invasive species. Which of the following best predicts the consequences of the introduction of *C. taxifolia* to the California coast?

8 Ecology MCQ

(A) Without natural herbivores or competitors,*C. taxifolia* will grow rapidly and crowd out native species of producers. ✓

(B) *C. taxifolia* will have a hard time establishing itself because it will have to compete against native species of sea grasses, which are better adapted to the environment.

(C) *C. taxifolia* will grow rapidly, leading to an increase in the diversity of producers.

(D) Because it is not in the Indian Ocean, its natural environment, *C. taxifolia* will not be able to grow efficiently.

25. There is strong evidence that two species of cliff-dwelling birds have nested in the same environment and fed in the same water for thousands of years, with each having stable population sizes. Based on this information, which of the following statements is the best conclusion?

(A) One species is a better diver than the other species is.

(B) The two birds compete for food.

(C) One species will displace the other.

(D) The two birds have different ecological niches that allow coexistence. ✓

(E) One species eats a more diverse diet than the other does.

26. Which of the following communities is likely to be most stable?

(A) One where a keystone predator has been removed

(B) One with uniformly spaced vegetation

(C) One with high species diversity ✓

(D) One that lacks decomposers

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27. Researchers studied the diversity of macroscopic invertebrates in different areas of watershed ecosystems. They collected samples from 58 sites along 11 streams in the same geographical area. Their data were classified by stream type and can be found in Figure 1. The stream-type classifications range from the perennial stream type, which always has water, to the intermittent stream type, which is dry for more than nine months during the year.

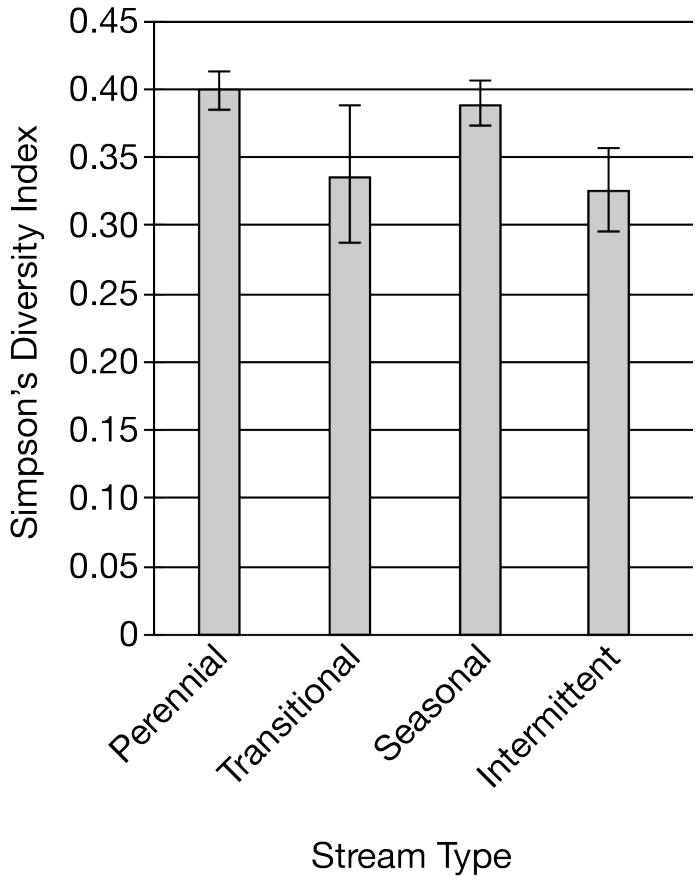


Figure 1. Diversity of macroscopic invertebrates in different stream types. Error bars represent a 95% confidence interval (± two Standard Errors from the mean).

Which of the stream types had a statistically lower macroscopic invertebrate species diversity than the seasonal streams?

(A) Transitional streams only

(B) Intermittent streams only ✓

(C) Perennial and intermittent streams only

(D) Transitional and intermittent streams only

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Answer B

Correct. In order to determine whether two sample means are statistically significant, the standard errors of the means $(SE_{\overline{x}})$ must be considered. The 95% confidence interval for intermittent streams does not overlap with the 95% confidence interval for seasonal streams.

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The coral species *Acropora millepora* is a marine animal that has a symbiotic relationship with algae species in the genus *Symbiodinium*. An increase in the concentration of dissolved CO_2 in oceans has a negative impact on this relationship and the health of the coral species. Scientists investigated the effect of CO_2 concentrations on the number of algae cells found in the coral (Figure 1) and on both the gross (total) and net rates of photosynthesis of those algae (Figure 2). The net rate of photosynthesis is equal to the gross rate of photosynthesis minus the rate of respiration by the algae.

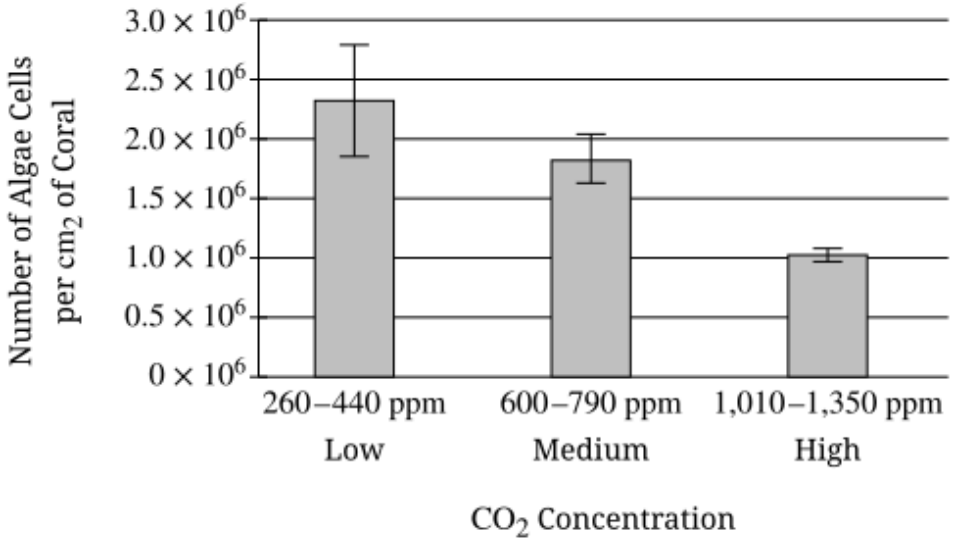


Figure 1. Effect of CO_2 concentration on the number of algae per cm^2 of coral. Error bars represent $\pm 2SE_{\bar{x}}$.

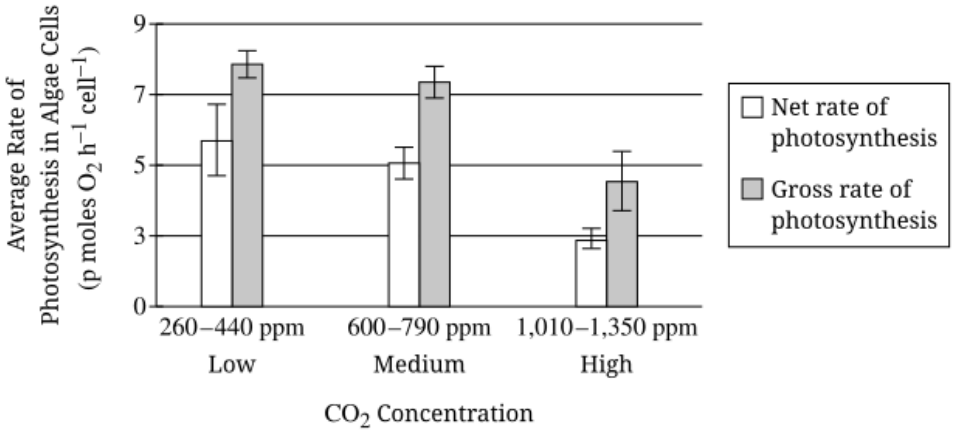


Figure 2. Average gross and net rates of photosynthesis in algae under various concentrations of CO_2. Error bars represent $\pm 2SE_{\bar{x}}$.

28. Which of the following best explains the symbiotic relationship between the coral and the algae?

8 Ecology MCQ

(A) The relationship is an example of commensalism because the algae add CO_2 to the oceans and the coral use CO_2.

(B) The relationship is an example of commensalism because the coral provide shelter for the algae and the algae provide chlorophyll to the coral.

(C) The relationship is an example of mutualism because the coral use sugars made by the algae and provide shelter for the algae. ✓

(D) The relationship is an example of mutualism because the algae perform photosynthesis for the coral and the coral perform respiration for the algae.

Answer C

Correct. Through photosynthesis, the algae produce sugars that the coral can use, and the coral provide living space for the algae. Because both species benefit from the interaction, this symbiosis is mutualism.

---

29. Based on Figure 1, which of the following values best represents the change in the number of algal cells at the high concentration of CO_2 as compared with the number at the low concentration of CO_2 ?

(A) 17.4% increase

(B) 43.5% increase

(C) 56.5% decrease ✓

(D) 65.2% decrease

Answer C

Correct. The percent change is calculated by $[(1.0 \times 10^6 - 2.3 \times 10^6)/2.3 \times 10^6] \times 100 = -56.5\%$. This indicates a 56.5% decrease.

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30. Interactions among populations determine how they access resources within a community. In Costa Rica, two different species of mites, *Rhinoseius colwelli* and *Rhinoseius richardsoni*, exhibit a high degree of interspecies competition. Both mite species live within the flowers of plants, and adult male mites tend to kill mites if they belong to a different species. An experiment was carried out to determine whether these aggressive interactions have resulted in niche partitioning by *R. colwelli* and *R. richardsoni*, of different flower species, with each mite species living in different plant species. The results are summarized in Figure 1.

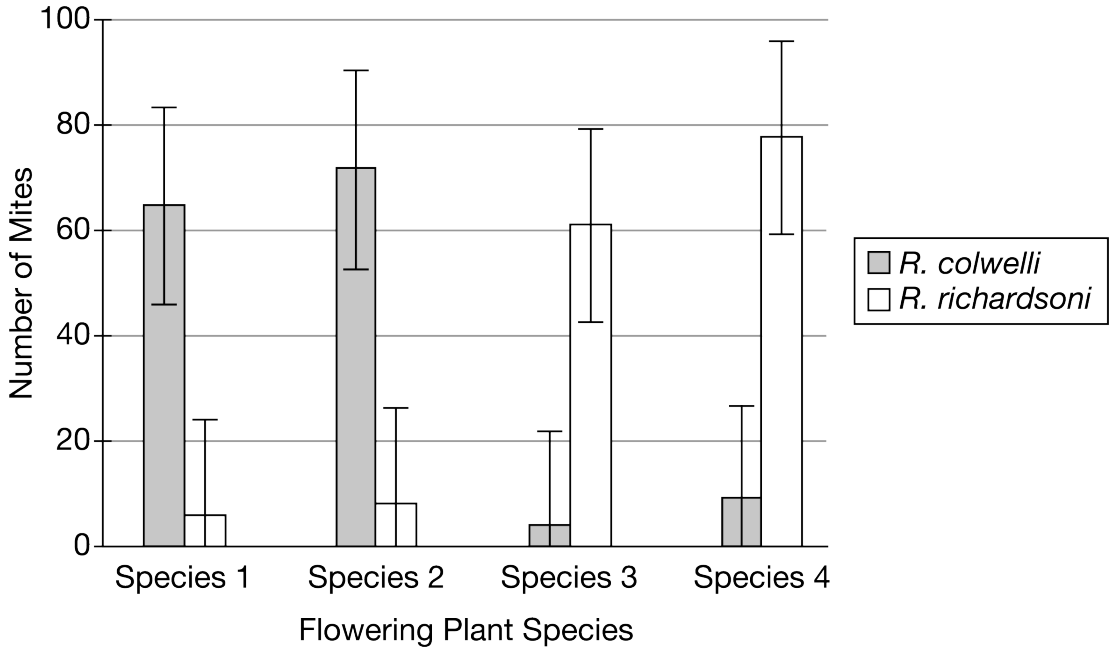


Figure 1. Mite Habitat Preferences

Which of the following is an accurate interpretation of the data in Figure 1 ?

(A) *R. richardsoni* is found significantly more frequently on Species 4 compared to Species 3.

(B) *R. colwelli* is found significantly more frequently on Species 1 compared to Species 2.

(C) *R. richardsoni* is found significantly more frequently on Species 3 and Species 4 compared to Species 1 and Species 2. ✓

(D) *R. colwelli* is found significantly more frequently on Species 3 and Species 4 compared to Species 1 and Species 2.

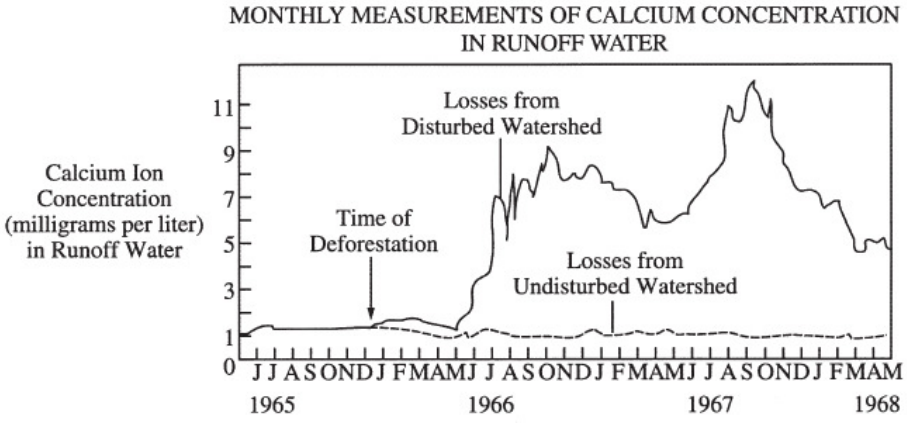
Answer C

Correct. The high degree of interspecific competition between these two interacting populations of mites has influenced community structure. Specifically, the aggressive adult male mites have promoted species-specific separation of mites on to flowers of different plant species. In general within this community, *R. richardsoni* is restricted to Species 3 and Species 4 and *R. colwelli* is restricted to Species

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1 and Species 2.

31. In an experiment on the effects of deforestation of a portion of a northern temperate deciduous forest, all the trees were removed from a small mountainside watershed area and herbicides were applied for three years to prevent regrowth. Measurements of the concentration of calcium (an important nutrient) in runoff water were taken both before and after the deforestation. Monthly measurements also were made in a similar watershed in another part of the same forest that had been left undisturbed. The findings are displayed in the graph below.



Which of the following is best supported by the data?

(A) Deforestation does not change the amount of water flowing through a watershed.

(B) The time of year during which deforestation occurs has no impact on the negative effects related to it.

(C) The loss of calcium affects several other variables in the forest ecosystem.

(D) Calcium loss can be rated as the most damag-ing aspect of mountainside deforestation.

(E) The presence or absence of plants greatly affects the loss of calcium from the soil. ✓

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32. Climate change can affect oceanic ecosystems and their food webs. In the East China Sea (ECS), three major climactic events were considered:

the Pacific Decadal Oscillation (PDO),
the East Asia winter monsoon (EAWM), and
the El Niño Southern Oscillation.

Figure 1 shows the relationships among these climactic events, surface sea temperature (SST), amount of phytoplankton, and copepod abundance. Copepods are important primary consumers in this ecosystem, and many species of fish depend on copepods as a food source. Solid arrows on the diagram indicate a statistical effect. The thicker the arrow, the greater the effect. Dashed arrows indicate statistically insignificant relationships.

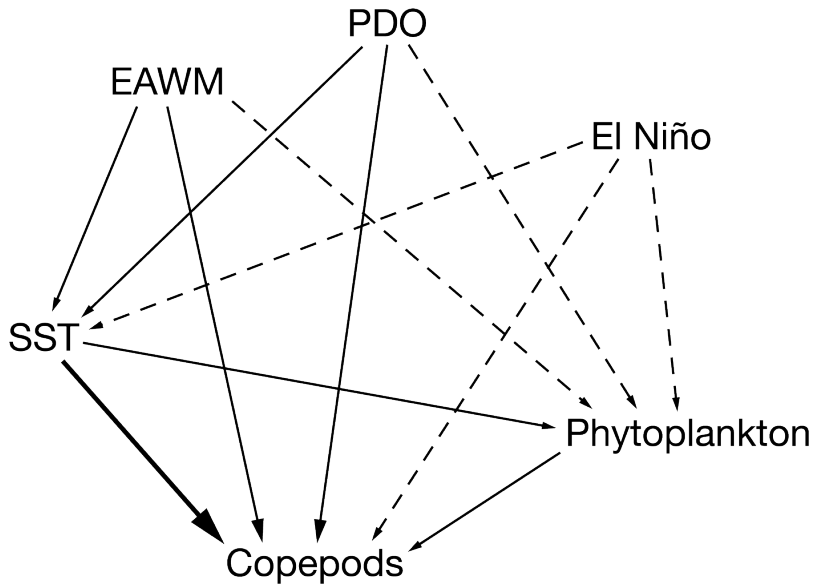


Figure 1. Structural equation model deciphering interactions among climate, phytoplankton, and copepods in the southern East China Sea. Solid arrows indicate statistical effects, with the thickness of the arrow reflecting the relative effect. Dashed arrows indicate inconsequential effects.

What do the data in Figure 1 suggest about a null hypothesis that the East China Sea ecosystem is unaffected by climactic events?

(A) The null hypothesis is rejected since there are statistical effects of EAWM and PDO on surface sea temperature and copepod abundance. ✓

(B) The null hypothesis is rejected because the El Niño Southern Oscillation has a statistical impact on the abundance of copepods in the ECS.

(C) The null hypothesis is rejected because the EAWM and PDO do not have statistical effects on the abundance of copepods and phytoplankton in the ECS.

(D) The null hypothesis is rejected because the El Niño Southern Oscillation does not have a statistical impact on the abundance of copepods in the ECS.

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Answer A

Correct. The null hypothesis should be rejected because the data indicate that the EAWM and the PDO have significant effects on surface sea temperature as well as copepod abundance.

33. Figure 1 shows a food web that describes the relationships within a community containing nine species. Each circle represents a separate species. The size of the circle represents the population size of the species relative to other species in the same trophic level.

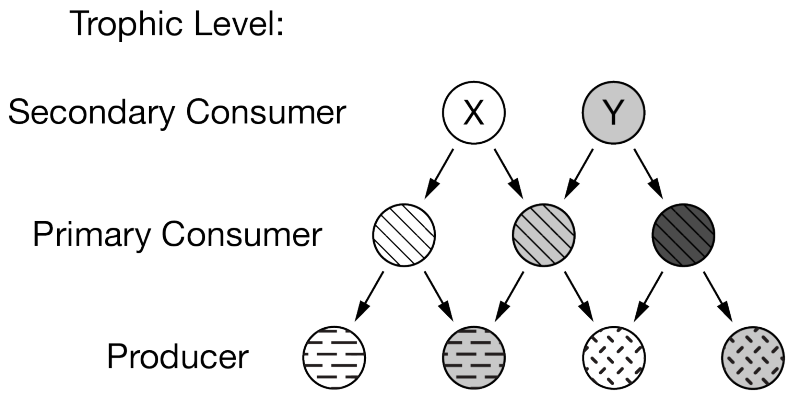
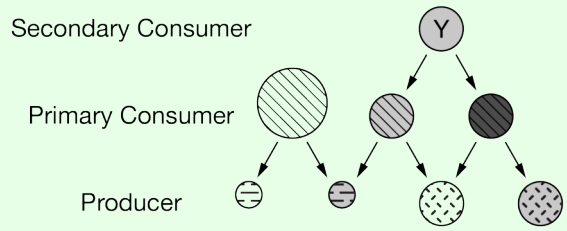


Figure 1. Trophic relationships among organisms in a community

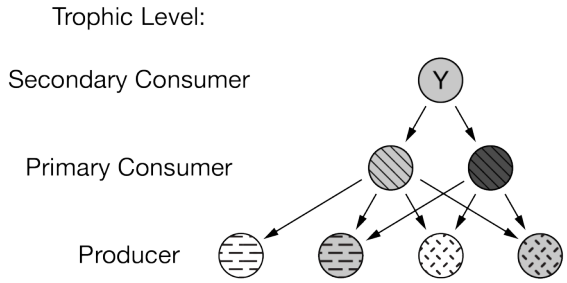
Based on the information in Figure 1, which of the following best predicts a short-term effect of removing secondary consumer X from the community?

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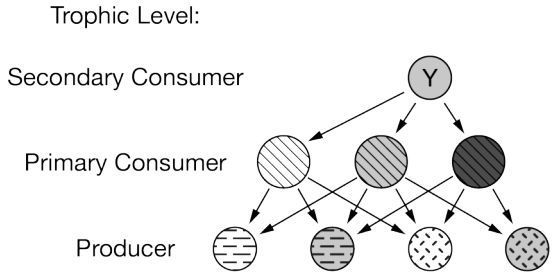
(A)



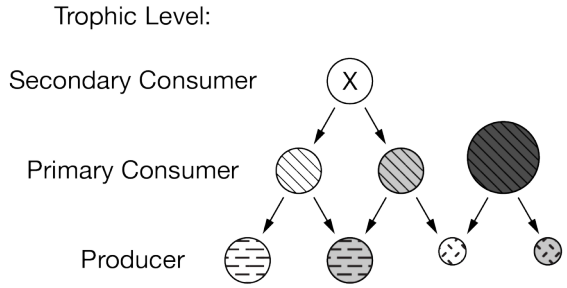
(B)



(C)



(D)



Answer A

Correct. If secondary consumer X is removed, then the leftmost primary consumer will increase in number. Consequently, the two producer species that this primary consumer eats will decrease in number. Note that disturbance events that remove a secondary consumer species can cause large shifts in community composition when the trophic web consists of only a few strong links instead of many weak links.

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34. Phytoplankton (dashed line) are important because they occupy the bottom of the food web absorbing CO_2, functioning as a carbon sink. A scientist suggests that an El Niño event in early 1998 caused a significant decrease in the abundance of phytoplankton that year in the equatorial Pacific Ocean.

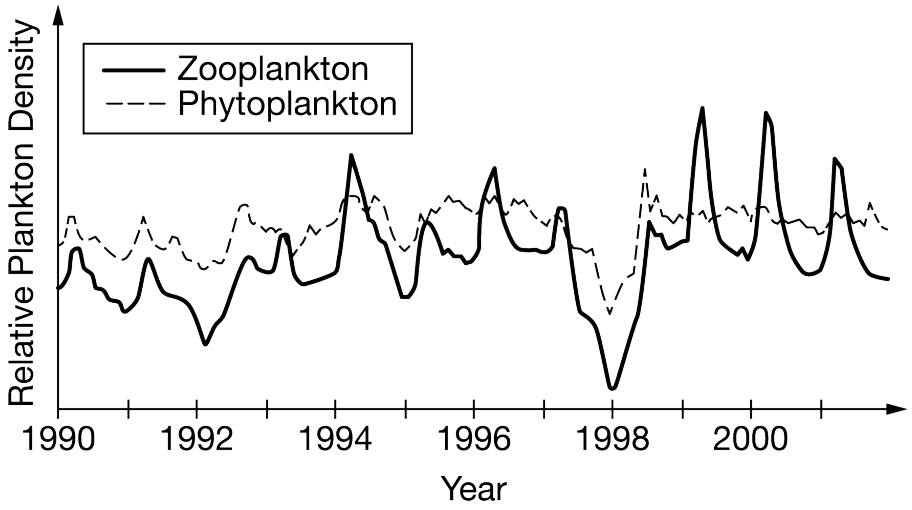


Figure 1. Relative abundance of plankton over time in the eastern equatorial Pacific Ocean

Which of the following observations would best support this alternative hypothesis?

(A) Previous dips in plankton abundance coincided with other El Niño events. ✓

(B) Primary consumers also experienced a dip in abundance in early 1998.

(C) After mid-1998, there is no correlation between phytoplankton and zooplankton abundance.

(D) CO_2 levels in surface waters increased significantly in this region during 1998.

Answer A

Correct. Observing that each dip in phytoplankton abundance occurred during an El Niño event is strong evidence of a cause-and-effect relationship between the two events.

35. The marine iguana, *Amblyrhynchus cristatus*, is endemic to the Galápagos Islands and is especially vulnerable to El Niño events. These iguanas feed on red and green algae. During an El Niño climate event, surface waters and their currents in the central and eastern Pacific Ocean become significantly warmer than usual. Also during El Niño years, land masses experience increased rainfall.

A researcher suggests that El Niño conditions cause a significant decrease in the size of the Galápagos marine iguana population. Which of the following would best support this alternative hypothesis?

8 Ecology MCQ

(A) Increased rainfall on land leads to increased nutrient runoff into the ocean and increased food resources for marina iguanas.

(B) Warm surface water contains fewer nutrients needed by red and green algae than cooler surface waters does. ✓

(C) Warm surface water shortens the incubation time of marine iguana embryos.

(D) Cool surface water decreases the metabolic rate of marine iguanas, so they need more food to compensate for the lowered metabolic rate.

Answer B

Correct. Upwelling brings cold, nutrient-rich waters to the surface. During El Niño years this is inhibited, so algae populations decrease because nutrients are limiting, resulting in decreased marine iguana populations.

In a laboratory population of fruit flies, body color is determined by a single gene with a wild-type allele associated with black body color and a mutant allele associated with yellow body color. A group of researchers hypothesizes that the mutation in fruit flies that causes the yellow body is sex-linked. They also hypothesize that this mutation results in a change in the structure of the sex combs, bristle-like structures on the front legs of male flies that facilitate mating.

To determine the inheritance pattern of the mutant allele, researchers crossed yellow-bodied females with wild-type males. Flies from the F_1 generation were then crossed. The distribution of phenotypes observed in the F_1 and F_2 generations is shown in Table 1.

TABLE 1. DISTRIBUTION OF PHENOTYPES OBSERVED IN THE P, F_1, and F_2 GENERATIONS

	Wild-Type Females*	Yellow-Bodied Females*	Wild-Type Males**	Yellow-Bodied Males**
P Generation	0	3	3	0
F_1 Generation	38	0	0	41
F_2 Generation	117	124	121	130

*Female fruit flies have two X chromosomes.

**Male fruit flies have one X chromosome and one Y chromosome.

8 Ecology MCQ

36. Which of the following best supports the claim that the fitness of yellow-bodied flies is lower than the fitness of wild-type flies?

(A) Males with the yellow-body phenotype are selected by females for mating more often than are wild-type males.

(B) Males with the yellow-body phenotype blend in with their environment and avoid detection by predators better than do wild-type males.

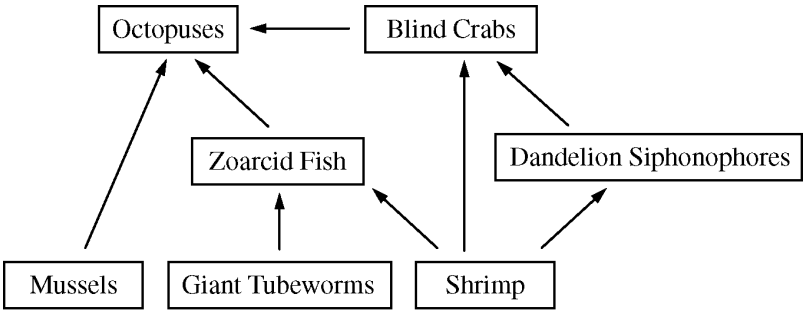
(C) Males with the yellow-body phenotype reproduce less but produce a greater number of viable offspring than do wild-type flies.

(D) Males with the yellow-body phenotype have weaker sex-comb bristles and decreased mating success compared with the bristles and mating success of wild-type males. ✓

Answer D

Correct. Evolutionary fitness is measured by an individual's or species' ability to successfully produce viable offspring. If yellow-bodied males have altered sex-comb bristles that reduce their reproductive success, they have lower fitness.

- **Directions:** Each of the questions or incomplete statements below is followed by four suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.



The food web above represents feeding relationships in a biological community near a deep-sea hydrothermal vent. Hydrothermal vents are geysers on the seafloor that gush super-heated, mineral-rich water. The seawater surrounding hydrothermal vents typically contains carbon dioxide (CO_2), molecular hydrogen (H_2), hydrogen sulfide (H_2S), and methane (CH_4). Sunlight, however, fails to reach the seafloor where deep-sea hydrothermal vents are located.

As part of an investigation, researchers collected living specimens from an area near a deep-sea hydrothermal vent. Mussels in the collection were found to be dependent on molecular hydrogen in seawater. Also, the researchers discovered multiple species of bacteria living in the gills of the mussels. Mussels use gills for filter-feeding and gas exchange with the surrounding seawater. On the basis of their experimental results, the researchers hypothesized that some bacteria living in the gills of the mussels are capable of chemosynthesis.

8 Ecology MCQ

37. On the basis of the food web, which of the following members of a deep-sea biological community is most likely to also have a symbiotic relationship with chemosynthetic organisms?

(A) Octopuses

(B) Blind crabs

(C) Zoarcid fish

(D) Shrimp ✓

38. To refine their model of deep-sea biological communities, the researchers investigated areas of the seafloor that are distant from any active hydrothermal vents. Which of the following is the best interpretation of the observation that some octopus species are only found near active hydrothermal vents?

(A) Octopuses depend indirectly on inorganic compounds in the seawater surrounding hydrothermal vents. ✓

(B) Organisms capable of moving across the seafloor are more difficult to observe than are organisms that are rooted in place.

(C) Genetic bottleneck events in isolated areas of the deep ocean adversely affect some octopus populations but not others.

(D) Biomass in a typical marine food web is expected to be greater in higher trophic levels than in lower trophic levels.

39. Based on an analysis of the food web, an observation that deep-sea bacteria consume molecular hydrogen (H_2) is most relevant to resolving which of the following apparent contradictions?

(A) Water gushing from deep-sea hydrothermal vents can be as hot as 400°C, which is a lethal temperature for most organisms.

(B) Some deep-sea organisms appear to be primary consumers, but no plants live near the hydrothermal vents. ✓

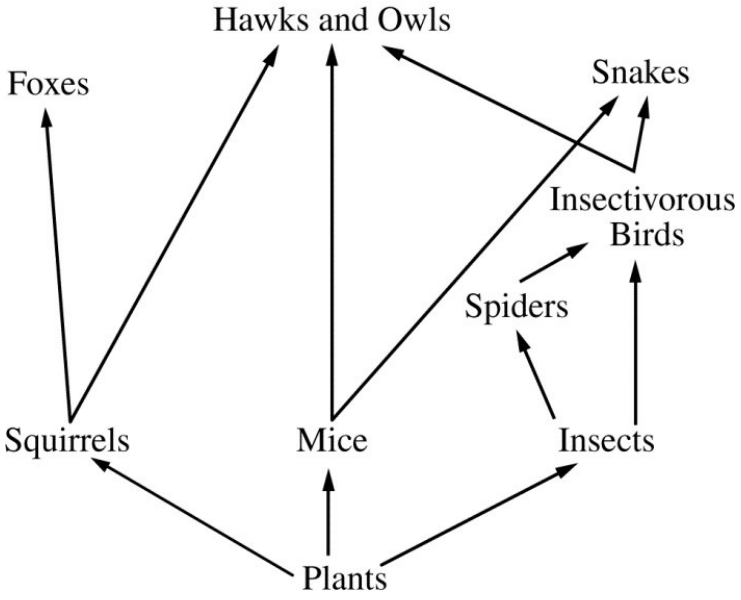
(C) Zoarcid fish are thought to be aggressive predators, but they are frequently described as being lazy swimmers.

(D) Some tissues of tubeworms contain hemoglobin, which is an oxygen-carrying molecule, but there is little free oxygen at this depth.

8 Ecology MCQ

40.

FOOD WEB IN A PARTICULAR MEADOW COMMUNITY



Which of the following changes to the environment will most likely lead to more energy entering the meadow community represented above?

(A) Increasing the number of nesting sites for hawks and owls

(B) Removing squirrels from the area

(C) Increasing the light available to the plants ✓

(D) Applying a chemical pesticide that is specific for spiders

41. The fossils in a layer of rock are significantly different from those in the rock layer directly above it. One hypothesis to explain the difference is that a major extinction event occurred at the point in time represented by the transition between the two rock layers.

Which of the following is the best plan for collecting data to use in a test of the hypothesis?

(A) Examining the upper part of the lower rock layer for evidence of a catastrophic event, such as the presence of shocked quartz, iridium, or soot particles ✓

(B) Digging farther into the hillside to look for a section of sedimentary rock that has a third layer below the two layers that were already identified

(C) Comparing the genomes of organisms from different evolutionary lineages to estimate the relative geologic age of the upper rock layer

(D) Searching fossil collections in several museums for examples of the types of features once shared among organisms that are now extinct

8 Ecology MCQ

42. Fur color is a genetically controlled trait in mice. A population of a species of mouse that typically lives in habitats with light-colored soil was studied for several generations. The background color of the habitat of this population became darker after the second generation due to human activity. A researcher sampled approximately 120 mice in each generation and recorded the fur color phenotypes as shown in Figure 1.

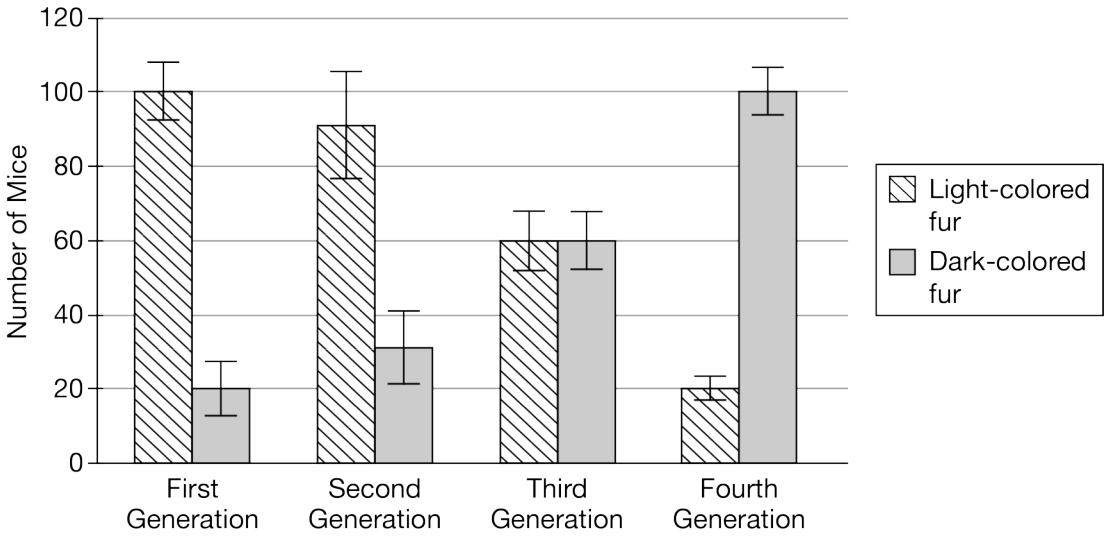


Figure 1. Fur color phenotype distribution of mice over four generations

The researcher claims that there is no selective advantage to fur color, so light and dark fur color phenotypes are present in similar frequencies in the fourth generation of this mouse population. Based on the data in Figure 1, which of the following best evaluates this null hypothesis?

(A) The null hypothesis is rejected, because the change to a darker background color led to an increase in the dark-colored fur phenotype. ✓

(B) The null hypothesis is rejected, because there is no significant difference between the frequencies of the two phenotypes right after the human-caused change to the habitat in the third generation.

(C) The null hypothesis cannot be rejected, because there is no significant difference between the frequencies of the two phenotypes right after the human-caused change to the habitat in the third generation.

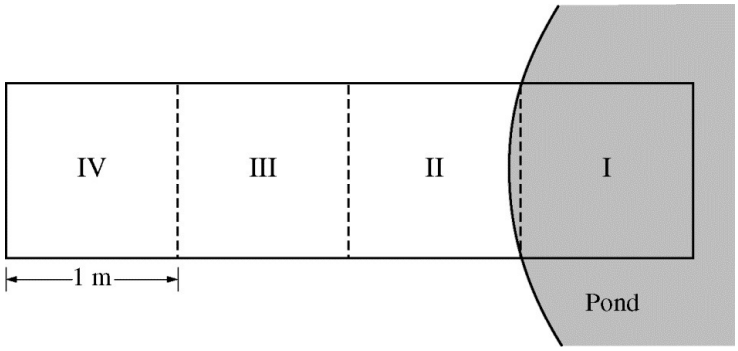
(D) The null hypothesis cannot be rejected, because the darker background color caused by the human-caused change to the habitat led to a decrease in the light fur color phenotype.

Answer A

Correct. The null hypothesis is rejected because there are significantly more dark-colored mice than light-colored mice in the fourth, and most recent, generation. This occurs because the mice with dark fur are more effectively camouflaged on the dark background created by the change in habitat background color.

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Researchers investigated the habitat preferences of two species of garter snakes, *Thamnophis sirtalis* and *Thamnophis atratus*. To create a choice chamber, the researchers built a meshed enclosure and positioned one end of the enclosure at the edge of a small pond. Zone I of the enclosure was located in the water, whereas zone IV of the enclosure was located 2–3 meters away from the water, as represented in the figure below. Snakes inside the enclosure were able to move freely between zones.



In a series of experiments, the researchers introduced a single snake into zone IV of the enclosure at 7:00 A.M. The researchers recorded the location of the snake at six time points throughout the day. In a related experiment, the researchers introduced two snakes, one of each species, into the enclosure at the same time and observed the location of each of the two snakes at the same six time points as before. The researchers repeated both the one-snake and two-snake experiments using different individual snakes of each species. The results are presented in the table.

ZONES MOST FREQUENTLY OCCUPIED BY GARTER SNAKES IN A MESHED ENCLOSURE

	Species Introduced Inside the Meshed Enclosure			
	T. atratus Only	*T. sirtalis* Only	*T. atratus* and *T. sirtalis*	
Time of Day	Zone Most Frequently Occupied by *T. atratus*	Zone Most Frequently Occupied by *T. sirtalis*	Zone Most Frequently Occupied by *T. atratus*	Zone Most Frequently Occupied by *T. sirtalis*
8:00 A.M.	I	II	III	II
10:00 A.M.	II	I	II	II
12:00 P.M.	I	I	III	I
2:00 P.M.	I	I	IV	I
4:00 P.M.	II	I	IV	I
6:00 P.M.	II	II	IV	I

43. Which of the following additions to the experimental design will best help test whether the observed habitat preferences were the result of competition between species?

8 Ecology MCQ

(A) Placing two individuals from the same population together inside the enclosure ✓

(B) Introducing different types of plants into the enclosure together with the snakes

(C) Doubling the number of repetitions for the experimental treatment with *T. atratus* alone

(D) Repeating the investigation with an enclosure that is twice as long as the original

44. The researchers discovered that both species of garter snakes feed almost exclusively on amphibians found in or near the pond. Based on the results of the investigation, which of the following is most likely to occur if the amphibians become a limited source of food?

(A) Individuals of *T. atratus* will no longer prefer to be near the water.

(B) *T. sirtalis* will shift its food preference to herbivory.

(C) The reproduction rate of *T. sirtalis* will increase.

(D) The population size of *T. atratus* will decrease. ✓

45. Based on the data in the table, which of the following best predicts the results of a study in which natural populations of *T. atratus* and *T. sirtalis* are observed together in an environment that includes a freshwater pond?

(A) Neither *T. atratus* nor *T. sirtalis* will be observed near the water.

(B) *T. atratus* will be observed near the water whether or not *T. sirtalis* is present.

(C) *T. sirtalis* will be observed near the water whether or not *T. atratus* is present. ✓

(D) *T. atratus* and *T. sirtalis* will be observed together near the water.

46. Based on the data in the table, which of the following best describes the habitat preference of *T. atratus* when introduced alone inside the meshed enclosure?

(A) *T. atratus* exhibited equal preferences for all four zones.

(B) *T. atratus* exhibited a preference for the zones in or near water. ✓

(C) *T. atratus* exhibited a preference for the zones most distant from the water.

(D) *T. atratus* exhibited a preference to remain within the zone in which it was initially introduced.

8 Ecology MCQ

47. A study recorded changes in the distribution of some commercial marine species off the East Coast of the United States over a period of several decades. The results of the study are summarized in Figure 1.

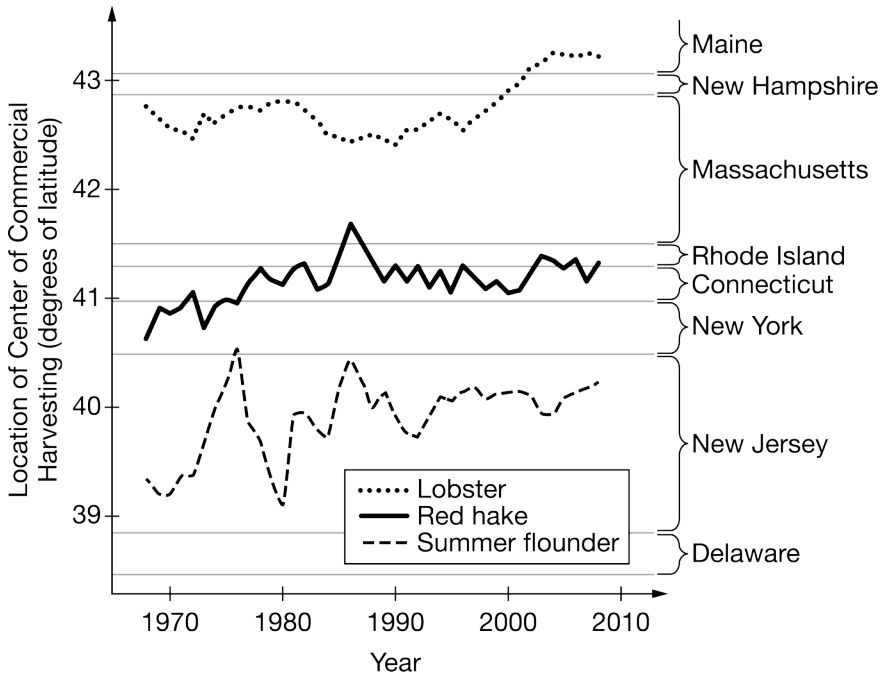


Figure 1. Latitude that is the center of commercial harvesting of specified marine species

Some researchers link changes in the distribution of these species to warming of the oceans due to climate change. Which of the following pieces of data best supports this alternative hypothesis?

(A) Decreased acidity of ocean water in the more southern latitudes caused a migration of these species northward.

(B) These species evolved new adaptations over time, allowing them to survive in warmer ocean environments.

(C) The prey of these species require colder waters to survive, and owing to warming in the oceans, they are moving farther north, to higher latitudes, where the water is colder. ✓

(D) Diseases devastated the prey of these species, forcing them to move northward as insufficient food was available in their original range.

48. The area covered by tropical rain forest is reduced by millions of hectares per year due to agriculture and logging. Which of the following best describes a likely result of tropical rain forest deforestation?

(A) Populations of plants and animals will decrease as more rain forest disappears, leading to a decrease in biodiversity. ✓

(B) An increase of soil moisture will lead to a rapid increase in new vegetation coverage.

(C) An increase in atmospheric carbon dioxide will lead to higher levels of ultraviolet radiation reaching the surface of Earth.

(D) More oxygen will be available to other organisms as plant numbers decrease.

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Answer A

This option is correct. It demonstrates an understanding of the human impact on ecosystems and correctly predicts that the loss of habitat from deforestation likely will lead to a loss of species diversity due to a decrease of available niches.

49. Henslow's sparrow is a songbird found in North American grasslands. The logging industry created vast meadow habitats for the sparrow in the late 1800s. Henslow's sparrow populations came to a peak in the mid-1900s. A recent survey of ten Midwest counties are shown in Figure 1.

Table 1. Year of Last Documented Observations of Henslow's Sparrow in Ten Selected Midwest Counties

County	Number of Confirmed Observations	Last Documented Observation
1	8	2007
2	1	2007
3	2	2007
4	2	2015
5	6	2015
6	3	2006
8	1	2004
9	4	2007
10	4	2007

Human activities can drive changes in ecosystems that affect the risk of extinction. An experiment is designed to help prevent the further decline and extinction of the Henslow's sparrow in which controlled burns will be set in the each of the ten Midwest counties in order to prevent the regrowth of the forest.

Which of the following represents an alternate hypothesis tested by this experiment with regard to the population sizes of Henslow's sparrows?

(A) There will be no difference in the populations observed in the controlled fire sites and the rest of counties 5 and 6.

(B) Population numbers will continue to decline even in the controlled fire sites because human activities always lead to extinction.

(C) Controlled fires in counties 6 and 8 will have no effect on the sparrow populations.

(D) Controlled fires in counties 4 and 5 will lead to an increase in the sparrow population. ✓

8 Ecology MCQ

Answer D

Correct. Counties 4 and 5 show evidence of existing sparrow populations, so the claim that controlled burns will affect these populations in an appropriate alternate hypothesis for this experiment.

50. Zebra mussels are an invasive species that has become widely established throughout the United States. Figure 1 shows the percent change in the population sizes of selected groups of organisms in the Hudson River since the introduction of zebra mussels. In this study chlorophyll-containing bacteria are considered phytoplankton and all other bacteria are considered bacterioplankton.

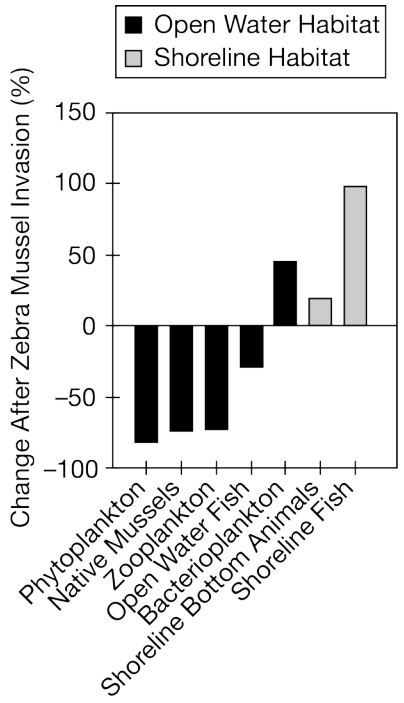


Figure 1. Change in biomass of selected Hudson River organisms after the introduction of zebra mussels

Which of the following hypotheses about the effect of zebra mussels on the Hudson River community is best supported by the data in Figure 1 ?

(A) Shoreline fish have replaced open water fish as top predators in a stable community.

(B) Zebra mussels have integrated into the existing community with little effect on the community stability.

(C) The reduction in primary productivity has resulted in an unstable community that will eventually collapse. ✓

(D) Bacterioplankton have replaced phytoplankton as the energy harvesters in the community.

8 Ecology MCQ

Answer C

Correct. Phytoplankton are primary producers that have experienced a large decline in number and have not been replaced by another primary producer. Thus, energy is not being harvested or distributed in the community, which contributes to instability and eventually the inability of the consumers to be supported.

Pacific salmon and black bears have often been cited as examples of keystone species. Pacific salmon spawn in freshwater streams but spend most of their lives at sea. When mature salmon return to the freshwater streams to spawn, they are preyed upon by bears and other predators. When salmon migrate from their marine habitat to the freshwater streams, they bring nitrogen and other marine-derived nutrients that subsequently remain in the areas surrounding the streams—a process called nitrogen influx.

In an investigation, the relationship between black bears, salmon, and influx of marine nitrogen into the area around a southwestern Alaskan stream was studied. The investigators established several test plots of the same size along the stream with the following species composition: no salmon or black bears (N), bears but not salmon (B), salmon but not bears (S), and a plot where salmon and bears interact (SB). Nitrogen influx in the different sampling areas was measured as a means of assessing the impact of the different species on the health of the ecosystem. The data are plotted in Figure 1.

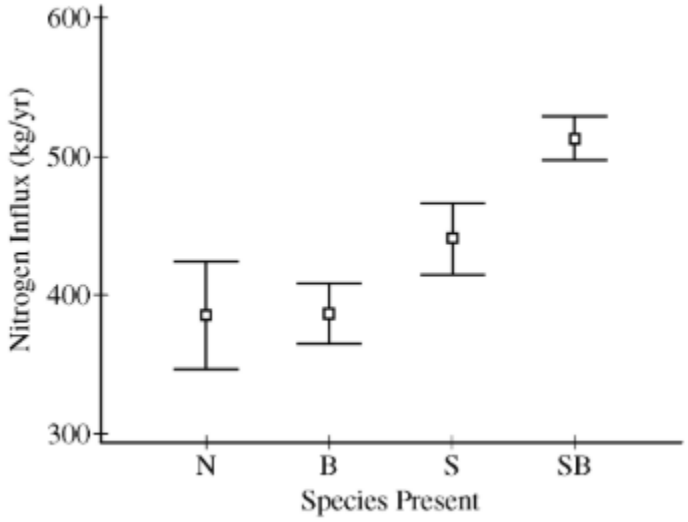


Figure 1. Mean nitrogen influx $\pm SE_{\bar{x}}$ as a function of species present. N = neither salmon nor bears present, B = only bears present, S = only salmon present and SB = both salmon and bears present.

51. If a dam is built downstream and prevents salmon migration to the test sites, which of the following most accurately predicts the impact on nitrogen influx?

8 Ecology MCQ

(A) Nitrogen influx will increase because the bears will no longer store nitrogen from the salmon.

(B) Nitrogen influx will decrease because there will be less bear-salmon interaction. ✓

(C) Nitrogen influx will remain stable because organisms other than bears and salmon will mobilize nitrogen in the environment.

(D) Nitrogen influx will increase because bears will no longer lose nitrogen to the salmon.

52. Which of the following statements is best supported by the data?

(A) Black bears are a keystone species.

(B) Salmon are a keystone species.

(C) Black bear urine is a significant source of nitrogen in the environment.

(D) The presence of black bears and salmon correlates with a significant increase in nitrogen influx. ✓

53. Which of the following most likely describes how the interaction between bears and salmon influences nitrogen dynamics in the environment?

(A) When bears consume salmon, they leave parts of the carcasses on the ground, which decompose, releasing nitrogen into the environment. ✓

(B) When salmon swim upstream to spawn, bears migrate to the rivers and deposit nitrogen from the forest with their excrement.

(C) Bears are a reservoir for nitrogen because they are the top consumer in the environment.

(D) Bears urinate in rivers and streams, which provides nitrogen for the salmon.

54. Which of the following pieces of additional data would help further investigate the relationship between bears, salmon, and influx of nitrogen into the local environment?

(A) The experimenters should remove the remains of salmon carcasses immediately after the salmon are eaten by the bears and determine the nitrogen content of the carcasses. ✓

(B) The experimenters should increase the number of bears in the area and measure the amount of nitrogen available for uptake by plants.

(C) The experimenters should set up a net to catch salmon before they enter the area and then measure nitrogen influx.

(D) The experimenters should clear-cut the trees from the area and determine how much nitrogen remains in the streams.

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55. Which of the following statements correctly describes a typical trend during the initial stages of succession in a terrestrial ecosystem?

8 Ecology MCQ

(A) Food webs become less complex.

(B) Diversity of plant species increases. ✓

(C) The amount of biomass and nonliving organic matter decreases.

(D) The litter layer becomes thinner because of increased assimilation of nutrients by shrubs and trees.

(E) The storage of inorganic nutrients within individual plants decreases.

The three-spined stickleback (Gasterosteus aculeatus) is a small fish found in both marine and freshwater environments. Marine stickleback populations consist mostly of individuals with pronounced pelvic spines, as shown in Figure 1. Individuals in freshwater stickleback populations, on the other hand, typically have reduced pelvic spines, as shown in Figure 2.

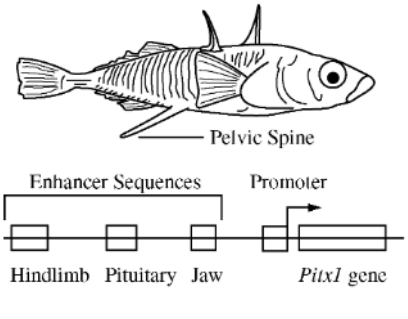


Figure 1. Marine stickleback

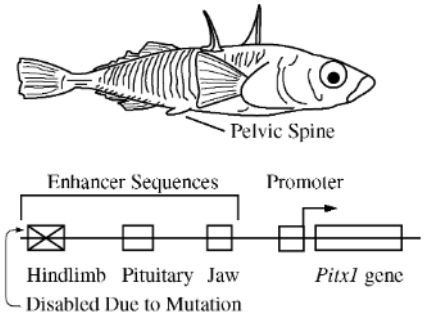


Figure 2. Freshwater stickleback

As represented in Figure 1 and Figure 2, the phenotypic difference between marine and freshwater sticklebacks involves Pitx1, a gene that influences the formation of the jaw, pituitary gland, and pelvic spine. Enhancer sequences upstream of the Pitx1 genetic locus regulate expression of the Pitx1 gene at the appropriate times and in the appropriate tissues during development. Previous studies have found that a mutation in the hindlimb enhancer interferes with the formation of a pronounced pelvic spine.

56. In sticklebacks, which of the following is most likely to occur if the jaw enhancer is disabled instead of the hindlimb enhancer?

(A) The jaw and a pronounced pelvic spine develop normally because the Pitx1 gene is expressed in both developing tissues.

(B) Neither the jaw nor a pronounced pelvic spine develop normally because there is no Pitx1 gene expression in either developing tissue.

(C) The jaw develops normally, but a pronounced pelvic spine does not develop because the Pitx1 gene is expressed in the developing jaw but not in the developing pelvis.

(D) The jaw does not develop normally, but a pronounced pelvic spine does develop because the Pitx1 gene is expressed in the developing pelvis but not in the developing jaw. ✓

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8 Ecology MCQ

57. Researchers were studying species diversity in a savanna rangeland community in southwest Queensland, Australia. Researchers counted the numbers of plants species in four different categories at two nearby sites within the savanna rangeland; one lightly grazed by herbivores and one heavily grazed by herbivores (Figure 1).

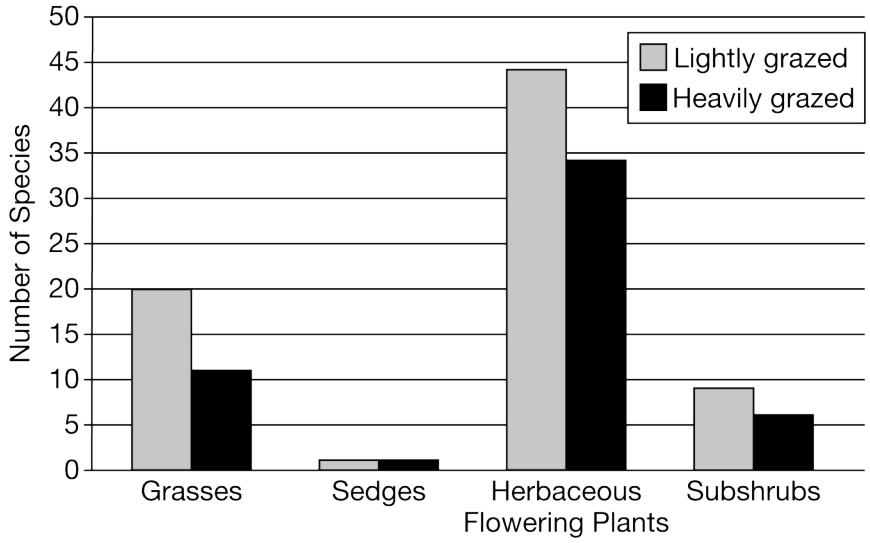


Figure 1. Plant species richness in lightly and heavily grazed savanna communities.

Which of the following best predicts the results of a drought in the savanna research area with regards to community resilience to this environmental change?

(A) The heavily grazed area is more likely to recover more quickly from the drought due to reduced competition between species.

(B) The lightly grazed area is likely to recover more quickly from the drought due to greater biodiversity. ✓

(C) The lightly and heavily grazed areas are likely to recover at equal rates due to the overlapping categories of plant species.

(D) The lightly and heavily grazed areas are likely to recover at equal rates due to the similar climates and locations.

Answer B

Correct. As shown in Figure 1, the lightly grazed area has a greater number of species of grasses, herbaceous flowering plants, and subshrubs. Higher levels of biodiversity in an ecosystem are associated with the greater ability of an ecosystem to respond to environmental changes. After a drought, some species will recover more quickly than others and the greater the biodiversity, the faster the ecosystem is likely to recover.

8 Ecology MCQ

58. The lionfish is a venomous fish found primarily in the Red Sea and the Indian Ocean. In the 1990s, lionfish were accidentally released into the Atlantic Ocean, where they found abundant resources and favorable environmental conditions. Which of the following scenarios is most likely to result in the lionfish having a major impact on the communities into which they were introduced?

(A) With no natural predators, the lionfish population will become very large. ✓

(B) Some native species of invertebrates will develop a resistance to lionfish venom.

(C) Random mating will allow the lionfish population to reach Hardy-Weinberg equilibrium.

(D) A virus that specifically infects lionfish will become more prevalent.

59. Scientific evidence has revealed five mass extinctions in the history of life on Earth.

Table 1. Five Mass Extinctions

Time period (million years ago)	Percent of species lost	Probable cause
65 mya	75%	Asteroid impact and volcanism
200 mya	75%	Asteroid impact and volcanism
250 mya	95%	Cold climate, formation of Pangea, Siberian volcanism
355 mya	70-80%	Global cooling
435 mya	22%	Global cooling

Some scientists propose that due to human activity and climate change, a sixth mass extinction has already begun. Select the best alternate hypothesis to support this claim.

(A) Current temperature changes will correlate with temperature changes during previous extinctions.

(B) Observed modern extinction rates will resemble the rates found early in previous extinctions. ✓

(C) The number of species present on Earth will equal the number of species on Earth before the last extinction.

(D) The number of species present on Earth will equal the number of species on Earth after the last extinction.

Answer B

Correct. By testing this hypothesis, we could establish a similarity to known extinctions. The null hypothesis would state that there is no correlation between the current extinction rate and those of previous extinctions.

8 Ecology MCQ

60. Scientists want to determine whether the rapid economic growth in China between 1950 and 2000 caused a mass extinction of animal species during that time period. A mass extinction event is considered to occur when the rate of species extinction far exceeds the background extinction rate. The scientists collected data on the number of extinctions of selected vertebrate specie in China between 1950 and 2000, as shown in Table 1. Extinction rates were ascertained by measuring the average percent of species lost (PSL).

Table 1. The average percent of extinctions of 252 protected vertebrate species in 2,365 counties in China from 1950 t 2000.

	Mammals	Birds	Amphibians and Reptiles	Background Rate
Average PSL ± 1 standard deviation	47.4 ± 33.59	19.75 ± 17.70	28.80 ± 28.86	27.21 ± 16.26

Which of the following is the null hypothesis most relevant for evaluating the data in Table 1 ?

(A) There is no significant difference between the extinction rate of animals between 1950 and 2000 in China and the background extinction rate. ✓

(B) There is no significant difference between the extinction rates of mammals, birds, amphibians, and reptiles in China.

(C) There is a significant difference between the extinction rate for mammals in China and the background extinction rate for mammals.

(D) Because of climate change, habitat loss, pollution, and excessive exploitation of resources, China experienced a mass extinction event between 1950 and 2000.

Answer A

Correct. The null hypothesis is that there is no significant difference between two groups of data. In this experiment, the null hypothesis would therefore be that there is no difference between the rates of extinction between 1950 and 2000 in China and the background rate of extinction.

8 Ecology MCQ

61. Potatoes as a food crop originated in South America. There are over 4,000 variations of potato grown there. Potatoes were brought to Europe by Spanish colonists in the late 1500s. One variety became the chief crop grown in much of Ireland by the early 1800s. Table 1 summarizes the spread of potatoes as a food crop.

Table 1. Spread of potatoes as food crop from origin in South America

Time Period	Location	Number of Potato Varieties Grown (approximate)
Pre–1567	South America	4,000
1567–1800	Europe	12
1840–1850	Ireland	1

The Irish Lumper variety of potato was introduced into Ireland in the early 1800s. The Lumper soon became the only variety of potato grown in much of Ireland. The potato blight destroyed most of the Lumper crop between 1845 and 1852 resulting in the death of about 1 million Irish from starvation.

A researcher suggests that the Irish monoculture (growing a single crop) of the Lumper potato did not significantly change the overall resistance of the potato crop in Ireland to diseases. Which of the following observations best refutes this null hypothesis?

(A) Before the arrival of the potato blight, there were already two other potato plant diseases in Ireland.

(B) The number of fatalities from the Irish potato famine was greater than the rest of the European countries combined. ✓

(C) During the early nineteenth century, the vast majority of people in Ireland depended on the Lumper potato for food.

(D) Growing more varieties of potatoes increases competition for available resources, which may yield fewer potatoes.

Answer B

Correct. Table 1 shows that twelve different varieties of potatoes were grown across Europe, as compared to the Lumper variety in Ireland. This suggests that the European potato crop, with more varieties, was less affected by the potato blight. This observation would support the alternative hypothesis that the variability of potato crops did affect the impact of the potato famine in Europe compared to Ireland.

62. Which of the following observations best represents a mutualistic relationship?

8 Ecology MCQ

(A) Some bacteria and fungi obtain their nutrients by enzymatically digesting larger organisms.

(B) Chloroplasts in green algae provide sugars for use in cellular metabolism.

(C) Hermit crabs utilize empty sea snail shells to protect themselves from predators.

(D) Flowers produce nectar that bees gather to make honey, and in the process the bees pollinate the flowers. ✓

63. Sea otters living along the Pacific coast were hunted to near extinction in the nineteenth century. After being protected from hunting in the early 1900s, a remnant population of otters near Adak Island, Alaska, recovered rapidly. Otters did not return to the environmentally similar nearby island, Alaid Island. Sea otters eat sea urchins, which eat kelp, a brown alga. Researchers surveyed both islands in 1988, to measure sea urchin biomass and kelp density. The data are presented in Table 1.

Table 1. Kelp and sea urchin data from two Alaskan islands in 1988

	Alaid Island	Adak Island
Understory Kelp Density (Number $\cdot 0.25\mathrm{m}^{-2}$)	0.23	6.56
Sea Urchin Biomass ($\mathrm{g} \cdot 0.25\mathrm{m}^{-2}$)	336.4	38.4

In 1991, researchers at Adak Island observed the first attack by a killer whale on a sea otter in historical times. The researchers hypothesized that the population sizes of the larger marine mammals that the killer whales normally prey on declined, so the killer whales were starting to prey on different prey, including the smaller sea otters.

Which of the following best predicts the effects of an increase in killer whale predation on sea otters on the Adak Island ecosystem?

(A) Both the sea urchin population and the kelp population will increase, leading to greater total biomass and diversity in the Adak Island ecosystem.

(B) Both the sea urchin population and the kelp population will decrease, leading to reduced total biomass and diversity in the Adak Island ecosystem.

(C) The sea urchin population will increase and the kelp population will decrease, leading to greater total biomass and diversity in the Adak Island ecosystem.

(D) The sea urchin population will increase and the kelp population will decrease, leading to reduced total biomass and diversity in the Adak Island ecosystem. ✓

Answer D

Correct. Increased predation by killer whales will lead to a decrease in the sea otter population at Adak Island. A decline in the sea otter population will allow the sea urchin population to increase without predation as a limiting factor. An increasing sea urchin population will lead to overgrazing and reduced kelp densities. Kelp forests provide important habitat and food for many other organisms, and their decline would cause the entire ecosystem to decline. Due to this cascade effect, sea otters are described as a keystone species because when they are removed from an ecosystem, the ecosystem declines

rapidly.

64.

TABLE 1. OZONE DAMAGE IN SIX PLANT SPECIES

Species	Number of Plants Observed	Number of Damaged Plants	Percent of Plants with Damage
Ash	544	41	7.5
Cherry	30	2	6.7
Dogwood	41	2	4.9
Sassafras	132	3	2.3
Sweet gum	170	2	1.2
Wild grape	124	20	16.1

To investigate the effects of ground-level ozone pollution on different species of plants, researchers recorded the number of plants in a certain area that had ozone-induced leaf damage. Based on the results of the survey shown in Table 1, which of the following plant species was most affected by ozone pollution?

(A) Ash

(B) Cherry

(C) Sweet gum

(D) Wild grape ✓

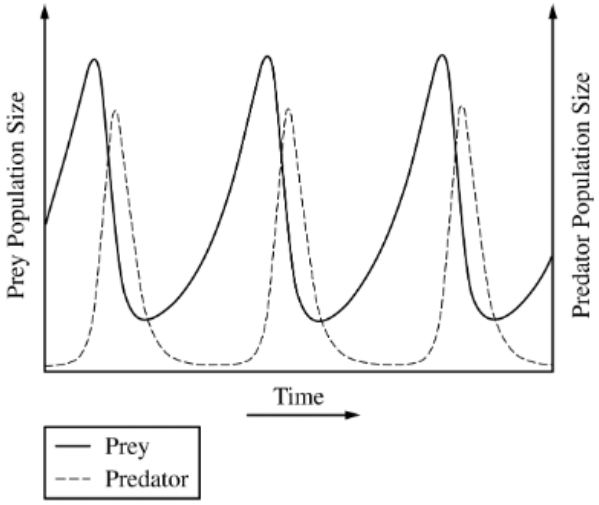
Answer D

Correct. Wild grape plants had the second-highest number of damaged plants observed. However, when calculated as a percentage of total plants observed, wild grape plants had the highest percentage of plants

8 Ecology MCQ

damaged.

65. The graph below shows a model of population changes over time in a predator and its prey.



Which of the following data would be most helpful in predicting possible disruptions to the relationship between the predator and its prey?

(A) Time of day, because as days become shorter, nighttime hunting increases

(B) Average birth weight of predators, because heavier offspring will begin hunting at younger ages

(C) Numbers of additional predator species, because additional predators will increase competition for prey ✓

(D) Prey metabolic rates, because predators are less successful against prey with higher metabolic rates

8 Ecology MCQ

66.

	Diet	
	Before Change	After Change
Principal diet	Hay	Grain
Main type of carbohydrate supplied by diet	Cellulose	Starch
Abundance of *Streptococcus bovis* in the rumen	Low	High
Abundance of *Ruminococcus albus* in the rumen	High	Low
pH of rumen fluid	6 – 7	< 5.6
Clinical warning signs in cattle	None	Weakness

Researchers conducted a controlled experiment to investigate the effect of diet on the health of beef cattle. The initial hypothesis was that an abrupt change in diet will benefit beef cattle by reducing the sizes of bacterial populations living in the digestive systems of the cattle.

In the experiment, the researchers determined the relative abundance of two bacterial species found in the rumen of cattle. The rumen is a part of the stomach that acts as a fermentation chamber in cattle and other ruminants. Shown in the table are the results from before and after an abrupt change in the cattle's diet. Based on the results, which of the following best explains why the initial hypothesis should be revised?

(A) The diets were too similar, since cellulose and starch are both carbohydrates obtained from eating plants.

(B) The abundance of one of the bacterial species increased in response to the change in diet. ✓

(C) The change in the diversity of microorganisms had no observable effect on cattle physiology.

(D) The ability of rumen fluid to resist changes in pH counteracted the changes in microorganism diversity.

8 Ecology MCQ

In the Florida Everglades, Burmese pythons are an invasive species. They were introduced into southern Florida in 1992. These pythons feed on many of the native Florida species, establishing the pythons as the top predator in the environment. By the year 2000, their population had increased dramatically.

Figures 1 and 2 display data collected by ecologists studying the results of the Burmese python invasion. Figure 1 shows counts of animals collected from nighttime road surveys in southern Florida, which are used to estimate population size. Figure 2 shows data collected from mosquitoes captured from the wild. DNA sequencing was used to identify the species of blood that the mosquitoes had in their stomachs, identifying various hosts used by the mosquitoes.

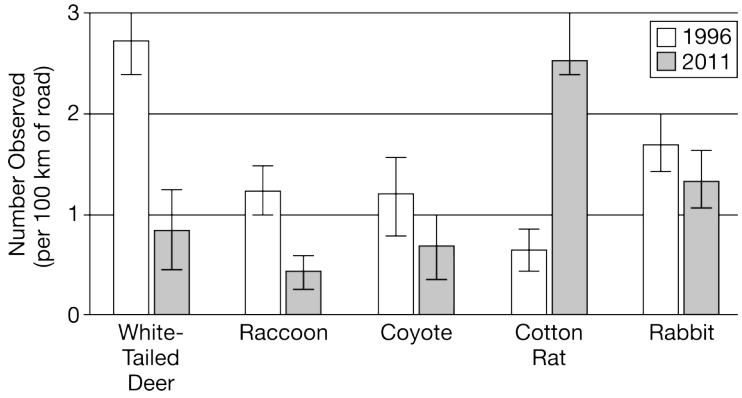


Figure 1. Comparison of observations of selected mammals in 1996 and 2011

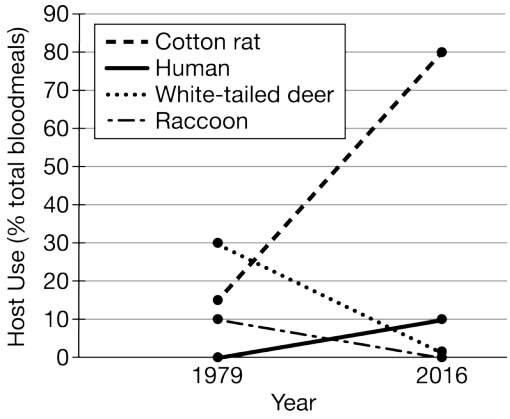


Figure 2. Change in host preference by *Culex cedecei* between 1979 and 2016. Numbers do not add up to one hundred percent because these represent a subset of all the host species.

In 1996, the native *Culex cedecei* mosquitoes in southern Florida preferentially took blood meals from white-tailed deer and raccoons. It was predicted that changes in host population size would alter these host preferences. Additionally, it is known that cotton rats are often infected by the Everglades virus, which normally exists in animals, but is capable of infecting humans. Ecologists predict that increased feeding on cotton rats by *C. cedecei* may significantly increase the tendency of this virus to infect humans.

8 Ecology MCQ

67. Which of the following most accurately explains an impact of Burmese pythons on the Everglades community in southern Florida using the data provided?

(A) Python parasitism has reduced the population of white-tailed deer and raccoons.

(B) Python predation has caused a trophic cascade that changed the mosquitoes' niche. ✓

(C) Pythons have a mutual symbiosis with cotton rats, allowing their population to increase.

(D) Intraspecific competition for food between white-tailed deer and raccoons has increased due to pythons.

Answer B

Correct. Since predation by invasive Burmese pythons has decreased the number of white-tailed deer and raccoons, the mosquitoes are forced to use alternative hosts, including cotton rats.

68. Which of the following populations have significantly decreased in size between 1996 and 2011 ?

(A) White-tailed deer only

(B) Cotton rats and rabbits only

(C) White-tailed deer and raccoons only ✓

(D) White-tailed deer, raccoons, and cotton rats

Answer C

Correct. The error bars do not overlap for the 1996 and 2011 population estimates for white-tailed deer and for raccoons. This indicates that there is a significant decrease between the 1996 and 2011 population sizes.

69. Based on the data, which of the following most accurately matches a potential change with its predicted effects on the southern Florida ecosystem?

(A) Introduction of a new top predator species that preys solely on Burmese pythons would cause an increase in the raccoon population. ✓

(B) If the population of Burmese pythons were to decrease, then the cotton rat population would increase.

(C) If more Burmese pythons are introduced in the Everglades, then mosquitoes will begin feeding on white-tailed deer frequently again.

(D) Introduction of a new snake species that competes with the Burmese python for food would lead to an increase in the white-tailed deer population.

8 Ecology MCQ

Answer A

Correct. If the Burmese python population were reduced, then there would be less predation on raccoons, and the raccoon population would be able to grow larger. The same would be true of the white-tailed deer population.

70. Scientists hypothesize that the invasive Burmese pythons are responsible for recent increases in human Everglades virus infections. Evaluate this alternate hypothesis using the data provided.

(A) The alternate hypothesis is supported because invasive Burmese pythons caused an increase in the mosquito population. This enabled them to transmit the Everglades virus at a higher frequency if they were infected.

(B) The alternate hypothesis is supported because the invasive Burmese pythons caused mosquitoes to switch to feeding on cotton rats. This means that mosquitoes are more likely to carry the Everglades virus when they bite humans. ✓

(C) The alternate hypothesis is refuted because humans are used as hosts for only a small percentage of mosquito feedings. Humans therefore will not have an increase in Everglades virus infection.

(D) The alternate hypothesis is refuted by the data because the population of cotton rats increased. Since there are more cotton rats, they will contain the Everglades virus and prevent it from infecting humans.

Answer B

Correct. As mosquitoes change hosts from white-tailed deer and raccoons to cotton rats, they will more frequently be carriers of the Everglades virus. Thus, interactions with humans are more likely to result in transmission of the Everglades virus.

8 Ecology MCQ

71. Soluble reactive phosphate (SRP) is a term used to describe the amount of phosphate available for uptake by living things. A study of river plants species and water phosphate levels was completed in Great Britain. Recently, in a section of one of the rivers studied, a population of fennel-leaved pondweed was replaced by spiked water-milfoil, even though these two plants are associated with different environmental conditions as indicated in the table below.

Table 1. Concentrations of phosphate and river plant correlations in Britain

SRP **Concentration** (mg/L)	**Plant Species Observed**
< 0.1	Bog pondweed *(Potamogeton polygonifolius)* and river water-crowfoot *(Ranunculus fluitans)*
$0.1 - 0.4$	Fennel-leaved pondweed *(Potamogeton pectinatus)*
0.4–1.0	Yellow water-lily *(Nuphar lutea)* and arrowhead *(Sagittarias agittifolia)*
> 1.0	Spiked water-milfoil *(Myriophyllum spicatum)*

A group of researchers suggest that the concentration of available phosphate in this particular section of the British river does not have a significant impact on which plant species grow. Alternatively, another group of researchers hypothesize that levels of available phosphate do affect which plant species grow at this location. Which of the following observations would best support this alternative hypothesis?

(A) Reforestation efforts in Britain near this section of the river have caused changes in the plant community structure.

(B) Increased fertilizer use near this section of the river has led to increased runoff of phosphates into the river. ✓

(C) Spiked water-milfoil adapted to lower phosphorous conditions has been observed to outcompete fennel-leaved pondweed in this section of the river.

(D) Measured levels of SRP in this section of the river have remained constant over a period of time.

Answer B

Correct. Human activities such as the addition of organic or chemical fertilizers to lawns, golf courses, and agricultural lands and the observed increase in runoff of the phosphorus in these fertilizers into the rivers is the most likely contributor to the increased SRP level in the water that resulted in the change in plant species.

8 Ecology MCQ

Researchers investigated the effects of nonnative trout on the behavior of native salmon inhabiting a freshwater stream. Juvenile trout compete with juvenile salmon for resources, and adult trout prey on juvenile salmon.

Researchers built enclosed channels in a freshwater stream. A transparent plastic mesh screen divided each channel into two sections, an upstream predator section and a downstream competitor section. Similar screens placed at the ends of each channel permitted the flow of water and small invertebrate organisms, eaten by juvenile members of both species of fish, through the channel. Juvenile salmon were placed in only the competitor section of each channel, in the absence or presence of juvenile trout, and in the absence or presence of adult trout placed only in the predator section (Figure 1). The researchers then recorded the percent of juvenile salmon that were active during the day and night, under the different conditions (Figure 2).

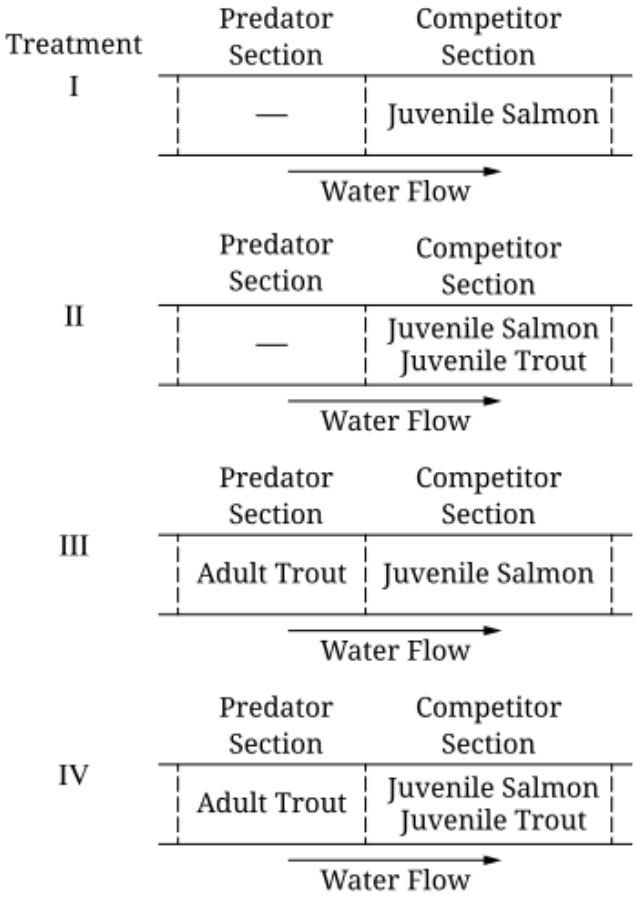


Figure 1. Experimental design of four freshwater stream channels used to monitor juvenile salmon behavior in the presence or absence of predator and/or competitor fish

8 Ecology MCQ

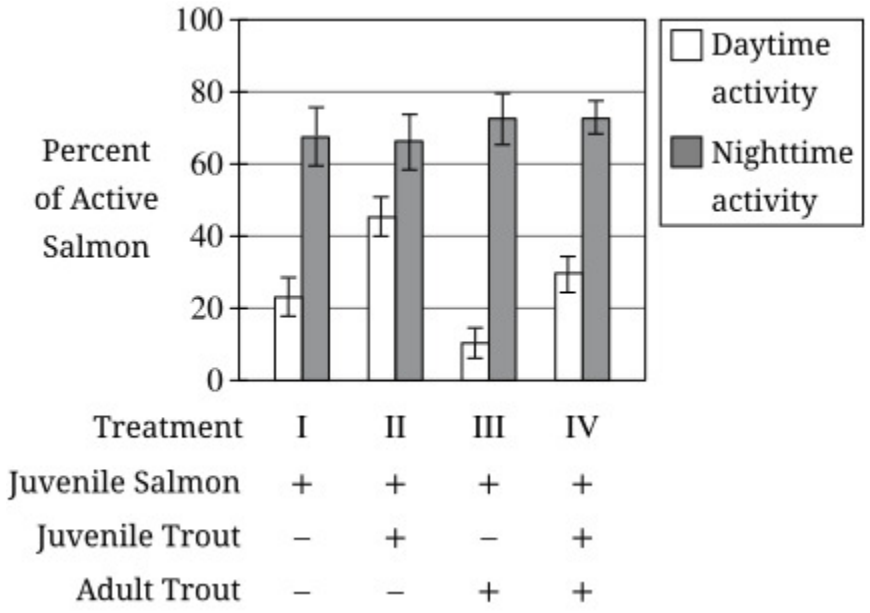


Figure 2. Activity of juvenile salmon in freshwater stream channels in the presence (+) or absence (−) of juvenile and/or adult trout. Error bars represent $\pm 2\mathrm{SE}_{\bar{x}}$.

72. Which of the following descriptions of juvenile salmon activity is best supported by the data in Figure 2 ?

(A) Daytime activity increases when only a predator is present.

(B) Nighttime activity decreases when only a competitor is present.

(C) Daytime activity increases when only a competitor is present. ✓

(D) Nighttime activity increases in the presence of competitors and predators.

Answer C

Correct. Juvenile salmon daytime activity increased when only a competitor was present (Treatment II) in comparison with daytime activity in the absence of a competitor (Treatment I) and was additionally the highest daytime activity observed in the experiment.

73. The researchers claim that the risk to juvenile salmon from predation is about the same as the risk due to competition. Which of the following best supports the researchers' claim?

8 Ecology MCQ

(A) The daytime activity in treatment IV is about the same as the daytime activity in treatment I and intermediate between the daytime activity in treatments II and III. ✓

(B) In treatment III, the daytime activity is the lowest of all of the treatments and the nighttime activity is the highest of all of the treatments.

(C) In treatment IV, the nighttime activity is twice as great as the daytime activity.

(D) The nighttime activity is about the same in all four treatments.

Answer A

Correct. The presence of both the competitor and the predator in treatment IV results in the juvenile salmon exhibiting a daytime activity level that is intermediate between the daytime activity in the presence of the competitor only (treatment II) or the predator only (treatment III), thus resembling the daytime activity in treatment I. In treatment IV, the juvenile trout must increase their activity because their food competitor is present (as in treatment II), but they are also cautious because the predator is present (as in treatment III).

74. A researcher claims that the predator avoidance behavior of juvenile salmon observed in the study has evolved because of natural selection. Which of the following data could best be used to support the claim?

(A) The salmon that avoid predators have longer life spans as compared with salmon that do not avoid predators.

(B) The salmon that avoid predators can grow to a larger size as compared with salmon that do not avoid predators.

(C) The salmon that avoid predators are able to access more resources as compared with salmon that do not avoid predators.

(D) The salmon that avoid predators have a greater number of viable offspring as compared with salmon that do not avoid predators. ✓

Answer D

Correct. Natural selection is the result of increased survival and reproduction of individuals with beneficial phenotypes. Traits of such individuals are then passed on to offspring, allowing the population to evolve over time.

75. The researchers hypothesize that juvenile salmon use chemical cues to detect predators that are nearby. Which of the following modifications to the experiment will help the researchers test their hypothesis?

8 Ecology MCQ

(A) Remove the mesh screen from the channel so that the predators have access to the juvenile salmon.

(B) Change the light schedule so that there are twice as many nighttime hours as daytime hours.

(C) Add rocks to the channel that the juvenile salmon can use to hide from predators.

(D) Replace the mesh screen with a solid, transparent barrier that allows light to pass through. ✓

Answer D

Correct. A solid barrier will prevent the movement of chemical signals in the water and therefore allow researchers to determine the influence of chemical cues on juvenile salmon behavior.

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76. Students in a class were divided into five groups to analyze community structure. Each group of students assessed the species of plants in four environments and determined the Simpson's Diversity Index for each environment. The Simpson's Diversity Index takes into account the number of species present as well as the abundance of each species. The index values are shown in Figure 1.

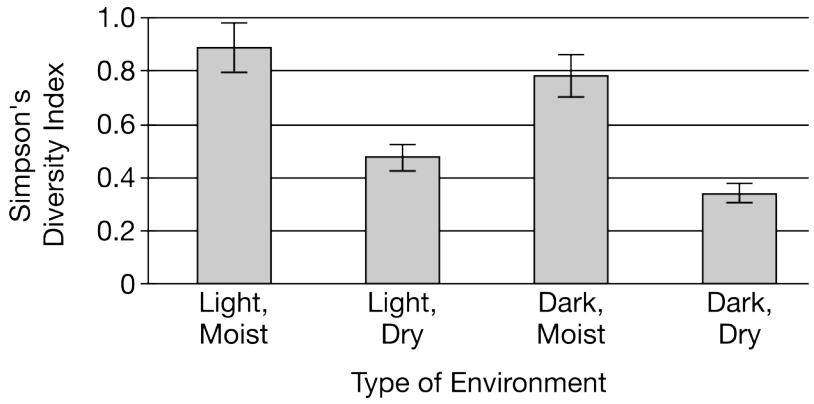


Figure 1. Simpson's Diversity Index for four environments

Based on the data in Figure 1, which of the following statements is most accurate about the species diversity of the four different environments that the students analyzed?

(A) The light, moist environment is the most ideal environment for plants.

(B) The most diverse environment is the dark, dry environment because plants are best able to protect themselves from drying out during the summer months.

(C) Based on the data, there is no statistical difference between the species diversity of plants in the light, moist environment and in the dark, moist environment. ✓

(D) Based on the data, there is no statistical difference between the species diversity of plants in light, dry and dark, dry environments.

8 Ecology MCQ

Answer C

Correct. Because the error bars overlap, the light, moist and dark, moist environments show no significant statistical difference in diversity.

77.

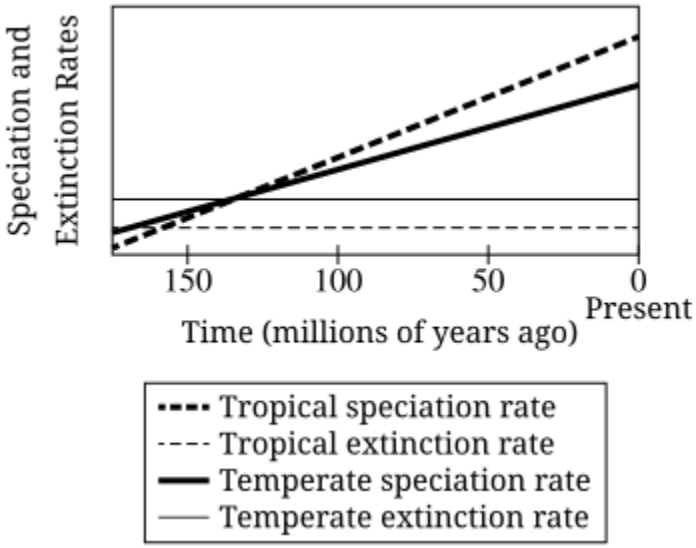


Figure 1. Estimate of rates of speciation and rates of species extinction in temperate and tropical biomes over the last 175 million years

Assuming both temperate and tropical biomes had equivalent biodiversity 175 million years ago, which of the following claims is best supported by the data in Figure 1 ?

(A) Tropical biomes in the present have less biodiversity than do temperate biomes.

(B) Tropical biomes in the present have greater biodiversity than do temperate biomes. ✓

(C) The rate of extinction is lower in temperate biomes than in tropical biomes.

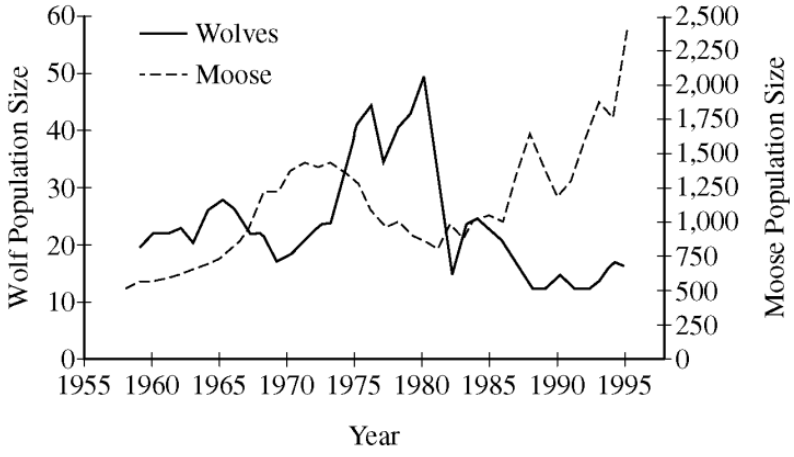
(D) The rate of speciation in both temperate and tropical biomes has not changed over time.

Answer B

Correct. Based on the data, tropical biome speciation has increased more rapidly than has temperate biome speciation and has a lower constant extinction rate than that of temperate biomes. Given this difference in overall net speciation, tropical biomes are expected to have more biodiversity than temperate biomes.

8 Ecology MCQ

78.



The graph above represents the number of individuals in a population of wolves and in a population of moose observed in the same isolated geographic area over a 40-year period, from 1955 through 1995. Which of the following statements about the two populations is best supported by the information presented in the graph?

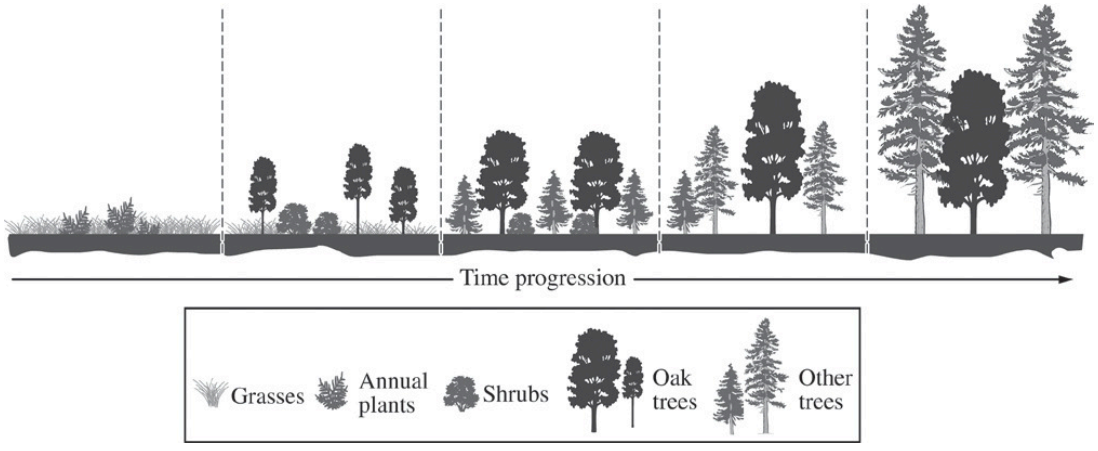
(A) The reproductive rate of the wolves was greater than the reproductive rate of the moose.

(B) Mutualism allowed the two populations to reproduce while occupying the same ecological niche.

(C) Speciation occurred when the two populations became reproductively isolated from each other.

(D) The wolves were predators of the moose, which were otherwise reproductively successful. ✓

79.



The diagram above shows the progression of ecological events after a fire in a particular ecosystem. Based on the diagram, which of the following best explains why the oak trees are later replaced by other trees?

(A) Eventually the other trees grow taller than the oak trees and form a dense canopy that shades the understory. ✓

(B) Oak trees alter the pH of the soil, making the forest better suited for shrubs and other trees.

(C) Roots of shrubs proliferate in the soil of the forest and prevent the oak trees from obtaining water.

(D) Oak trees succumb to environmental pollutants more readily than do either the shrubs or the other trees.

8 Ecology MCQ

Answer A

This option is correct. The other trees shade out the oak trees. Oak seedlings are relatively intolerant to shade. Oak trees need sunlight for photosynthesis, and although they do survive in poor sunlight, they do not proliferate enough to be the dominant plant in the forest. Some oak species need sunlight for full development.

80. Thrips are insects that feed on rose pollen. Scientists noted that the thrips population increased in the spring and decreased dramatically during the summer. The researchers hypothesized that food abundance was the limiting factor for the population. Which of the following types of data would be most useful for the scientists to collect at regular intervals on a designated test plot of rose plants?

(A) Amount of sunlight (hours/day)

(B) Mean temperature (^{o}C)

(C) Density of rose pollen produced (g/m^2) ✓

(D) Amount of pollen produced by each flower (g/flower)

8 Ecology MCQ

81. Figure 1 shows the changing pattern of human population sizes in rural and urban areas from 1960 to 2016 with regard to the global population.

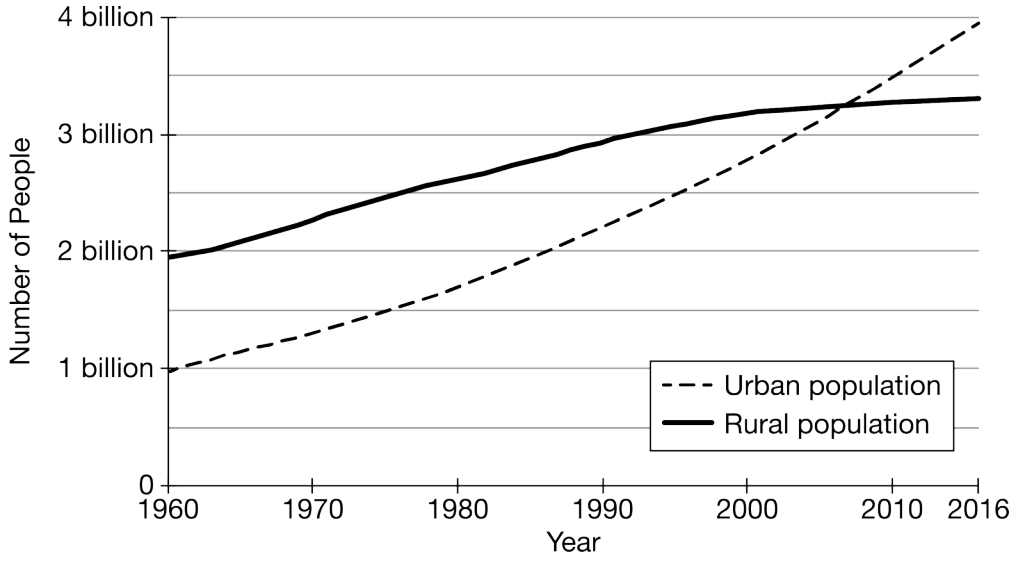


Figure 1. Comparison of population size in urban and rural areas from 1960 to 2016

A scientist interested in investigating how human population trends might affect local animal species' richness stipulates a null hypothesis that animal biodiversity in rural areas will be unaffected in the future by the human population trends shown in Figure 1. Which of the following would best refute this null hypothesis?

(A) More rural areas will become available for animal habitats over time.

(B) Urbanization causes habitat fragmentation. ✓

(C) Urbanization decreases the per person fossil fuel use in urban areas.

(D) Water in urban areas is treated before it is released into the rivers that flow into rural areas.

Answer B

Correct. As humans infringe on species' habitats, causing fragmentation, the organisms must adapt to new, smaller habitats, and biodiversity typically decreases.

8 Ecology MCQ

82. Water in a pond contaminated with the weed killer atrazine is suspected of inhibiting metamorphosis in northern leopard frogs. A team of scientists collected fertilized northern leopard frog eggs from a different pond that is not contaminated. Which of the following is the best experimental design to determine whether atrazine is responsible for inhibiting metamorphosis in northern leopard frogs?

(A) Place half of the fertilized eggs in a pool of water with the same concentration of atrazine as the contaminated pond and place the other half of the fertilized eggs in a pool of water that has no atrazine. Monitor the development of the embryos through metamorphosis into adulthood. ✓

(B) Place all of the fertilized eggs in a pool of pond water with the same concentration of atrazine as the contaminated pond and compare the number of frogs that reach metamorphosis to those that reach adulthood in the contaminated pond.

(C) Allow all fertilized eggs to develop into adults. Expose one-third of the frogs to one-half of the concentration of atrazine in the contaminated pond and expose another one-third of the frogs to the same concentration of atrazine as the contaminated pond. Leave the last one-third of the frogs in water with no atrazine and note any adverse changes in the physical condition of the atrazine-treated frogs in three months.

(D) Divide the fertilized eggs into three groups and expose each group to a different concentration of atrazine. Release the eggs back into the contaminated pond and check for metamorphosis after three months.

8 Ecology MCQ

In the mid-1920s, gray wolves, *Canis lupus*, were eliminated from Yellowstone National Park in the northwestern United States. Over the course of 1995 and 1996, 31 wolves from Canada were relocated to the park. Researchers have tracked the growth of the wolf population from 21 in 1995 to 108 at the end of the study in 2016 (Figure 1).

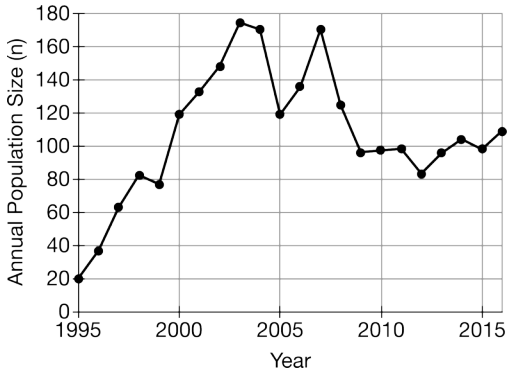


Figure 1. Yellowstone National Park annual wolf population from 1995 to 2016

The wolves prey on large herbivores, primarily elk. The carcasses of wolf kills are a valuable food source for coyotes, *Canis latrans*. When coyotes are feeding on carcasses, they may be chased off the carcass and/or killed by wolves. Coyotes also hunt small mammals for food.

Researchers were studying the effect of wolves' presence on the behavior of coyotes. Researchers observed coyotes in a specific area of Yellowstone National Park and recorded the proportion of the coyotes' time spent on five different activities when wolves were present in the study area and when wolves were absent (Figure 2).

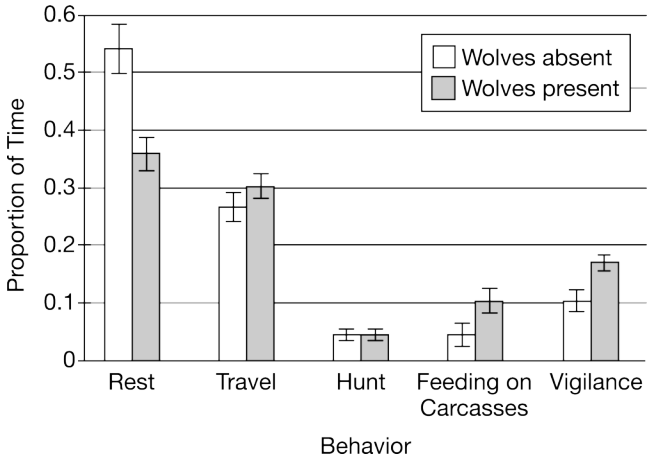


Figure 2. Coyote behavior in the presence and absence of wolves. Error bars represent a confidence interval of 95% (± 2 standard errors of the mean).

83. Which of the following best describes the change in coyote behavior in the presence of wolves?

8 Ecology MCQ

(A)

Rest	Travel	Hunt	Feeding on Carcasses	Vigilance
Increases	No effect	No effect	Decreases	Decreases

(B)

Rest	Travel	Hunt	Feeding on Carcasses	Vigilance
Increases	Decreases	No effect	Decreases	Decreases

(C) ✓

Rest	Travel	Hunt	Feeding on Carcasses	Vigilance
Decreases	No effect	No effect	Increases	Increases

(D)

Rest	Travel	Hunt	Feeding on Carcasses	Vigilance
Decreases	Increases	No effect	Increases	Increases

Answer C

Correct. According to Figure 2, in the presence of wolves (white bars), the proportion of time spent on rest is less, the proportion of time spent feeding on carcasses is more, and the proportion of time spent on vigilance is more than in the absence of wolves (gray bars). Although the mean proportion of time spent on travel increases in the presence of wolves, the error bars overlap, showing that the difference is not statistically significant.

8 Ecology MCQ

84. Following the reintroduction of wolves to Yellowstone National Park, researchers measured the mean height of young aspen trees in upland and streamside areas.

Table 1. Mean Height of New Aspen Growth in Early Spring

	Year	
	1998	2010
Upland	31 cm	176 cm
Streamside	33 cm	237 cm

Which of the following is the most likely cause of the change in mean aspen height from 1998 to 2010 ?

(A) The reintroduction of wolves led to a decrease in the population of elk, allowing aspen trees to grow taller. ✓

(B) The reintroduction of wolves led to a decrease in the population of small mammals, allowing aspen trees to grow taller.

(C) The reintroduction of wolves led to an increase in the number of carcasses that provided fertilizer for young aspen trees.

(D) The reintroduction of wolves led to an increase in the number of carcasses, so coyotes no longer ate young aspen trees.

Answer A

Correct. Top carnivores are keystone species, meaning they have a larger role in maintaining an ecosystem than would be expected by their population size. After wolves were eliminated from Yellowstone National Park, the population of elk rose dramatically. The high elk population led to the overgrazing of plant species, including aspen. With the reintroduction of wolves, the elk population has declined, allowing plant species to recover.

85. Identify the independent variable for the experimental results shown in Figure 2.

(A) The presence of wolves ✓

(B) The proportion of time

(C) Coyote behavior

(D) The study area

8 Ecology MCQ

Answer A

Correct. The independent variable is selected by the researchers, and the researchers test its effect on the dependent variable. In this experiment, the researchers were looking at the effect of wolves' presence on the behavior of coyotes. Therefore, the presence of wolves is the independent variable, and the behavior of coyotes is the dependent variable.

86. Based on information in Figure 1, which of the following is closest to the annual growth rate of the wolf population in Yellowstone National Park from 1996 to 2004 ?

(A) 4 wolves per year

(B) 17 wolves per year ✓

(C) 87 wolves per year

(D) 135 wolves per year

Answer B

Correct. The equation for calculating a rate is $\frac{dY}{dt}$, which is the difference in the variable on the y-axis divided by the difference in the variable on the x-axis. In this case, that equation becomes annual population growth rate = (final population size - initial population size)/(number of years). According to Figure 1, in 1996 the population was 36 wolves, and 8 years later, in 2004, the population was 171 wolves. Therefore, $\frac{171-36}{8} = 16.875$, which can be rounded to 17 wolves per year.

87. Based on the data in the figure, a student claimed that since 2007, the carrying capacity of wolves in Yellowstone National Park has been 100 wolves, and the maximum annual per capita growth rate of gray wolves is 0.6 wolves per wolf per year.

Which of the following is closest to the calculated wolf population size in 2017 based on the student's claim?

(A) 100

(B) 103 ✓

(C) 113

(D) 173

8 Ecology MCQ

Answer B

Correct. According to the logistic growth model, the change in population size $\left(\frac{dN}{dt}\right)$ can be calculated using the equation $\frac{dN}{dt} = r_{\max}N\left(\frac{K-N}{K}\right)$. According to the student's claim, $r_{\max} = 0.6$ wolves/year and $K = 100$ wolves. From Figure 1, $N = 108$ wolves in 2016. Therefore, $\frac{dN}{dt} = 0.6 \times 108 \times \left(\frac{100-108}{100}\right) = -5.184$. The –5.184 needs to be added to the 2016 population size to get the population size for 2017: $108 + (-5.184) = 102.816$, which is rounded to 103 to get the final answer.

---

88. Researchers were studying the effects of microbes on the growth of the yellow bedstraw plant, *Galium verum*. They grew yellow bedstraw plants under four treatments: in the presence of soil bacteria only, arbuscular mycorrhizal (AM) fungi only, both bacteria and AM fungi, or neither. After 6 months, they measured the biomass of the plants (Figure 1).

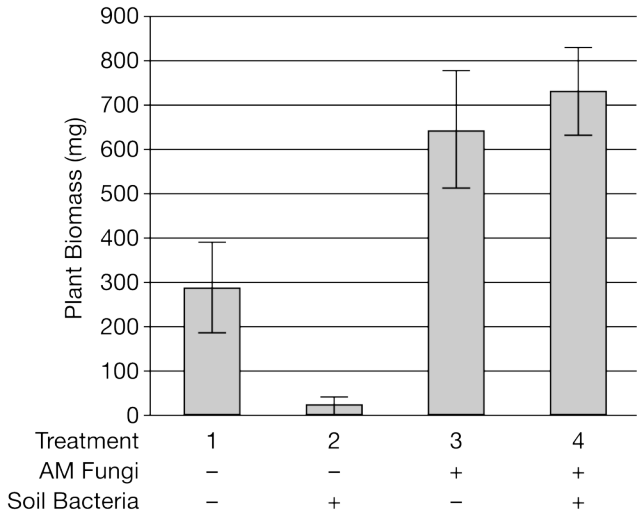


Figure 1. The Effect of Soil Microbes on the Biomass of *Galium verum* $(\text{Mean} \pm 2SE_{\overline{x}})$

Which of the following best describes a statistical relationship indicated in Figure 1 ?

(A) Yellow bedstraw plants grow significantly more in the presence of both soil bacteria and AM fungi (treatment 4) than in the presence of soil bacteria alone (treatment 2). ✓

(B) Yellow bedstraw plants, in the presence of AM fungi (treatment 3), grow significantly more if soil bacteria (treatment 4) are also present.

(C) Yellow bedstraw plants grow significantly more when soil bacteria (treatment 2) are present compared to when there are no soil bacteria (treatment 1).

(D) Yellow bedstraw plants grown in soil without bacteria and AM fungi (treatment 1) grow significantly more than when only AM fungi is present in the soil (treatment 3).

8 Ecology MCQ

Answer A

Correct. The error bars in treatments 2 and 4 do not overlap, which means that the differences between the two data sets are statistically significant. Thus, AM fungi significantly increases yellow bedstraw growth when these plants are exposed to soil bacteria.

89. Zebra mussels, *Dreissena polymorpha*, were introduced into the Great Lakes on shipping vessels, first appearing in Lake St. Clair in the 1980s. Since that time, they have rapidly expanded in range, as shown in Figure 1.

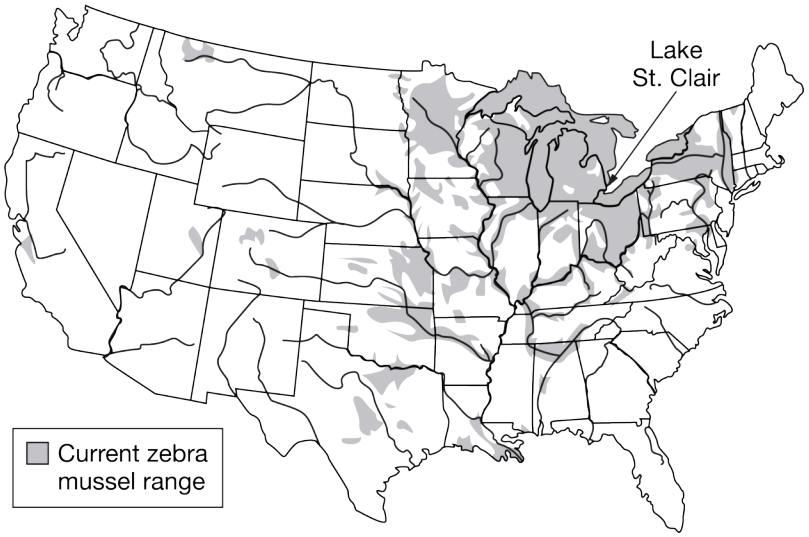


Figure 1. Map showing the current range of zebra mussels in the United States

Which of the following best supports the alternate hypothesis that zebra mussels display characteristics of an invasive species?

(A) Lacking natural predators and able to outcompete native species of mussels and other freshwater species for food, zebra mussels have experienced exponential population growth and rapid expansion into new habitats. ✓

(B) Since their arrival, zebra mussels have undergone rapid adaptive radiation, producing new species in many different habitats.

(C) Throughout their current range, zebra mussels coexist with native mussel species, whose populations have been unaffected by the arrival of zebra mussels.

(D) Zebra mussels can crossbreed with native mussels, producing hybrids that are better adapted in the range shown in Figure 1.

8 Ecology MCQ

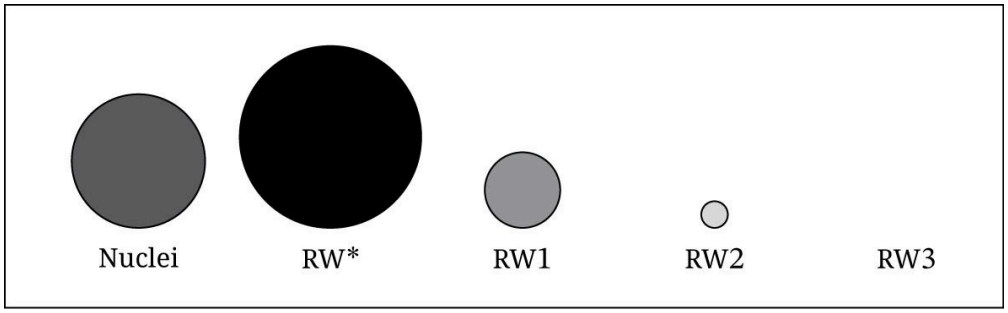
Answer A

Correct. Because invasive species typically do not have natural predators in the habitats into which they are introduced, they can experience rapid exponential population growth. In addition, because these mussels are able to outcompete native species, zebra mussel range has expanded rapidly.

90. A researcher conducted an experiment in a brightly lit aquarium containing a large population of single-celled green algae. Water containing radioactive phosphorus was added to the aquarium. After 7 days, the researcher took a water sample containing the algae and separated the algae from the radioactive water (RW*).

The researcher added deionized water to rinse the algae and saved the rinse water and then repeated that process twice. The rinse water from each of the three rinses was saved as RW1, RW2, and RW3, respectively. The researcher also removed and kept the nuclei of the algal cells. The nuclei were mixed with deionized water, and a drop of the mixture was placed on a piece of filter paper. A drop of the radioactive water and a drop of each of the rinse water samples were also placed on the same paper. These samples were exposed to X-ray to determine their radioactivity. The results after 24 hours are shown in Figure 1.

Figure 1. Radioactivity of saved algal water samples. Darker color and larger circle size indicate a higher level of radioactivity.



Based on Figure 1, which of the following statements best explains why the nuclei sample had a higher level of radioactivity than any of the rinse water samples?

(A) Phosphorus is used in the building of nucleic acids in the nucleus. ✓

(B) The nuclei sample was contaminated with radioactive water.

(C) Radioactive phosphorus is dissolved from rocks and becomes part of the sugar molecules created during photosynthesis.

(D) Radioactive phosphorus is dissolved from rocks and is used to create cholesterol stored in the nucleus.

8 Ecology MCQ

91.

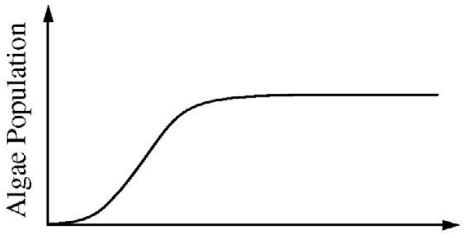


Figure I

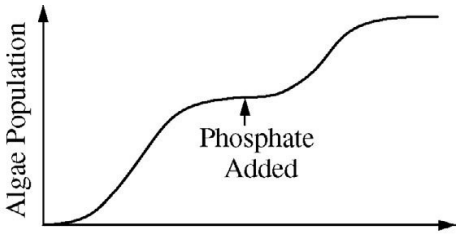
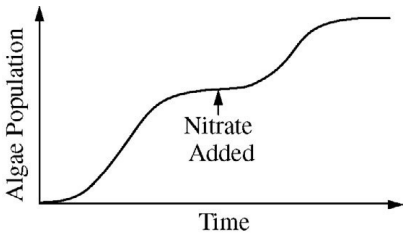
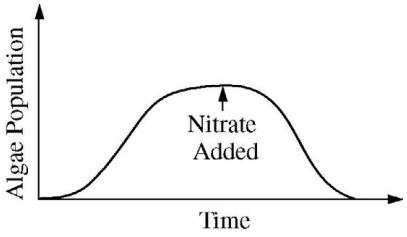


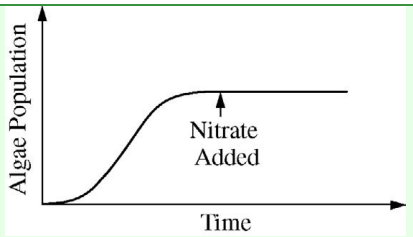
Figure II

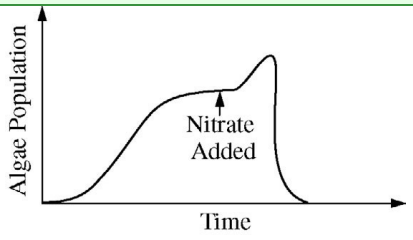
Figure I shows the growth of an algal species in a flask of sterilized pond water. If phosphate is added as indicated, the growth curve changes as shown in Figure II.

Which of the following is the best prediction of the algal growth if nitrate is added instead of phosphate?

(A)


(B)


(C)


(D)


8 Ecology MCQ

92.

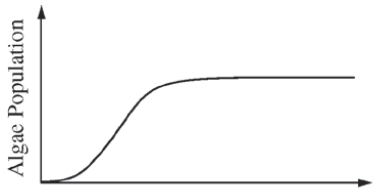


Figure I

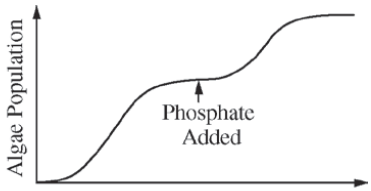
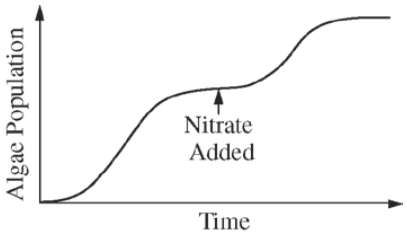
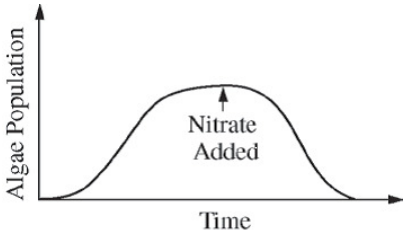


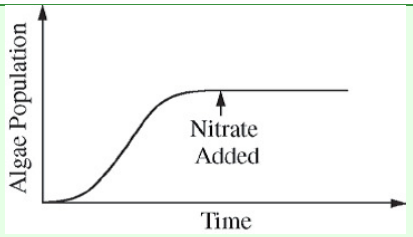
Figure II

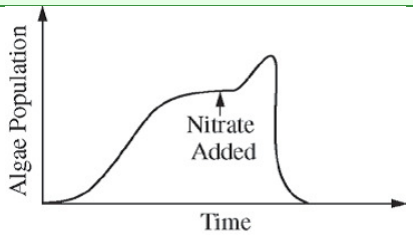
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Which of the following is the best prediction of the algal growth if nitrate is added instead of phosphate?

(A)


(B)


(C)


(D)


8 Ecology MCQ

In the American Southwest, annually emerging cicadas are dependent upon the cottonwood trees in the floodplain. Adult male cicadas perch in the cottonwood trees and chirp to attract females. Females lay their eggs in the branches of the cottonwoods, and, upon hatching, the cicada nymphs drop to the ground, burrow into the soil, feed on the tree roots, and later emerge as adults. Other organisms in the floodplain community prey on the cicadas, as shown in the food web.

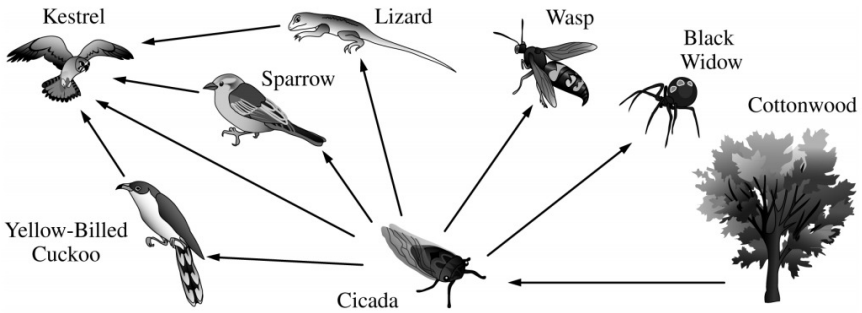


Figure 1. Food web in floodplain community

Human activity has caused a decrease in the amount of flooding and an increase in the incidence of wildfires in the floodplain. In an investigation into the recent changes in the floodplain ecosystem, researchers monitored the soil temperature, amount of cottonwood ground coverage (i.e., area of the ground that is shaded by leaves), and cicada emergence for a period from mid-June until late July. The results of the investigation are represented in Figure 2 and Figure 3.

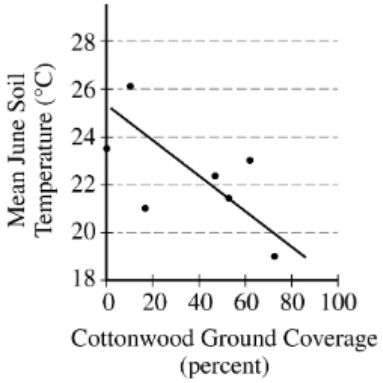


Figure 2. Effect of cottonwood ground coverage on soil temperature

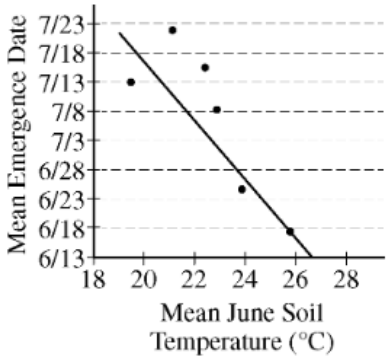


Figure 3. Relationship between soil temperature and cicada emergence date

To assess the impact of wildfires on soil temperature and cicada emergence, the researchers compared mean emergence dates for two natural sites where portions had been affected by wildfire. In addition, cicada emergence was monitored at an experimental site where the soil temperature was experimentally maintained. The data are shown in the table.

8 Ecology MCQ

MEAN CICADA EMERGENCE DATES AT SITES AFFECTED BY WILDFIRES OR WITH EXPERIMENTALLY CONTROLLED SOIL TEMPERATURE

	Natural Site 1		Natural Site 2		Experimental Site	
	Unburned	Burned	Unburned	Burned	24°C	27°C
Mean Emergence Date	July 8	June 18	July 15	July 3	July 14	July 5

93. Based upon the data, which of the following best describes the effect on the timing of cicada emergence if the cottonwood ground cover decreases from 50 percent to 25 percent?

(A) Cicadas will emerge approximately 10 days earlier. ✓

(B) Cicadas will emerge approximately 2 days earlier.

(C) Cicadas will emerge approximately 10 days later.

(D) Cicadas will emerge approximately 2 days later.

94. Based on the data, which of the following describes the most likely relationship between wildfires, soil temperature, and the timing of cicada emergence in the same year?

(A) Wildfires caused a reduction in ground cover, which in turn caused increased soil temperatures leading to earlier emergence of cicadas. ✓

(B) Wildfires caused an increase in ground cover, which in turn caused increased soil temperatures delaying the emergence of cicadas.

(C) Wildfires caused an increase in growth of new plants, which in turn caused a decrease in soil temperatures which led to earlier emergence of cicadas.

(D) Wildfires caused a decrease in ground cover, which in turn caused a decrease in soil temperatures leading to later emergence of cicadas.

95. The data from the temperature-controlled experimental plots can best be used to support which of the following conclusions about cicada development?

(A) Adult cicadas lay more eggs when the soil is warmer than when the soil is cooler.

(B) Adult cicadas mate more frequently when the soil is warmer than when the soil is cooler.

(C) Cicada nymphs grow larger in warmer soil than in cooler soil.

(D) Cicada nymphs mature to adults faster in warmer soil than in cooler soil. ✓

96. Yellow-billed cuckoos typically hatch in midJuly. Emerging cicadas are a primary food source for nesting cuckoos. Which of the following best predicts the effect of wildfires on yellow-billed cuckoo populations?

8 Ecology MCQ

(A) The yellow-billed cuckoo population will decline because the decreased ground cover will allow lizards to prey on cuckoo nests.

(B) The yellow-billed cuckoo population will decline because the cicadas will emerge before the hatching season begins. ✓

(C) The yellow-billed cuckoo population will grow because the adults will more easily see and eat the cicada nymphs.

(D) The yellow-billed cuckoo population will remain unchanged because cuckoos do not nest in areas affected by wildfires.

97. Which of the following statements best predicts the effect of increased cottonwood growth on the food web?

(A) Increased cottonwood growth will lead to an increase in the number of primary consumers, resulting in more competition between kestrels and sparrows for food.

(B) Increased cottonwood growth will decrease the habitat available for lizards, resulting in a decline in the number of kestrels.

(C) Increased cottonwood growth will provide more resources for cicadas, resulting in an increase in the kestrel population. ✓

(D) Increased cottonwood growth will lead to a large accumulation of dead wood that serves as fuel for wildfires, resulting in the collapse of the food web.

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98. In some species of bats, adult females regurgitate (bring up food from the stomach) to feed their young. In addition, adult bats regurgitate to feed other adults that have not recently eaten. The adults receiving the food may be unrelated bats who have previously participated in food sharing and are members of the same social group. Scientists claim that sharing among unrelated adult bats increases the fitness of the species.

Which of the following best supports the scientists' claim?

(A) Bats that do not find food over a 24-hour period may die, preventing their alleles from being passed on within the social group. ✓

(B) Bats that do not find food may adapt to eating other types of food, adding new alleles to the social group.

(C) Food sharing reduces the number of unrelated animals that cluster in one area, thus increasing the genetic diversity in the social group.

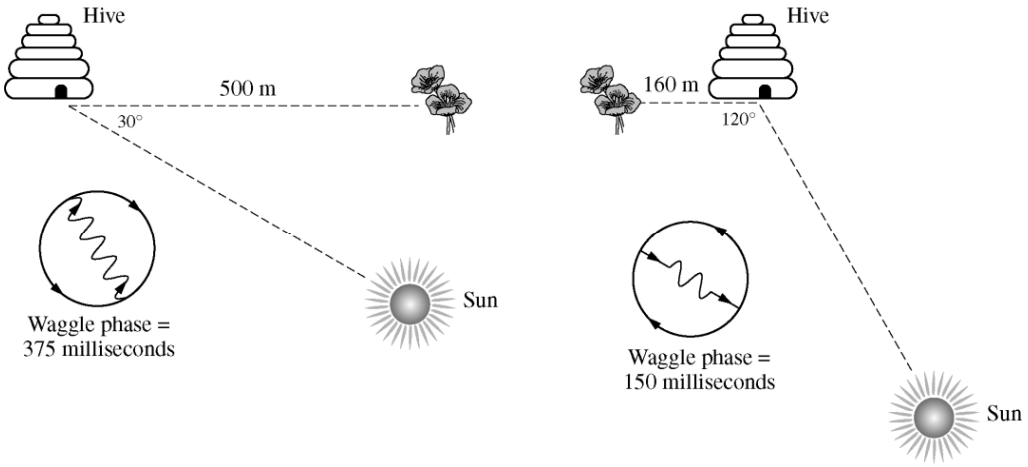
(D) Food sharing increases the number of unrelated animals that cluster in one area, thus decreasing competition for food among members of the social group.

Answer A

Correct. The death of individuals within a social group could potentially reduce the overall genetic diversity of the entire group. Loss of genetic diversity is directly proportional to loss of overall fitness for a species.

8 Ecology MCQ

99.



As depicted in the diagram, honeybees communicate the location of flower patches to members of their hives with waggle dances that give information about the direction and distance to the flowers. Which of the following statements about how honeybees communicate the position of flower patches is most consistent with the model?

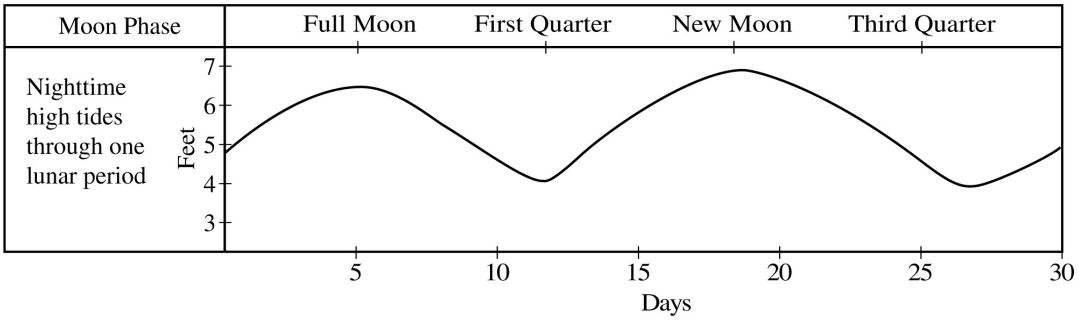
(A) The number of repetitions of the waggle dance indicates the quality of the pollen source.

(B) The total area covered by any waggle dance leads bees to the target flowers.

(C) The angle of the waggle phase relative to the vertical plane indicates the position of the target flowers relative to another beehive.

(D) The farther the target flowers from the hive, the longer the waggle phase. ✓

100.



The California grunion (*Leuresthes tenuis*) is a small marine fish that lives in shallow waters near the ocean shore. Grunions swim as far onto the beach as possible to mate and lay their eggs (spawn).

A researcher proposes that the spawning behavior takes place when the nighttime tides are highest during the month. Which of the following pieces of evidence would best support the researcher's claim?

(A) Grunion spawning occurs every two weeks during the spawning season. ✓

(B) Grunion spawning occurs when nighttime illumination levels are the lowest.

(C) High tide occurs one time per lunar cycle.

(D) The light intensity of the moon varies within the lunar cycle.

8 Ecology MCQ

101. A group of students designed an experiment to determine the effect of compost on the germination and growth of plants. The students set up experimental plots by mixing different ratios of soil and compost. They planted 20 pea and 20 melon seeds in each plot and watered each plot regularly. The students recorded the number of seeds that germinated, and as the plants grew, they recorded plant length, number and size of leaves, and general health observations.

The following observations were recorded.

- In all treatments, more pea seeds germinated than melon seeds.
- There was a week of rainy weather at three weeks.
- Melon plants developed a fungal growth at four weeks.
- Melon plants grew longer than pea plants, but many melon leaves showed signs of yellowing as the weeks passed.

The students concluded that pea plants grew better in compost than did melon plants.

Which of the following best addresses the validity of the conclusion made by the students?

(A) The conclusion is valid because the independent and dependent variables in the experiment were controlled.

(B) The conclusion is valid because the experimental design included a large sample size.

(C) The conclusion is invalid because other variables in the experiment (both biotic and abiotic) affected the results. ✓

(D) The conclusion is invalid because the units of measurement were not given in the experimental design.

102. The pesticide DDT was widely used in the 1940s as a method of insect control. In the late 1950s the first DDT-resistant mosquitoes were discovered, and eventually DDT-resistant mosquitoes were found globally. When DDT is used now, the development of DDT resistance in mosquito populations occurs in months rather than years.

Which of the following best explains the observations concerning DDT resistance in mosquitoes?

(A) Competition for limited resources causes mosquitoes to migrate to geographical areas that have richer supplies of DDT.

(B) The proportion of DDT-resistant mosquitoes in a population remains constant due to the metabolic costs of DDT utilization

(C) Natural selection favors DDT-resistant mosquitoes that are already present in a population when DDT exposure occurs. ✓

(D) DDT is a chemical signal that delays normal reproductive cycles in many mosquito populations.

8 Ecology MCQ

Diapause is the interruption of an organism's life cycle in response to environmental cues. The soil nematode *Caenorhabditis elegans* is capable of entering adult reproductive diapause (ARD) when food is scarce. In *C. elegans*, individuals normally become reproductively mature 2 days after hatching and remain fertile for 18 days. They reproduce either by self-fertilization or by mating with another individual.

In an investigation, researchers examined the survival and reproductive success of *C. elegans* following different times in ARD. In the first experiment, groups of *C. elegans* were held in ARD without food for 0-30 days. Upon reintroduction of food, average brood sizes (average numbers of offspring per adult) were determined following either self-fertilization or mating with a well-fed male. The results are shown in Figure 1.

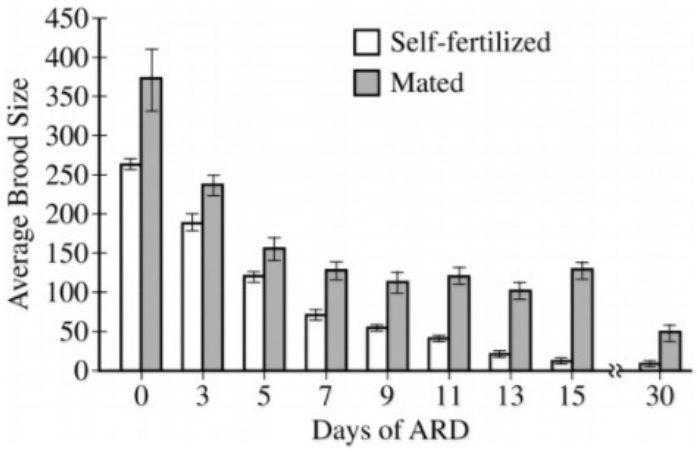


Figure 1. Mean brood sizes $\pm 2SE_{\overline{X}}$ after different times in ARD.

Individual *C. elegans* were held in ARD and subsequently

allowed to reproduce either via self-fertilization (unshaded bars)

or by mating with well-fed males (shaded bars).

In a second experiment, individuals were held in ARD without food for $0-30$ days and monitored for average survival times following reintroduction of food (Figure 2).

8 Ecology MCQ

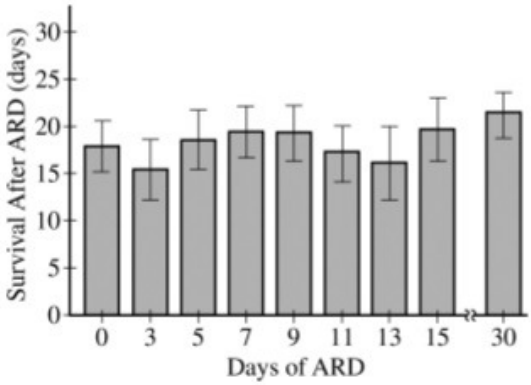


Figure 2. Mean survival $\pm 2SE_{\overline{X}}$ following different times in ARD. Individual *C. elegans* were held in ARD and subsequently given access to food, whereupon their survival times were determined.

103. Which of the following best describes the reproductive ability of *C. elegans* following the ARD induced in the first experiment?

(A) Mating with a well-fed male consistently produced more offspring than did reproduction via self-fertilization. ✓

(B) The numbers of progeny produced by self-fertilization and by mating with well-fed males were not statistically different.

(C) *C. elegans* stopped reproducing after 20 days without food.

(D) There was no relationship between days without food and average brood size.

104. The average brood size per mated individual upon reintroduction of food following 30 days of ARD is closest to which of the following?

(A) 10

(B) 50 ✓

(C) 250

(D) 400

105. Which of the following conclusions is most consistent with the data shown in Figure 2?

(A) Animals that spend $3\text{–}5$ days in ARD are more likely to survive periods of food scarcity than are animals that spend $13\text{–}15$ days in ARD.

(B) Animals that spend 30 days in ARD live significantly longer after reintroduction of food than animals that spend only 3 days in ARD.

(C) The number of days an animal spends in ARD does not significantly affect its time of survival after reintroduction of food. ✓

(D) The large standard errors of the means make conclusions from the data impossible.

8 Ecology MCQ

A student placed 20 tobacco seeds of the same species on moist paper towels in each of two petri dishes. Dish A was wrapped completely in an opaque cover to exclude all light. Dish B was not wrapped. The dishes were placed equidistant from a light source set to a cycle of 14 hours of light and 10 hours of dark. All other conditions were the same for both dishes. The dishes were examined after 7 days and the opaque cover was permanently removed from dish A. Both dishes were returned to the light and examined again at 14 days. The following data were obtained.

	Dish A		Dish B	
	Day 7 Covered	Day 14 Uncovered	Day 7 Uncovered	Day 14 Uncovered
Germinated seeds	12	20	20	20
Green-leaved seedlings	0	14	15	15
Yellow-leaved seedlings	12	6	5	5
Mean stem length below first set of leaves	8 mm	9 mm	3 mm	3 mm

106. The most probable cause for the difference in mean stem length between plants in dish A and plants in dish B is which of the following?

(A) Shortening of cells in the stem in response to the lack of light

(B) Elongation of seedlings in response to the lack of light ✓

(C) Enhancement of stem elongation by light

(D) Genetic differences between the seeds

Answer B

This option is correct. In response to the lack of light, the plant elongates cells in an attempt to reach light necessary for photosynthesis at the expense of leaf development.

8 Ecology MCQ

A student placed 20 tobacco seeds of the same species on moist paper towels in each of two petri dishes. Dish A was wrapped completely in an opaque cover to exclude all light. Dish B was not wrapped. The dishes were placed equidistant from a light source set to a cycle of 14 hours of light and 10 hours of dark. All other conditions were the same for both dishes. The dishes were examined after 7 days, and the opaque cover was permanently removed from dish A. Both dishes were returned to the light and examined again at 14 days. The following data were obtained.

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(A) Shortening of cells in the stem in response to the lack of light

(B) Elongation of seedlings in response to the lack of light ✓

(C) Enhancement of stem elongation by light

(D) Genetic differences between the seeds

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108. Meerkats live together in groups to help raise and protect their offspring. Meerkat pups are most vulnerable when they are away from their burrow foraging for food. Adult meerkats exhibit sentinel behavior, standing upright and making alarm calls to warn other members in the group that predators are nearby.

In a study of meerkat behaviors, scientists observed meerkat foraging groups and recorded the presence or absence of pups along with the frequency of adult sentinel behavior. The scientists determined that sentinel behavior increased when pups were present in the foraging group.

Which of the following correctly identifies the dependent and independent variables in the described study?

(A) The frequency of sentinel behavior is the independent variable, and the presence of pups is the dependent variable.

(B) The presence of pups is the independent variable, and the frequency of sentinel behavior is the dependent variable. ✓

(C) The number of predators is the independent variable, and the presence of pups is the dependent variable.

(D) The distance from the burrow is the independent variable, and the number of predators is the dependent variable.

8 Ecology MCQ

Answer B

Correct. The scientists are studying the relationship between the presence of pups (independent variable) and the frequency of sentinel behavior (dependent variable).

109. Hydrangea macrophylla is a species of plant with blue or pink flowers. Flower color in Hydrangea macrophylla plants is thought to be determined primarily by soil conditions rather than by inherited differences. Which of the following will provide the most direct evidence that flower color in Hydrangea macrophylla is due primarily to soil conditions?

(A) Comparing DNA sequence information for Hydrangea macrophylla to DNA sequence information for a different species of flowering plant

(B) Performing a genetic cross between a pink-flowered plant and a blue-flowered plant and then determining the ratio of phenotypes in the offspring

(C) Growing cuttings from the same Hydrangea macrophylla plant under controlled conditions that vary only with regard to soil pH ✓

(D) Analyzing fossilized remains of an ancestral species that grew in geographic areas similar to the habitats of modern Hydrangea macrophylla

8 Ecology MCQ

110. Scientists are studying the two primary vocalizations made by Belding's ground squirrels (*Urocitellus beldingi*). The trill call consists of a series of medium-pitched notes given rapidly, while the whistle call consists of a single high-pitched note. The objective of the experiment is to determine the function of each vocalization. Recordings of the calls were played to four different groups of squirrels, and the response of each individual squirrel was documented. The responses are summarized in Table 1.

Table 1. Response of ground squirrels to recorded vocalizations

	Percent of Squirrels Responding							
Alarm Call	Trill				Whistle			
Response*	Cr	Po	R/S	O	Cr	Po	R/S	O
Group A	12	65	22	1	42	2	51	5
Group B	9	58	31	2	38	0	60	2
Group C	15	64	18	3	48	4	44	4
Group D	11	66	21	2	1	1	57	1

*Responses

Cr = Crouching
Po = Posting (standing on hind legs)
R/S = Run to shelter
O = Other or no response

Which of the following correctly identifies the dependent variable and the independent variable for the experiment?

(A) The independent variable is the percent of squirrels responding, and the dependent variable is the type of alarm call.

(B) The independent variable is the percent of squirrels responding, and the dependent variable is the type of response.

(C) The independent variable is the group of squirrels, and the dependent variable is the percent of squirrels responding.

(D) The independent variable is the type of alarm, and the dependent variable is the percent of squirrels responding. ✓

8 Ecology MCQ

Answer D

Correct. The type of alarm is the independent variable because it is the factor manipulated by the scientists. The response of the squirrels is being studied to determine how it is affected by the nature of the alarm and is therefore the dependent variable.

Mice live in territorial social groups, and each group has a single dominant male with the rest of the males in the group being subordinate. Male mice mark the area in which they live with their scent by leaving urine wherever they walk. There are two classes of proteins that are found in mouse urine, major urinary proteins (MUPs) and major histocompatibility complex (MHC) proteins. The proteins provide a unique scent for each mouse.

In an experiment, wild male mice were kept isolated in separate cages and their urination patterns were recorded. They were then combined into a single cage that was divided into two compartments by a mesh screen, and again their urination patterns were recorded (Figure 1).

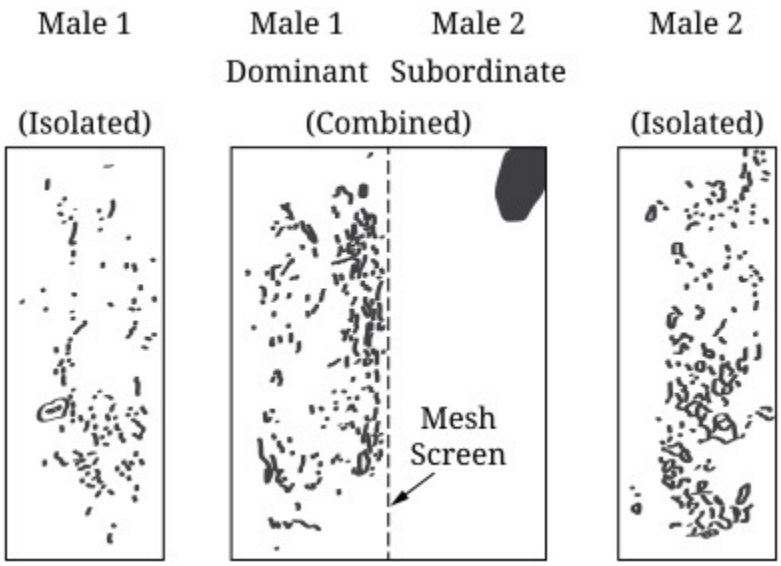


Figure 1. Urination patterns recorded during an experiment

111. Which of the following claims about communication and behavior in mice is best supported by the data from the experiment?

(A) Subordinate males use scent to avoid territory occupied by a dominant male. ✓

(B) Dominant males use urine to communicate sex differences.

(C) Subordinate males occupy more space when in the presence of a dominant male.

(D) Dominant males occupy less space when in the presence of a subordinate male.

8 Ecology MCQ

Answer A

Correct. The scent patterns produced in the combined experiment indicate that the subordinate male is occupying only one small area of his enclosure, which is located as far as possible from the dominant male's territory.

112. In addition to their role in urine scent marking, MHC proteins are used by body cells in both sexes of mice to distinguish self cells that originate in an individual's body from nonself cells that do not originate in an individual's body.

Which of the following best explains how MHC-related odor-producing molecules might be used to maintain genetic diversity in mouse populations?

(A) They could be used by female mice for mate selection to avoid mating with related individuals. ✓

(B) They could be used by subordinate males to attract dominant males to the area to enrich the gene pool.

(C) They could be used by subordinate males to decrease competition with other males resulting in increased genetic diversity.

(D) They could be used by females to decrease bacterial infections that occur during pregnancy.

Answer A

Correct. Close relatives are more likely to share the same MHC proteins. It is possible that female mice could detect and avoid the scent of MHC proteins coming from close relatives and instead choose to mate with males producing different MHC proteins. This would increase the genetic diversity in the mice population.

8 Ecology MCQ

113. A biologist was investigating the effects of herbivory on the amount of nicotine produced by the leaves of tobacco plants (*Nicotiana rustica*). Six groups of plants were grown from seed in the same greenhouse with identical environmental conditions. Three groups were then each exposed for a different interval of time to tobacco hornworms (*Maduca sexta*), which feed on the foliage of tobacco plants. The amount of nicotine present in the leaves was measured at the beginning and end of each time interval. As a control, the other three groups were not exposed to hornworms. The results of the experiment are summarized in Table 1.

Table 1. Response of tobacco plants to hornworm herbivory

Test Group	Time Interval (Weeks)	Hornworms	Initial Concentration of Nicotine$(\mathrm{g/mL})$	Final Concentration of Nicotine$(\mathrm{g/mL})$	% Change in Nicotine Concentration
1	1	Present	0.2	0.8	300
2	1	Absent	0.3	0.2	-33
3	2	Present	0.2	1.4	600
4	2	Absent	0.2	0.3	50
5	3	Present	0.3	2.5	733
6	3	Absent	0.4	0.5	25

Which of the following correctly identifies the dependent variable and the independent variable for the experiment?

(A) The dependent variable is the time interval the plants were exposed to the hornworms and the independent variable is the percent change in the amount of nicotine.

(B) The dependent variable is the absence or presence of hornworms, and the independent variable is the initial amount of nicotine.

(C) The dependent variable is the initial amount of nicotine, and the independent variable is the final amount of nicotine.

(D) The dependent variable is the percent change in the amount of nicotine, and the independent variable is the time interval the plants were exposed to the hornworms. ✓

Answer D

Correct. The change in the concentration of nicotine is the factor being studied and is therefore the dependent variable. The time interval the plants were exposed to the hornworms is changed by the biologist and therefore the independent variable.

8 Ecology MCQ

- **Directions:** Each of the questions or incomplete statements below is followed by four suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.

The three-spined stickleback (*Gasterosteus aculeatus*) is a small fish found in both marine and freshwater environments. Marine stickleback populations consist mainly of individuals with armor-like plates covering most of their body surface (completely plated). Approximately 10,000 years ago, some marine sticklebacks colonized freshwater environments. After many generations in the freshwater environments, the freshwater stickleback populations lacked the armor plating (low plated) typical of marine stickleback populations.

Over the period between 1957 and 2005, one freshwater population, in Lake Washington, a lake in a coastal region of the northwestern United States, changed from having a majority of individuals of the low-plated phenotype to having more individuals of the completely-plated phenotype than of the low-plated phenotype. Figure 1 shows the distribution of plated phenotypes in Lake Washington sticklebacks at four time points between 1957 and 2005.

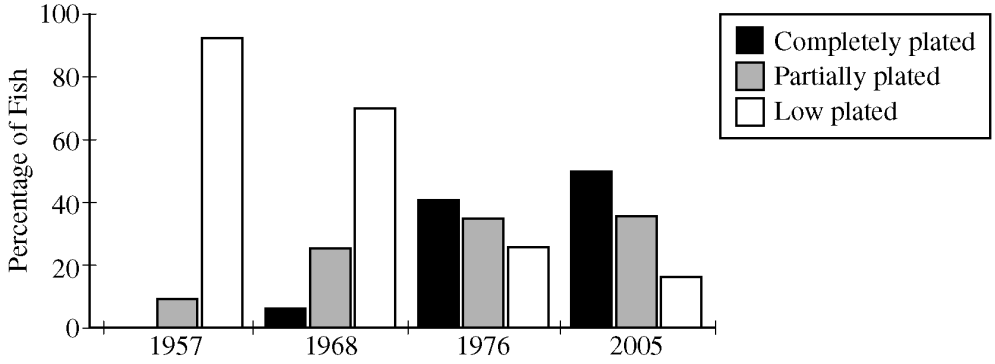


Figure 1. Armor plating phenotypes in Lake Washington stickleback population

A single gene, *ectodysplasin* (*EDA*), is thought to be responsible for the variation in the number of armor plates in sticklebacks. Figure 2 shows a phylogenetic tree constructed by comparing DNA sequences of the *EDA* gene from a number of stickleback populations with low-plated or completely plated phenotypes. Figure 3 shows a phylogenetic tree constructed by comparing the sequences of 25 genes that were randomly selected from the same populations as shown in Figure 2. In both figures, shaded populations display the completely plated phenotype.

8 Ecology MCQ

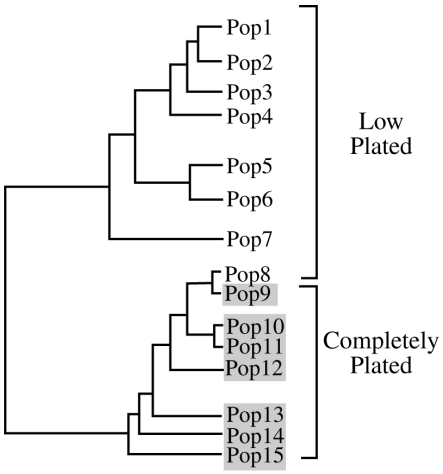


Figure 2. Phylogeny based on *EDA* gene only

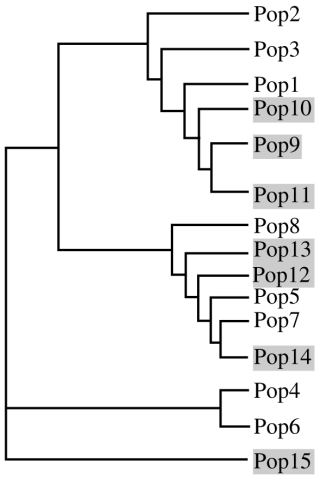


Figure 3. Phylogeny based on 25 random genes

114. Prior to 1960, Lake Washington was highly polluted and underwater visibility was limited to one or two meters. In the late 1960s, a large cleanup effort reduced pollution, resulting in visibility that increased to six to seven meters by 1976. Which of the following best explains how the change in underwater visibility affected armor plating in Lake Washington sticklebacks between 1957 and 1976?

(A) Higher visibility allowed sticklebacks to increase their food consumption to make armor production easier.

(B) Higher visibility made the sticklebacks more susceptible to large-toothed predators in the lakes, giving complete armor a selective advantage. ✓

(C) Clearer water allowed sunlight to penetrate to deeper depths, so the sticklebacks had to increase the amount of armor to protect themselves from the resulting increase in water temperature.

(D) Clearer water contained fewer of the molecular building blocks needed for armor production, so sticklebacks showed a reduction in the amount of armor.

8 Ecology MCQ

115. A group of biologists were investigating the effects of light color on phototropism (growing toward the light source) in tomato plants. Four groups of tomato plants were grown in the same greenhouse under identical conditions. Each group was then placed in a different room, each room having a different color of light, exposed to that single color of light from different directions for one week at a time. After each week, the biologists counted the number of tomato plants in the groups that were showing evidence of phototropism. The results are summarized in Table 1.

Table 1. Phototropic responses of plants to different colors of light

Test Group	Direction of Light	Percent of Plants Showing Phototropism
Group 1 (red light)	North	98
	South	99
	East	98
	West	96
Group 2 (green light)	North	4
	South	8
	East	5
	West	5
Group 3 (blue light)	North	65
	South	58
	East	65
	West	62
Group 4 (white light)	North	100
	South	100
	East	100
	West	99

Which of the following correctly identifies the dependent and independent variables in this experiment?

(A) The color of the light is the dependent variable, and the percentage of plants showing phototropism is the independent variable.

(B) The percentage of plants showing phototropism is the dependent variable, and the color of the light is the independent variable. ✓

(C) The direction of the light is the dependent variable, and the percentage of plants showing phototropism is the independent variable.

(D) The color of the light is the dependent variable, and the direction of the light is the independent variable.

8 Ecology MCQ

Answer B

Correct. The dependent variable is a variable whose value depends on that of another. For this experiment, this would be the percentage of plants showing phototropism. The independent variable represents the potential reason or cause for a given outcome. In this case, it is the color of the light.

116. Prairie dogs are small mammals that live in large colonies in burrows in the ground. Prairie dogs that are near their own relatives when a predator approaches are much more likely to issue a warning bark than those that are near unrelated prairie dogs. The prairie dogs that hear a warning bark are more likely to hide in their burrows than to remain above ground. However, the prairie dog that gives the warning bark is putting itself at increased risk of being identified and killed by the predator.

Which of the following presents the most likely evolutionary explanation for the behaviors described?

(A) The warning bark changes the behavior of the related prairie dogs nearby, allowing the prairie dog's family to have increased survival and reproductive success. ✓

(B) The barking prairie dog chooses to warn other prairie dogs, leading to more prairie dogs living above ground.

(C) The barking prairie dog is alerting unrelated prairie dogs to the predator, so it is not giving any advantage to its own relatives.

(D) The failure of the individual to bark when surrounded by unrelated prairie dogs ensures survival of the individual.

In a study on seed germination of radish seeds, five beakers were set up containing equal numbers of seeds. The beakers were subjected to the conditions indicated in the table below.

	Beaker 1	Beaker 2	Beaker 3	Beaker 4	Beaker 5
Incubation temperature	22°C	22°C	22°C	22°C	5°C
Treatment of radish seeds	None	Boiled	None	None	None
Addition of deoxygenated water	Yes	Yes	No	Yes	Yes
Addition of mineral oil to block oxygen diffusion	No	No	Yes	Yes	No

117. The experimental design allows testing of which of the following?

(A) Only the effect of temperature on germination

(B) Only the need for oxygen for germination

(C) Only the effect of light on germination

(D) The effects of both temperature and oxygen on germination ✓

(E) The effects of both temperature and light on germination

8 Ecology MCQ

118. Germination would be expected to occur first in which beaker?

(A) 1 ✓

(B) 2

(C) 3

(D) 4

(E) 5

119. Comparing results from which of the following pairs of beakers would help determine if enzymes are necessary for germination?

(A) 1 and 2 ✓

(B) 1 and 3

(C) 1 and 4

(D) 2 and 4

(E) 3 and 4

8 Ecology MCQ

To investigate the influence of predation risk on ray behavior, a student observed and counted the large marine animals swimming in a shallow, nearshore section of a coral reef ecosystem. The time of each observation was recorded relative to the time of high tide. The student noted that at low tide, when the water level is low, many of the large animals are forced out of the study area and into the deeper waters of the outer reef. During high tides, when the water level is high, the large animals are able to reenter the study area.

Over a three-day period, the student observed a total of 604 individual rays belonging to three species: cowtail rays, giant shovelnose rays, and black stingrays. For each ray that was sighted, its body length was estimated and its status as either alone (ungrouped) or found with other rays (grouped) was noted. Occasionally, rays were observed sifting through the sandy substrate of the study area to capture food items such as molluscs and crustaceans. In one instance, an injured ray with bite marks that were likely sustained in a shark attack was sighted. In addition to the rays, the student observed lemon sharks (n = 46) and blacktip reef sharks (n = 39). The results of the study are presented in the figures below.

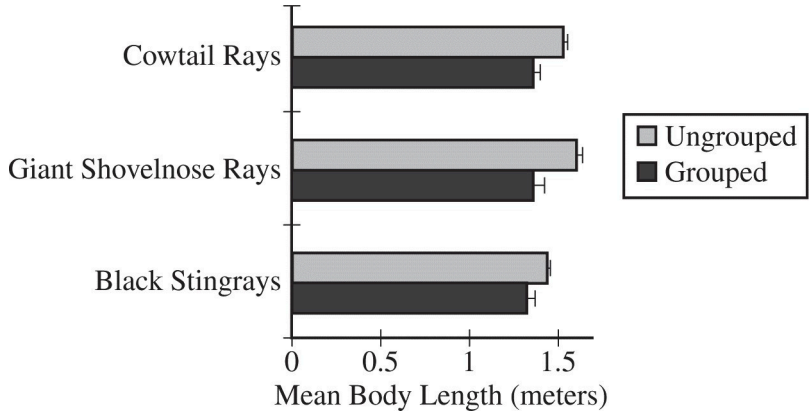


Figure 1. Comparison of mean body lengths of the grouped and ungrouped rays that were observed in a nearshore section of a coral reef ecosystem. Error bars represent $2SE_{\bar{x}}$

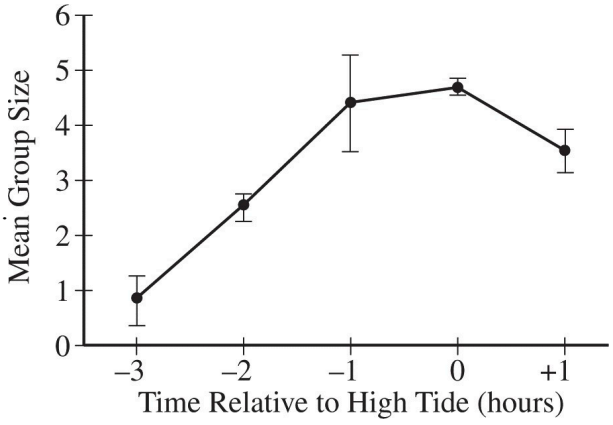


Figure 2. Mean numbers of rays per group in the study area at different stages of the tide cycle. High tide occurs at T = 0 hours.

8 Ecology MCQ

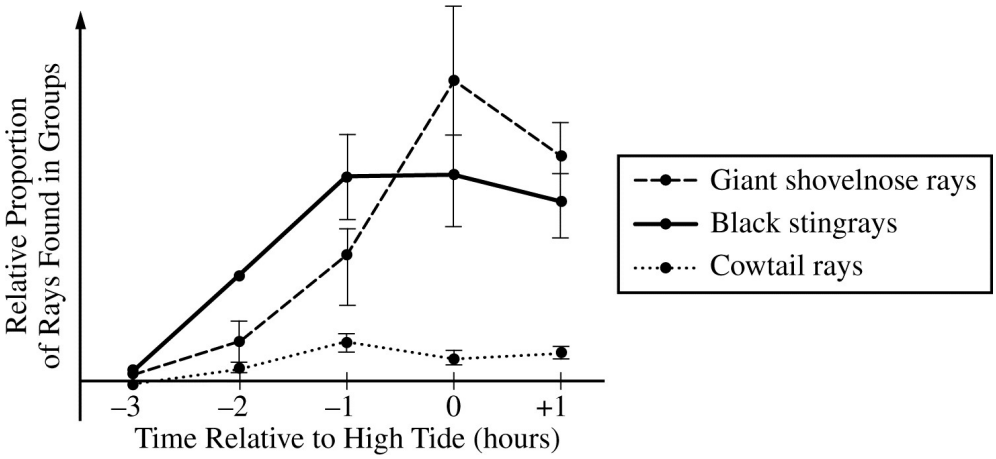


Figure 3. Relative proportions of rays in groups at different stages of the tide cycle for each of the three different populations. High tide occurs at T = 0 hours.

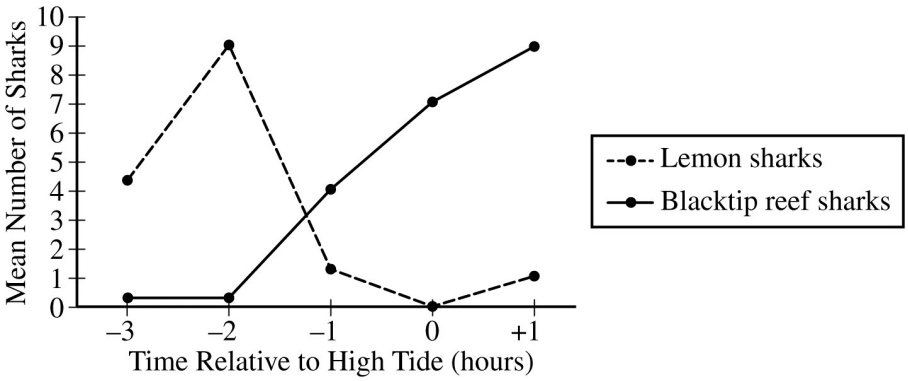


Figure 4. Mean numbers of lemon sharks and blacktip reef sharks in the study area at different stages of the tide cycle. High tide occurs at T = 0 hours.

120. Which of the following scientific claims about the survival strategies used by organisms in a coral reef ecosystem is best supported by the data presented in Figure 1?

(A) Smaller members of a vulnerable population are more likely to act collectively. ✓

(B) Smaller members of a vulnerable population are more likely to rely on defensive coloration.

(C) Larger members of a vulnerable population are more likely to exhibit aggression.

(D) Larger members of a vulnerable population are more likely to behave altruistically.

121. Based on the data presented in Figure 2, which of the following behavioral patterns do rays in a coral reef ecosystem typically exhibit three hours before high tide?

8 Ecology MCQ

(A) Group learning

(B) Mate selection

(C) Solitary foraging ✓

(D) Social interaction

122. Which of the following scientific claims about interacting populations of giant shovelnose rays and blacktip reef sharks is best supported by the results shown in Figures 3 and 4?

(A) Some parasitic species cause significant changes in the behavior of their hosts.

(B) Some invasive species outcompete native species for limited resources.

(C) Some migratory species transmit infectious diseases to resident populations.

(D) Some prey species form protective groups in the presence of predators. ✓

123. Which of the following best justifies the use of the study area to investigate how one species influences the behavior of another?

(A) Black stingrays were present in the study area in the presence and absence of blacktip reef sharks. ✓

(B) Giant shovelnose rays and cowtail rays spent long periods of time at rest while inside the study area.

(C) Natural barriers in the study area prevented lemon sharks from competing with blacktip reef sharks for limited resources.

(D) Cowtail rays in the study area formed groups at some stages of the tide cycle but remained alone at other stages.

124. Based on the results of the study, which of the following is the most likely connection between behavior and evolutionary fitness in a nearshore coral reef environment?

(A) Rays that search for food alone at low tide typically grow to larger sizes than do rays that search for food in groups.

(B) Rays that join groups during rising tides are reproductively more successful than are rays that do not join groups. ✓

(C) Rays that swim far from shore at high tide often encounter a greater variety of species than do rays that remain near the shore.

(D) Rays that roam across large distances during falling tides become stronger swimmers than do rays that spend more time at rest.

8 Ecology MCQ

125. Quorum sensing is a form of cell-to-cell communication used by some bacterial species to regulate their behavior and population size. *Staphylococcus aureus* uses quorum sensing to activate its virulence genes once its population reaches a minimum size. Virulence refers to the ability of a pathogen to cause disease.

S. aureus secretes autoinducing peptide (AIP), which binds to the AgrC receptor on the bacterial cell membrane. When the *S. aureus* population is sufficiently large and produces enough AIP, this concentration activates AgrC and stimulates a signal cascade resulting in the production of hemolysin, a toxin used to attack the host organism. In order to lessen the impact of the toxin produced by an *S. aureus* infection on patients, scientists created AIP mimics intended to act as competitive inhibitors of AIP − AgrC binding at the cell surface.

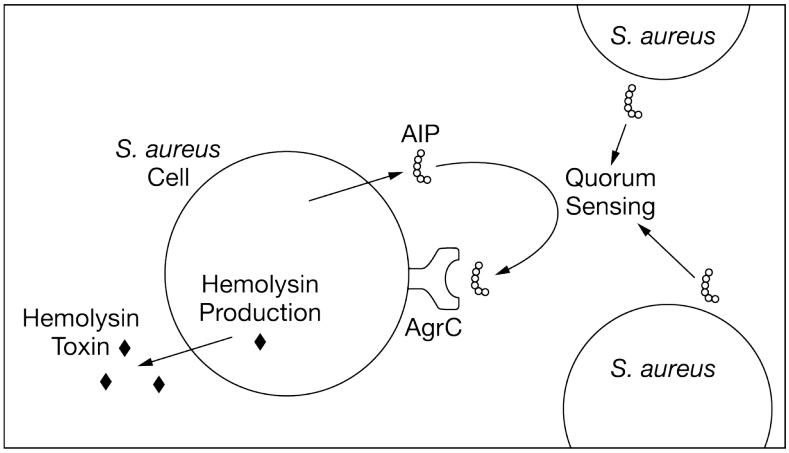


Figure 1. AIP cascade producing hemolysin toxin

Identify the experimental design that would determine which AIP mimic would most effectively inhibit the production of hemolysin toxin.

(A) Grow *S. aureus* in separate culture media containing one AIP mimic each and measure the amount of AIP mimic transported into the cells. The most effective AIP mimic will be absorbed the fastest.

(B) Grow *S. aureus* in separate culture media containing one AIP mimic each and measure the production of hemolysin toxin. The most effective AIP mimic will be in the culture producing the lowest amount of hemolysin. ✓

(C) Grow *S. aureus* in culture media containing hemolysin and measure the production of AIP mimic in response to hemolysin intake. The most effective AIP mimic will be produced in the highest amount.

(D) Grow *S. aureus* in separate culture media containing one AIP mimic each and measure the amount of live cells in each culture. The most effective AIP mimic will be the one that killed the most *S. aureus*.

Answer B

Correct. The most effective mimic will compete with AIP the best, prevent activation of the AgrC receptor, and decrease hemolysin production.

8 Ecology MCQ

126. In short-day (long-night) plants, the critical environmental factor that induces flowering is the

(A) duration of uninterrupted dark only ✓

(B) duration of uninterrupted light only

(C) total combined duration of the dark and light periods

(D) proportion of light to dark

(E) intensity of light during the light period

Snowshoe hares are small mammals that live in forests throughout Canada and the northern United States. Hares feed primarily on vegetation and are preyed on by lynx and owls.

In the fall, the hares shed their brown coat (molt), and a new white coat grows in over a period of 40 days. In the spring, the hares molt again and their white coat is replaced with a brown coat over a period of approximately 30 to 50 days.

In a recent study, researchers observed that the majority of hares in an isolated population began the molting process during the same week in the fall, even though the ground was not snow covered. The majority of hares also began the molting process during the same week in the spring, even though the ground was still snow covered. The researchers formulated a hypothesis that the timing of the molting process is influenced by seasonal changes in the daylight period.

127. Which of the following would best support the claim that the timing of the molting process in certain species is influenced by seasonal changes in the daylight period?

(A) More species that molt seasonally do so when the daylight period is the longest, than when the daylight period is the shortest.

(B) More species that molt seasonally do so when the daylight period is the shortest, than when the daylight period is the longest.

(C) More species that molt seasonally are found in the arctic, where the daylight period varies, than near the equator where the daylight period is relatively constant. ✓

(D) More species that molt seasonally are found near the equator, where the daylight period is relatively constant, than in the arctic, where the daylight period varies.

Answer C

Correct. Molt timing is strongly correlated with seasonal changes in the daylight period. Thus, these results would best support the claim.

128. To test their hypothesis, the researchers conducted a laboratory experiment in which groups of hares were exposed to gradually decreasing periods of daylight. The lengths of the daylight period when molting began were recorded.

Which of the following experimental controls should have been included to ensure that the results could be attributed to changes in the daylight period and not to some other variable?

8 Ecology MCQ

(A) Adjusting the temperature in the lab to simulate seasonal temperature changes

(B) Exposing one group of hares to a fixed daylight period for the duration of the experiment ✓

(C) Feeding the hares at various times of the day to simulate the availability of food in nature

(D) Using lights of different intensities to simulate different weather conditions

Answer B

Correct. The second group of hares in a fixed daylight period serves as a control. If all other environmental factors between the two groups of hares are the same and the first group shows a change in the start of molting, then this would suggest that the length of daylight is the cause.

129. If a viral infection significantly reduces the size of the snowshoe hare population, which of the following predicts the most likely effect on the community during that year?

(A) The biomass of producers will decrease.

(B) The biomass of the lynx will increase.

(C) New predators will feed on the owls.

(D) The lynx and owl populations will decrease. ✓

Answer D

Correct. Because hares are a primary food source for both lynx and owls, a decrease in the hare population size will result in fewer surviving lynx and owls; thus, these populations will also decrease.

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130. Which of the following statements most directly supports the claim that different species of organisms use different metabolic strategies to meet their energy requirements for growth, reproduction, and homeostasis?

(A) During cold periods pond-dwelling animals can increase the number of unsaturated fatty acids in their cell membranes while some plants make antifreeze proteins to prevent ice crystal formation in tissues.

(B) Bacteria lack introns while many eukaryotic genes contain many of these intervening sequences.

(C) Carnivores have more teeth that are specialized for ripping food while herbivores have more teeth that are specialized for grinding food.

(D) Plants generally use starch molecules for storage while animals use glycogen and fats for storage. ✓

8 Ecology MCQ

Answer D

This option is correct. Starch, glycogen, and fat are all molecules used for energy storage, yet plants have evolved to use starch while animals have evolved to use fat for long-term energy storage and glycogen for short-term energy storage.

131. Which of the following statements most directly supports the claim that different species of organisms use different metabolic strategies to meet their energy requirements for growth, reproduction, and homeostasis?

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132.

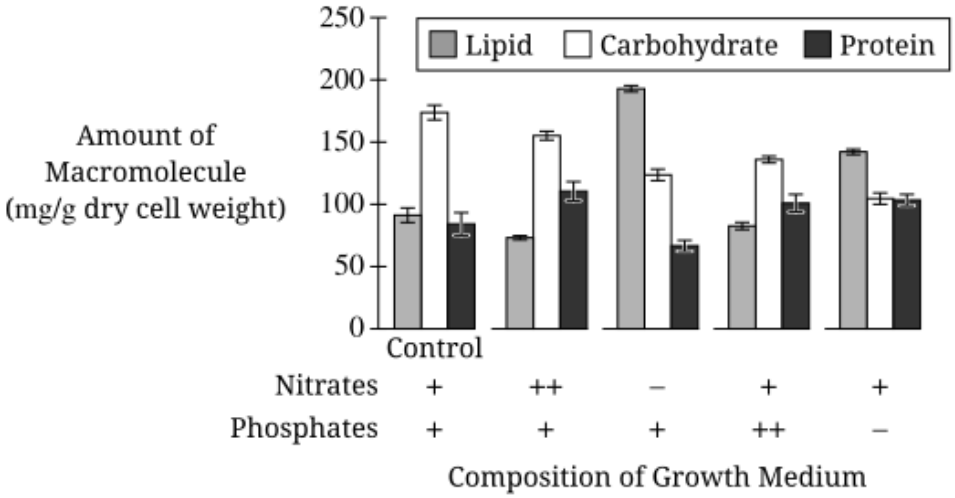


Figure 1. Effect of different abiotic factors on the macromolecule content of bacterial cells. Cells were grown in media with a standard amount (+), twice the standard amount (++), or a reduced amount (−) of each substance. Error bars represent $\pm 2\mathrm{SE}_{\bar{x}}$.

To test the effect of different abiotic factors on the macromolecule content of bacterial cells, researchers grew identical samples of bacterial cells on growth media in which the amount of nitrates and phosphates differed. The amounts of carbohydrates, lipids, and proteins in each sample of cells were then determined (Figure 1).

Based on Figure 1, use of which of the following growth media resulted in a decrease in the amount of protein but an increase in the amount of lipids compared with the control?

8 Ecology MCQ

(A) Medium with a standard amount of nitrates and a reduced amount of phosphates

(B) Medium with a reduced amount of nitrates and a standard amount of phosphates ✓

(C) Medium with twice the standard amount of nitrates and a standard amount of phosphates

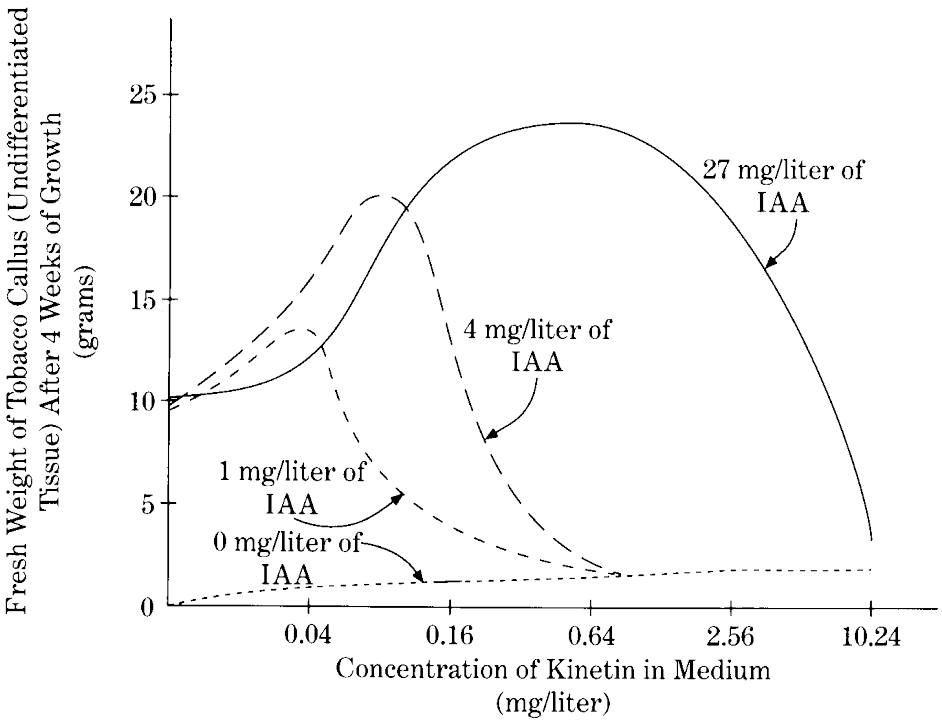
(D) Medium with a standard amount of nitrates and twice the standard amount of phosphates

Answer B

Correct. Based on the graph, this medium produced bacteria with an approximately 100% increase in lipid but an approximately 30% reduction in protein compared with the control.

Directions: Each group of questions below concerns an experimental or a laboratory situation. In each case, first study the description of the situation. Then choose the one best answer to each question following it.

The following questions refer to the data presented in the graph below of tobacco cells grown in tissue culture. The numbers on the curves indicate the concentrations of indoleacetic acid (IAA) in milligrams per liter.



8 Ecology MCQ

133. The optimum concentrations of hormones for promoting maximum tobacco cell growth are

(A) 27 mg/liter of IAA alone

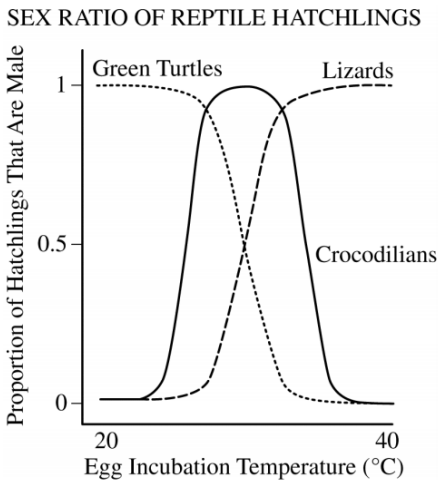
(B) 27 mg/liter of IAA and 2.56 mg/liter of kinetin

(C) 27 mg/liter of IAA and 0.64 mg/liter of kinetin ✓

(D) 4 mg/liter of IAA and 0.64 mg/liter of kinetin

(E) 1 mg/liter of IAA and 0.64 mg/liter of kinetin

134.



The graph above represents the proportions of hatchlings of certain types of reptiles that are male at different egg incubation temperatures. Which of the following claims is best supported by the data?

(A) The expression of sex-determination genes in certain reptiles is influenced by temperature. ✓

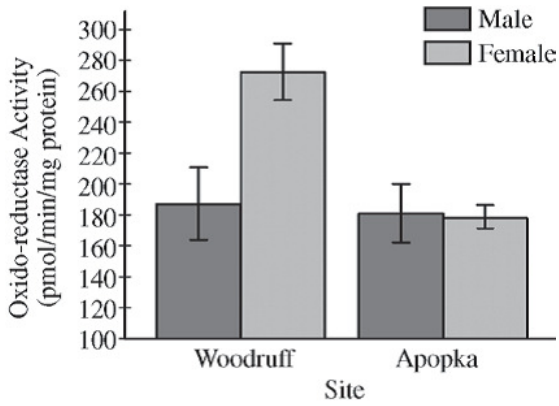
(B) Green turtle eggs do not hatch at temperatures above 35°C.

(C) In reptiles, sex determination is influenced by the sex ratio of the existing population.

(D) Incubating lizard eggs at constant low temperatures produces a large proportion of males.

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135.



Testosterone oxido-reductase is a liver enzyme that regulates testosterone levels in alligators. One study compared testosterone oxido-reductase activity between male and female alligators from Lake Woodruff, a relatively pristine environment, and from Lake Apopka, an area that has suffered severe contamination. The graph above depicts the findings of that study.

The data in the graph best support which of the following claims?

(A) Environmental contamination elevates total testosterone oxido-reductase activity in females.

(B) Environmental contamination reduces total testosterone oxido-reductase activity in females. ✓

(C) Environmental contamination elevates total testosterone oxido-reductase activity in males.

(D) Environmental contamination reduces total testosterone oxido-reductase activity in males.

Answer B

This option is correct. In Lake Woodruff, the lake that is relatively pristine, females had high levels of oxido-reductase enzyme activity, but in Lake Apopka, where environmental contamination conditions were severe, the enzyme activity was low, indicating that environmental factors had an influence on the organism.

8 Ecology MCQ

Students investigated the effect of light on the carbon cycle in aquatic ecosystems by performing the controlled experiment summarized below. The students placed equal amounts of water (pH 7.0) from a large aquarium in glass beakers. The students transferred aquatic plants from the aquarium to several of the beakers, and then they placed equal numbers of the beakers in the light or the dark (Figure 1: groups I and II). Similarly, the students transferred goldfish from the same aquarium to other beakers, and then they placed equal numbers of those beakers in the light or dark (Figure 1: groups III and IV). Finally, the students placed an equal number of beakers containing water only in the light or dark (Figure 1: groups V and VI).

After exposing the samples to light or dark for one hour, the students recorded the pH of the water in each beaker. Carbon dioxide dissolved in water will lower the pH of an aqueous solution. In the experiment, the students used changes in pH to monitor changes in the amount of carbon dioxide in the water. For each treatment group, the students calculated the mean pH and standard error, as documented in the table below.

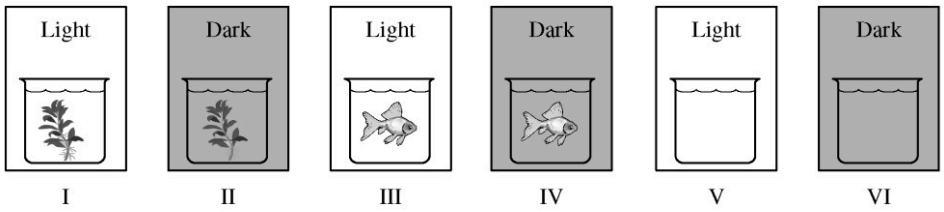


Figure 1. Treatment groups

MEAN pH OF TREATMENT GROUPS AFTER 1 HOUR

Treatment group (n = 10)	I	II	III	IV	V	VI
Mean pH	8.2	6.4	5.9	5.6	7.3	6.8
Standard error of the mean	0.3	0.3	0.2	0.3	0.5	0.4

136. Which of the following observations provides the best evidence that photosynthesis occurred in treatment group I?

(A) The specimens in the beakers were aquatic plants from a large aquarium.

(B) The beakers were placed in the light.

(C) The mean pH of the samples increased after one hour. ✓

(D) The standard error of the mean was smaller than that for treatment group V.

137. Which of the following modifications to the experimental design will best help reduce the standard errors of the means?

(A) Using pond water instead of aquarium water

(B) Exposing samples to light for a greater amount of time

(C) Increasing the sample size of each treatment group ✓

(D) Collecting organisms from a natural water source instead of an aquarium

8 Ecology MCQ

138. The results for treatment groups V and VI could suggest which of the following questions about the design of the experiment?

(A) Do the glass beakers prevent light from reaching the test samples?

(B) Is the method used for measuring pH harmful to aquatic organisms?

(C) Is the availability of carbon dioxide a limiting factor in aquatic ecosystems?

(D) Does the aquarium water contain living microorganisms? ✓

139. To investigate whether an organism in the study is capable of both photosynthesis and respiration, a comparison of which treatment groups is most appropriate?

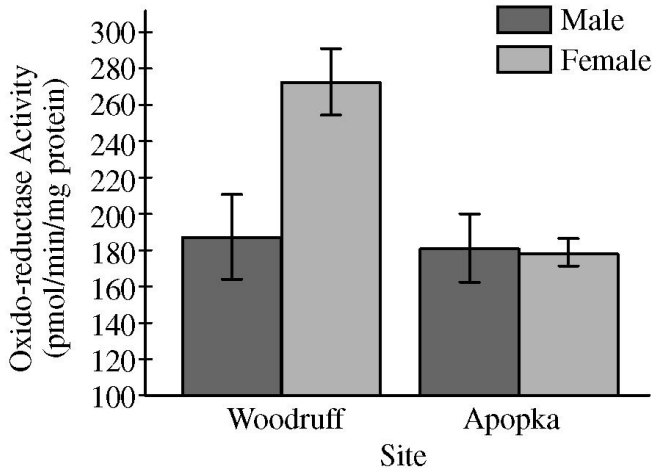
(A) I and II ✓

(B) II and IV

(C) III and V

(D) IV and VI

140.



Testosterone oxido-reductase is a liver enzyme that regulates testosterone levels in alligators. One study compared testosterone oxido-reductase activity between male and female alligators from Lake Woodruff, a relatively pristine environment, and from Lake Apopka, an area that has suffered severe contamination. The graph above depicts the findings of that study.

The data in the graph best support which of the following claims?

(A) Environmental contamination elevates total testosterone oxido-reductase activity in females.

(B) Environmental contamination reduces total testosterone oxido-reductase activity in females. ✓

(C) Environmental contamination elevates total testosterone oxido-reductase activity in males.

(D) Environmental contamination reduces total testosterone oxido-reductase activity in males.

141. Which of the following describes the mechanism by which a plant stem grows toward light?

8 Ecology MCQ

(A) The plant seeks light in order to maximize photosynthesis.

(B) Nervelike impulses stimulate contractile cells on the lighted side of the stem.

(C) Cells on the dark side of the stem elongate more than those on the lighted side. ✓

(D) The plant grows into an open area where its leaves will not be shaded by competing plants.

(E) The greater energy supply on the lighted side of the stem stimulates metabolism and growth on that side.

142.

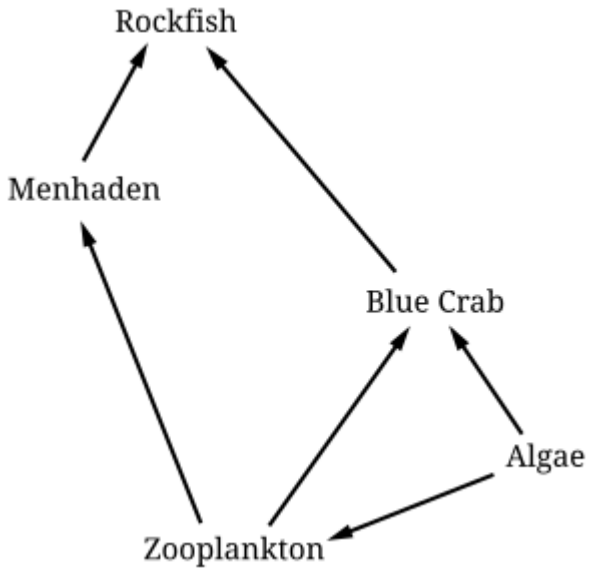


Figure 1. Feeding relationships between selected organisms in the Chesapeake Bay

Figure 1 represents the feeding relationships between selected organisms in the Chesapeake Bay. Decades of overharvesting have reduced the populations of both blue crabs and rockfish. However, recent efforts have increased the number of rockfish in the bay. Based on Figure 1, what is the most likely result of the increase in rockfish?

(A) The population of menhaden will increase because the energy available from zooplankton will increase.

(B) The population of menhaden will decrease because the energy resources available to them will decrease.

(C) The population of zooplankton will decrease because the energy available from algae will decrease.

(D) The population of blue crabs will decrease because predation by rockfish will increase. ✓

Answer D

Correct. Based on the figure, rockfish prey on blue crabs. An increase in the rockfish population will result in an increase in predation, which will reduce the blue crab population.

8 Ecology MCQ

143. The deep ocean floor communities contain very little life. In constant darkness, the water at these depths hovers around $2^{\circ}C$. Sparsely distributed in this environment are hydrothermal vent ecosystems. Superheated water streams out of the vents at temperatures approaching $400^{\circ}C$. These waters are rich in inorganic materials, including carbon dioxide and hydrogen sulfide. Hydrothermal vent zones have a density of organisms 10,000 to 100,000 times greater than found on the surrounding seafloor.

Scientists have studied these vents to determine the energy source for these communities and the identity of the primary producers in the perpetual darkness of the ocean floor. As expected, they found a total absence of photosynthetic organisms at all vents, identifying mats of bacteria as the producers of the carbon compounds that are used by the rest of the community for growth, repair, and as an energy source.

Which of the following describes how the vent bacteria harvest energy to produce organic compounds for use by the vent community?

(A) The bacteria use the heat from the hot water to assemble carbon-based materials from inorganic molecules.

(B) The bacteria function as chemoautotrophs, extracting energy from hydrogen-rich inorganic molecules. ✓

(C) Organic matter drifting down from the ocean's surface is used to make molecules with enough energy to sustain the vent community.

(D) The bacteria can recycle the energy from the waste products and dead material produced by the heterotrophs of the vent community into energy-rich carbon compounds that can be recycled.

Answer B

Correct. The bacteria use the energy stored in hydrogen sulfide to produce carbon compounds that are used by the heterotrophs in the community. This process is known as chemosynthesis.

144. Beaked whales feed at various depths, but they defecate at the ocean's surface. Nitrogen-rich whale feces deposited in surface waters supply nutrients for algae that are eaten by surface dwelling fish. Which of the following best predicts what would happen if the whale population decreased?

(A) There would be a reduction in surface nitrogen concentration, which would cause an algal bloom.

(B) The surface fish populations would decline due to reduced populations of algae. ✓

(C) The remaining whales would accumulate mutations at a faster rate.

(D) The remaining whales would be forced to forage in the deepest parts of the ocean.

145. ATP serves as a common energy source for organisms because

8 Ecology MCQ

(A) it is the smallest energy molecule

(B) it stores the least energy of any energy source

(C) its energy can be easily transferred to do cellular work ✓

(D) it is extremely stable and can be stored in the cell for long periods of time

(E) traces of it have been found in fossils of ancient organisms dating back to the beginning of life on Earth

146. A student created a terrarium by placing soil in the bottom of a large glass container. He added several species of mosses and tiny grasses to the soil. The student ensured that the soil and plants were damp by adding water from a nearby creek. He added a few dead, decaying leaves and twigs, some small insects, and a few small garden slugs to the terrarium. He placed an airtight seal on the container and placed the container near a window where it would receive some sunlight but would not overheat. After three years, the terrarium still contains living organisms.

Which of the following statements best explains how this terrarium could sustain living organisms for three years?

(A) After a short time, the seal began to leak, allowing gaseous forms of carbon, phosphorus, and nitrogen to enter the terrarium.

(B) Small amounts of carbon, phosphorus, and nitrogen atoms have been leaking through the terrarium's glass.

(C) Carbon, phosphorus, and nitrogen atoms have been cycling through the terrarium's organisms and environment. ✓

(D) New carbon, phosphorus, and nitrogen atoms have been created through natural processes inside the terrarium.

8 Ecology MCQ

147. In an experiment, three different animals were exposed to a series of environmental temperatures and their body temperatures were assessed at multiple temperature points between 10 and 40 degrees Celsius. Ten different individuals of each type of animal were included in the study to produce an average body temperature profile for each type of animal. Figure 1 shows the data collected.

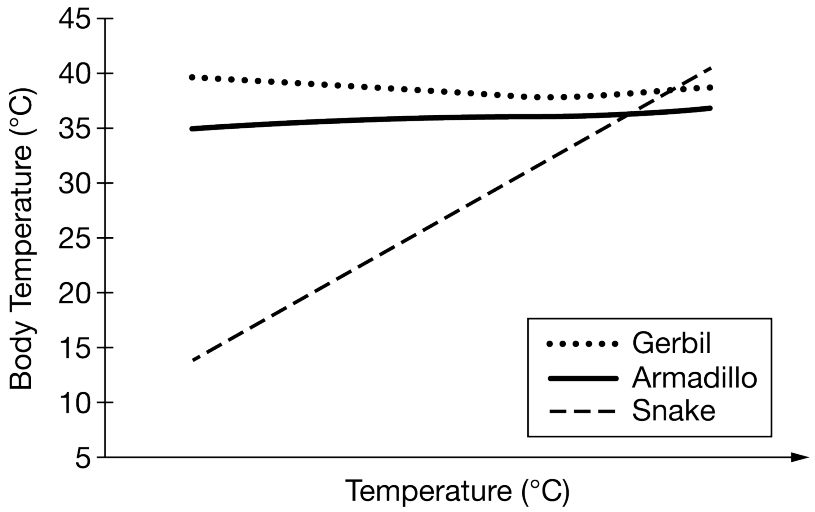


Figure 1. Effect of environmental temperature on body temperature of selected animals

Which of the following statements is most consistent with the data in Figure 1 ?

(A) Armadillos tend to be more metabolically active than either gerbils or snakes are.

(B) Armadillos cannot survive at environmental temperatures over $30^{\circ}C$.

(C) Snakes are found in terrestrial rather than aquatic environments.

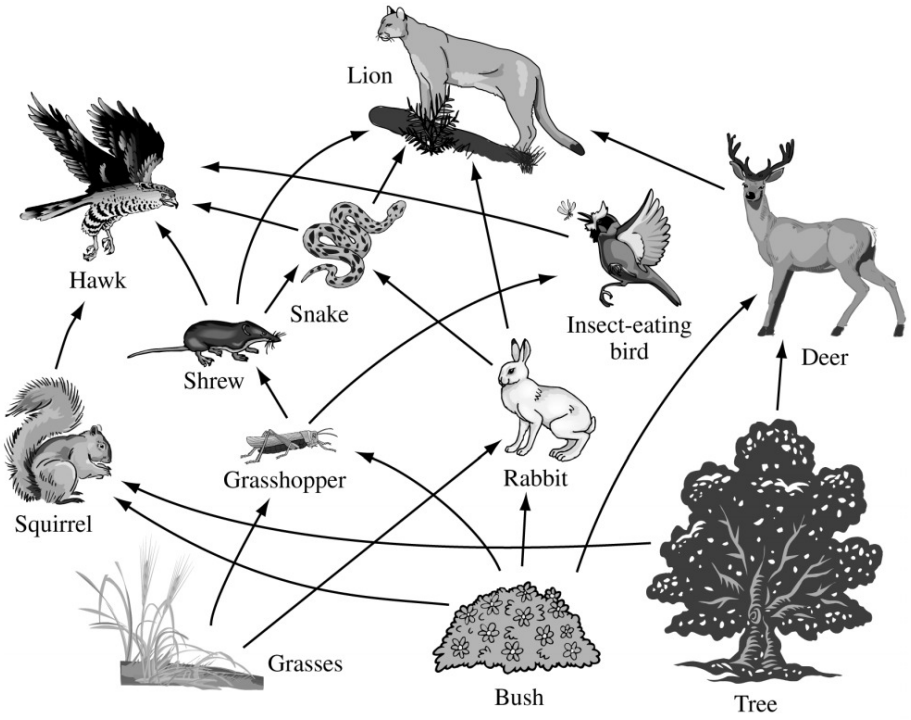
(D) Snakes display behaviors that allow them to absorb or radiate heat as required. ✓

Answer D

Correct. Figure 1 shows that snakes cannot control their internal temperatures through metabolic processes. In low temperatures, snake will bask on rocks and flatten their profiles to absorb heat to raise their body temperatures above ambient temperature.

8 Ecology MCQ

148.



The food web represented above does not include bacteria and fungi. Which of the following best describes a consequence of having no bacteria and fungi in the food web?

(A) Grasses will have no direct access to chemical components recycled from dead organisms. ✓

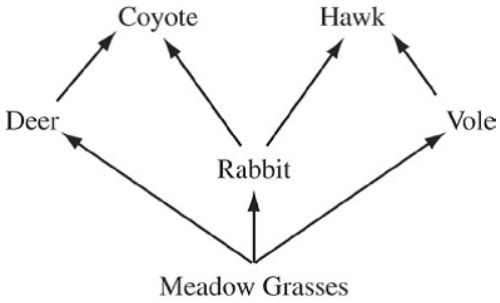
(B) Rabbits will have no direct access to energy absorbed from secondary consumers.

(C) Shrews will have no direct access to matter transferred from tertiary consumers.

(D) Hawks will have no direct access to nitrogen acquired from the atmosphere.

8 Ecology MCQ

149. The following is a food web for a meadow habitat that occupies 25.6 km^2. The primary producers' biomass is uniformly distributed throughout the habitat and totals 1,500 kg/km^2.



Developers have approved a project that will permanently reduce the primary producers' biomass by 50 percent and remove all rabbits and deer.

Which of the following is the most likely result at the completion of the project?

(A) The biomass of coyotes will be 6 kg, and the biomass of hawks will be 0.5 kg.

(B) The biomass of coyotes will be dramatically reduced. ✓

(C) The coyotes will switch prey preferences and outcompete the hawks.

(D) There will be 50 percent fewer voles and 90 percent fewer hawks.

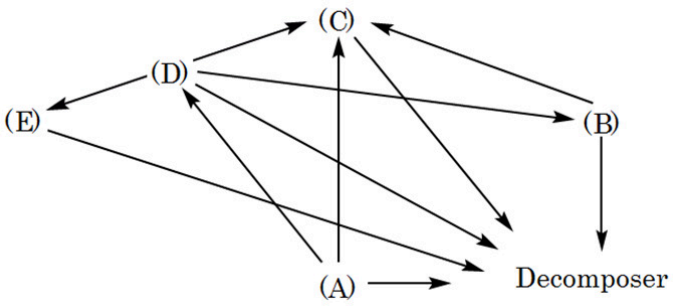
Answer B

This option is correct. It demonstrates an understanding of the components of a food web and interactions between all of the participants. Coyotes prey on deer and rabbits, and if developers remove them, coyotes will lose their primary source of nutrition/energy. With this loss of nutrition/energy they will experience a decrease in reproductive success and therefore a significant decline in their population.

8 Ecology MCQ

Directions: Each set of lettered choices below refers to the numbered words or statements immediately following it. Select the one lettered choice that best fits each word or statement. A choice may be used once, more than once, or not at all in each set.

The figure below represents a food web in a particular ecosystem. Each letter represents a species. The arrows indicate the direction of energy flow.



150. Which species would most likely represent humans if they were part of this ecosystem?

(A) A
(B) B
(C) C ✓
(D) D
(E) E

151. A photosynthetic organism would be represented by which species?

(A) A ✓
(B) B
(C) C
(D) D
(E) E

152. Members of which species are herbivores?

(A) A
(B) B
(C) C
(D) D ✓
(E) E

8 Ecology MCQ

153. Members of which species are most likely to be omnivorous?

(A) A

(B) B

(C) C ✓

(D) D

(E) E

8 Ecology MCQ

154. In an experiment to determine the effect of light availability on species richness in an environment, a grassland was divided into 26 plots that were assigned to one of two treatment groups:

1. Clipped – plots were mowed to simulate the grazing of herbivores that maintain the low height of grasses.

2. Enclosure – plots were enclosed in fences, and grasses were allowed to grow undisturbed to their maximum height.

After eight years, data was collected on the amount of light penetration to the soil (bottom light density) and the number of species in the two treatment groups. Figure 1 shows the averaged light penetration measurements, and Figure 2 shows the averaged number of species per plot type.

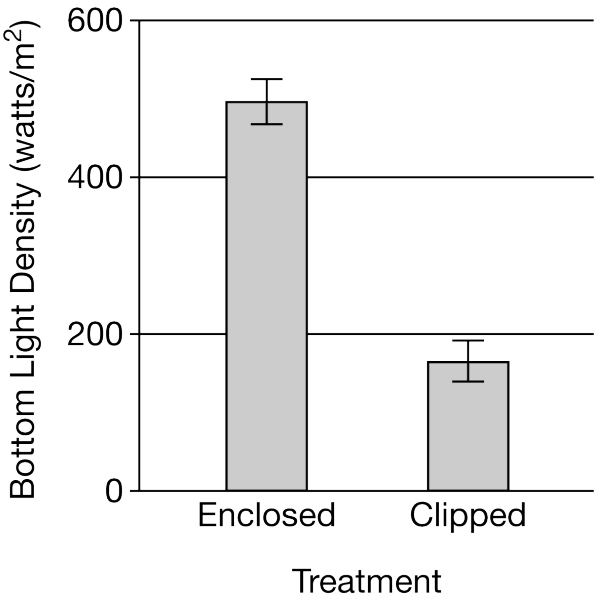


Figure 1. Average bottom light intensity in clipped plots versus enclosed plots

8 Ecology MCQ

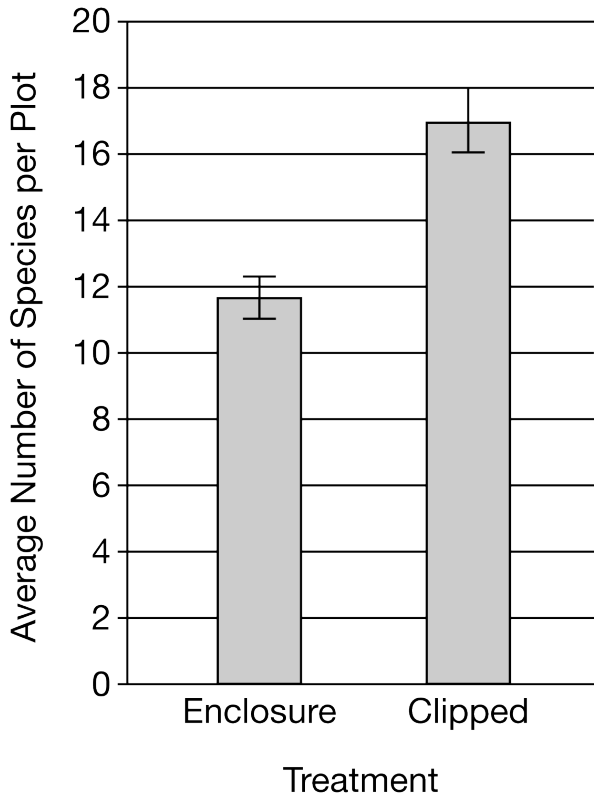


Figure 2. Richness comparison of two treatments

Which of the following statements is consistent with the data shown in the figures?

(A) Grazing activity in a field decreases the species richness of the field.

(B) Different patterns of energy availability in ecosystems will affect species richness. ✓

(C) Forests that develop a closed canopy with a shaded floor will exhibit increased species richness.

(D) The enclosed plots are more efficient at capturing and distributing energy than the clipped plots.

Answer B

Correct. When light energy is allowed to reach the soil surface of an ecosystem, species richness is greater than when most of the light is absorbed before it can reach the soil surface.

8 Ecology MCQ

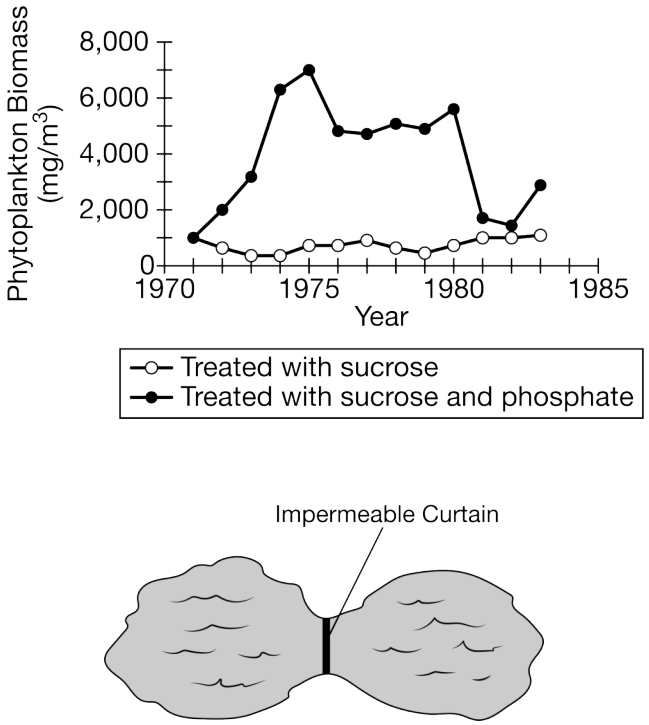


Figure 1. Phytoplankton biomass in two sides of a small lake that is divided by an impermeable curtain

In the early 1970s, researchers hypothesized that carbon was the limiting nutrient in many aquatic ecosystems. To test this hypothesis, the researchers divided a small lake in two roughly equal halves with an impermeable curtain that was fastened and sealed to the bedrock of the lake. Beginning in 1971 the researchers treated one side of the lake with sucrose and the other side with both sucrose and phosphate. From 1971 to 1983 the researchers monitored the phytoplankton biomass in both parts of the lake. The results are shown in Figure 1.

155. Which of the following was most likely a direct consequence of the addition of phosphate to the lake?

(A) The amount of biomass in the first trophic level decreased.

(B) The amount of biomass in the second trophic level decreased.

(C) The amount of energy available to producers in the lake increased.

(D) The amount of energy available to consumers in the lake increased. ✓

156. Which of the following treatments would have been the best control treatment for the experiment?

(A) An untreated section of the lake ✓

(B) A section of the lake that was treated with phosphate but not sucrose

(C) A different lake that was treated with sucrose and phosphate

(D) A small pool of the lake water maintained in a controlled laboratory environment

8 Ecology MCQ

157. The average growth rate of the phytoplankton population from 1971 to 1975 in the side of the lake treated with sucrose and phosphate is closest to which of the following?

(A) $125\ \frac{\text{mg}}{\text{m}^3}$ per year

(B) $1,000\ \frac{\text{mg}}{\text{m}^3}$ per year

(C) $1,500\ \frac{\text{mg}}{\text{m}^3}$ per year ✓

(D) $6,000\ \frac{\text{mg}}{\text{m}^3}$ per year

158. Which of the following claims is best supported by the data?

(A) Carbon was a limiting factor for phytoplankton in the lake.

(B) Phosphate was a limiting factor for phytoplankton in the lake. ✓

(C) Both carbon and phosphate were limiting factors for phytoplankton in the lake.

(D) Neither carbon nor phosphate was a limiting factor for phytoplankton in the lake.

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159. An ecologist collected a large sample of forest soil and distributed it evenly into two samples, sample A and sample B. Sample A was placed in a temperature control chamber that was maintained at the forest's average high temperature. Sample B was placed in a separate temperature control chamber that was maintained at $5^{\circ}C$ above the forest's average high temperature. Both samples were kept in their chambers for 60 days and were watered with only enough sterile water to keep the soil moist. After 60 days, soil from both samples was removed and tested for ammonia levels. Results are shown in Table 1.

Table 1. Ammonia Levels $(\text{mg/kg},\ \pm 2SE_{\overline{x}})$

Sample A	Sample B
11.4 \(±\) 0.7	3.3 \(±\) 0.4

Which of the following is the best explanation of the ammonia level in sample B shown in Table 1?

(A) The ammonia level was lower in sample B because the elevated temperature caused the ammonia to evaporate into the atmosphere.

(B) The ammonia level was lower in sample B because the added water washed ammonia out of the soil.

(C) The ammonia level was lower in sample B because the increased temperature reduced the number of nitrogen-fixing bacteria in the soil. ✓

(D) The ammonia level was lower in sample B because the elevated temperature caused the ammonia molecules to break down into simple atoms.

8 Ecology MCQ

Answer C

Correct. Nitrogen gas in the atmosphere is fixed into ammonia and ammonium in the soil by nitrogen-fixing bacteria.

160. Which of the following best explains why there are seldom more than five trophic levels in a food chain?

(A) Most carnivores function at more than one trophic level.
(B) Trophic levels above this number contain too many individuals.
(C) Top carnivores are too few in number to prey effectively.
(D) The ecosystem contains too much biomass.
(E) Energy is lost from each trophic level. ✓

161.

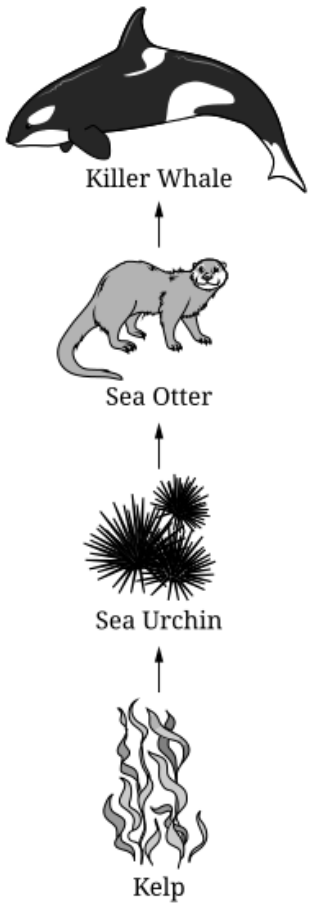


Figure 1. A food chain in the North Pacific Ocean

Based on Figure 1, which of the following is the most likely short-term effect of a decrease in the killer whale population?

8 Ecology MCQ

(A) The sea otter and kelp populations will increase, while the sea urchin population will decrease. ✓

(B) The sea otter and sea urchin populations will increase, while the kelp population will decrease.

(C) The sea otter population will stay the same, while the sea urchin and kelp populations will increase.

(D) There will be no change to the sea otter, sea urchin, and kelp populations because another predator species will fill the niche previously occupied by the killer whales.

Answer A

Correct. A decrease in killer whales will reduce predation on sea otters, allowing their population to increase. An increase in sea otters will decrease the population of sea urchins, which eat kelp. Therefore, the population size of kelp will also increase.

162. In a study of energy flow, the following data were collected:

- Each human requires 55 grams of protein per day.
- An acre of soybeans produces 200,000 grams of protein per year.
- A cow grazing on one acre of land produces 19,000 grams of protein per year.

Based on these data, an acre of land would support:

- 10 humans per year on a diet of soy, or
- 0.9 human per year on a diet of beef.

Which of the following statements best explains these relationships in terms of energy transfer?

(A) Humans are better adapted to make efficient use of soy protein than beef protein, so they can extract more energy from a gram of soy protein than from a gram of beef protein.

(B) A cow can harvest energy only at a certain rate, so adding another cow to the acre of land would double the number of humans supported by the acre.

(C) Preparing meat for human consumption takes more energy per gram of protein than preparing soy for consumption.

(D) Only 10% of the energy on each level of the trophic pyramid is available for use by the next level. The rest is lost as heat or used for growth and repair. ✓

Answer D

Correct. The 10% rule states that approximately 90% of the energy at any trophic level is lost to heat, growth, and repair of the organisms, and only 10% is available for harvesting by the next trophic level.

8 Ecology MCQ

163. Producers within very deep regions of the ocean most likely introduce energy into ecosystems through which of the following processes?

(A) Absorbing photons of light energy

(B) Capturing electrons from inorganic molecules ✓

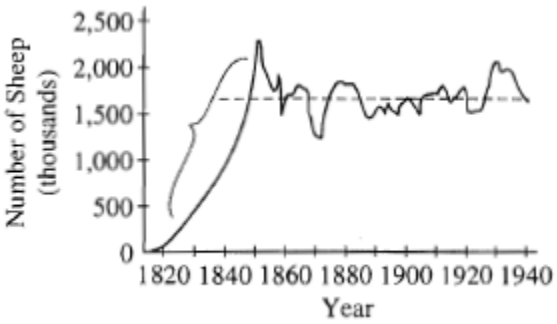
(C) Absorbing heat energy from hydrothermal vents on the ocean floor

(D) Living in other organisms

Answer B

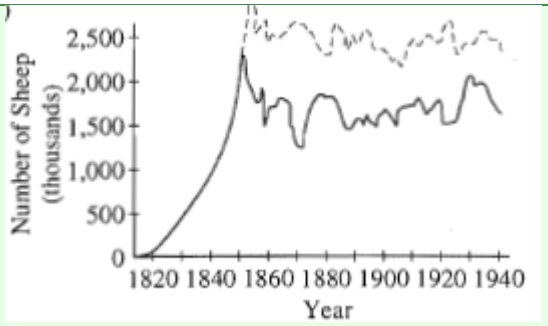
Correct. Breaking chemical bonds in inorganic molecules provides energy for the transfer of electrons in the bonds. Organisms that utilize electrons captured from the bonds of such molecules are classified as producers and are referred to as chemosynthetic organisms.

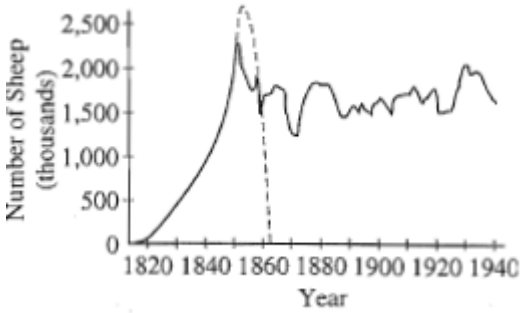
The graph below shows changes in a population of wild sheep that were introduced to the island of Tasmania in the early 1800's.

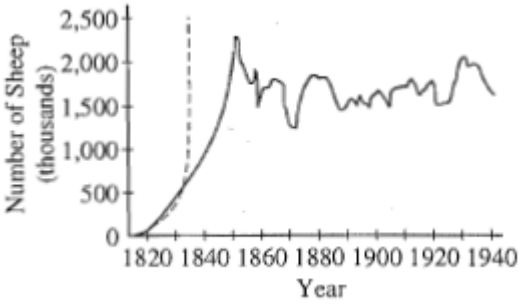


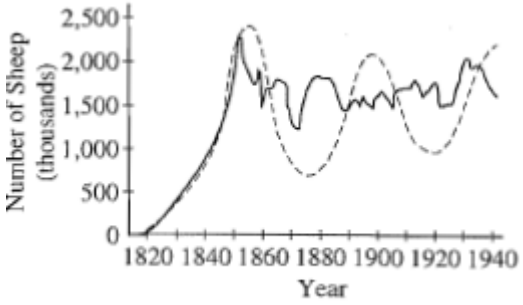
164. In the graphs below, the solid line represents the original population. The dotted line on which graph best represents the sheep population that would have resulted from a sustained increase in the primary productivity of the environment?

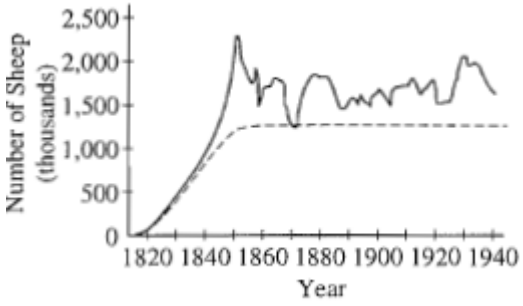
8 Ecology MCQ

(A) 

(B) 

(C) 

(D) 

(E) 

8 Ecology MCQ

165. The type of population growth represented by that portion of the graph line enclosed in the bracket is most accurately termed

(A) stable

(B) exponential ✓

(C) density-dependent

(D) arithmetic

(E) decelerating

166. The graph indicates that the sheep population most likely is

(A) growing in excess of its carrying capacity, since fluctuations in population size occurred after 1850

(B) headed for extinction because of the population explosion about 1930

(C) regulated by density-independent factors, because there appears to be about a 10-year cycle of sharp declines in size

(D) shifting from a *K*-selected strategy to an *r*-selected strategy

(E) stable after 1850 under the effects of density-dependent regulating factors ✓

167. The dashed line on the graph represents the

(A) maximum population size

(B) average birth rate

(C) biotic potential of the population

(D) carrying capacity of the environment ✓

(E) point of maximum effect for density-independent factors

8 Ecology MCQ

168. Figure 1 shows the population fluctuations of phytoplankton and zooplankton in the eastern Pacific Ocean over the course of six years. The major dip in phytoplankton was attributed to a lack of nutrients for phytoplankton.

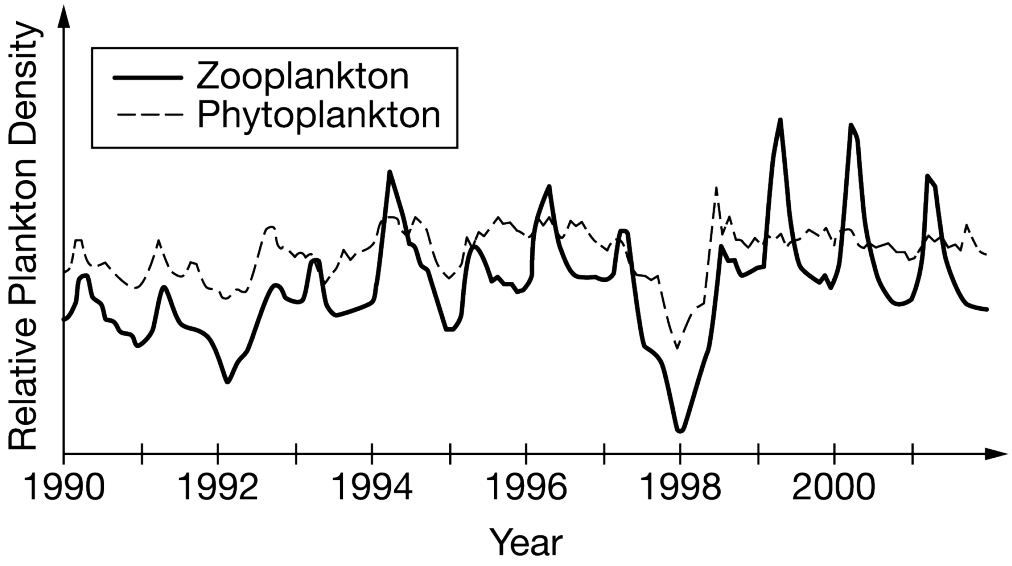


Figure 1. Relative abundance of plankton over time in the eastern equatorial Pacific Ocean

Of the following, which explains why the zooplankton also dipped during the same year and then seemed to cycle over the next three years?

(A) The zooplankton rely on the same nutrients for energy harvesting as the phytoplankton, so their population was also unable to acquire the necessary energy to maintain their population level. After that major dip there were enough nutrients for both categories of plankton.

(B) Zooplankton are more temperature-sensitive than the phytoplankton, so the annual seasonal changes will affect their population sizes in a cyclical fashion. There was a sudden, extreme temperature change in the first dip that also affected the phytoplankton.

(C) A migration of whales was blown off course by a storm one year and ate most of the phytoplankton and zooplankton. In the following years the whales returned to their normal migration route.

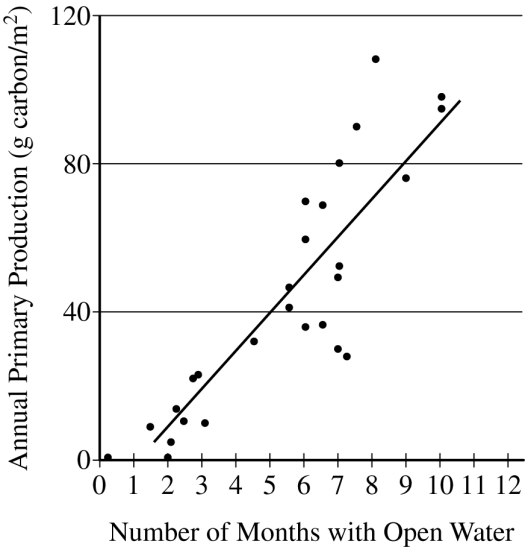
(D) The reduction in the phytoplankton population meant that the zooplankton's energy source was drastically reduced, leading to the decrease in the zooplankton population size. After the dip, the phytoplankton provided a steady energy source, and the zooplankton population fluctuated around its carrying capacity. ✓

Answer D

Correct. The energy source of zooplankton is the carbon compounds fixed by phytoplankton. Without the proper inorganic nutrients, the phytoplankton population is limited in its photosynthetic activity, leading to an energy deficit in the zooplankton populations. It is also correct that the annual fluctuations in zooplankton represent a stable population cycling around its carrying capacity.

8 Ecology MCQ

169.



In the Arctic Ocean, the predominant primary producers are phytoplankton. Phytoplankton are consumed by zooplankton, which in turn are eaten by codfish. In years when there is more open water (less ice coverage), there are more zooplankton and fish than in years with less open water (more ice coverage). Based on the graph above, the difference is most likely because

(A) when there is less open water, light is blocked from the zooplankton, so they cannot produce as much food for the fish

(B) when there is more open water, the temperature is warmer, so the zooplankton and fish populations increase in size

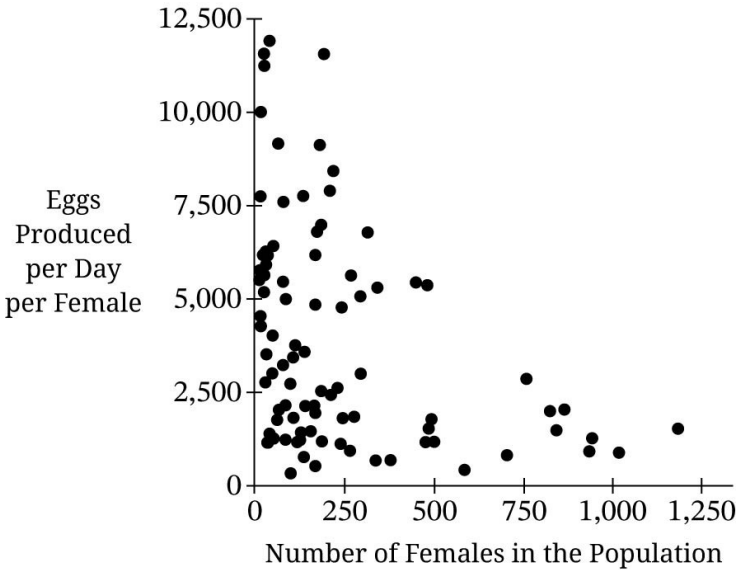
(C) the ice blocks the light, so in years with more ice coverage, there is less photosynthesis by the phytoplankton ✓

(D) the ice increases the light available for photosynthesis, so primary production increases and zooplankton populations increase in size

8 Ecology MCQ

170. Researchers studying parasitic hookworms investigated the effect of the size of the female hookworm population on the number of eggs produced per worm (Figure 1).

Figure 1. Eggs produced per hookworm in female populations of various sizes



Based on the data in Figure 1, which of the following statements best explains the relationship between numbers of females and eggs produced per day per female?

(A) As the female population increases, each individual expends more energy on competition, leaving less energy available for egg production. ✓

(B) Females in larger populations reduce their energy output for egg-laying to conserve resources for maintaining the overall population size.

(C) The decrease in eggs produced per female reflects a shift in energy allocation from reproduction to improved egg quality in denser populations.

(D) As the number of females grows, the total energy output of the population for egg production remains stable, resulting in fewer eggs produced per female.

Answer A

Correct. As the female population increases, competition for resources increases, leaving each individual with less energy to invest in egg production, resulting in fewer eggs per female.

8 Ecology MCQ

171. Baker's yeast, *Saccharomycetes cerevisiae*, is the species of yeast that is commonly used in baking, brewing, and making wine. It is a single-celled organism; each cell is capable of undergoing asexual reproduction by mitosis and sexual reproduction by meiosis, forming spores that combine with other spores.

To find out if energy and nutritional resource availability determines which form of reproduction is used, researchers grew twenty yeast colonies on an agar medium plate providing ideal nutrient environments. Each colony was composed of a population of yeast that originated from a single yeast cell that had reproduced asexually to form a population of millions of yeast clones. That plate was used to produce replicate plates of yeast colonies. Each replicate plate was a mirror image of the original cultures, so each colony could be subjected to each treatment group. The replicate plates contained different media as follows.

Robust media: Contains all nutrients required by yeast in excess amounts
Complete media: Contains all nutrients at a level promoting exponential growth at $r_{\max}$
Minimal media: Nutrients available at a low level, allowing growth, but stressing the yeast colonies
Deficient media: Nutrient content less than what is needed to sustain growth, extreme stress on yeast colonies

Eight replica plates were made, two plates of each media, and all plates were incubated at 30°C for 24 hours. Yeast from each colony were observed through a microscope and their means of reproduction determined. Table 1 summarizes these results.

Table 1: Reproductive Strategy of *S. cerevisiae* under different environmental stress levels (20 different colonies per medium $\times 2$)

Medium	Nutritional Conditions	Colonies Reproducing Sexually	Colonies Reproducing Asexually	Colonies Showing No Reproduction
Robust	Ideal	0	38	2
Complete	Adequate	2	35	3
Minimal	Mild Stress	24	13	3
Deficient	Extreme Stress	28	1	11

Which of the following best explains the data in Table 1 ?

(A) All forty colonies are genetically identical. Differences in stress responses are due to experimental error.

(B) Asexual reproduction allows stressed colonies to divide rapidly and compete for limited resources with other colonies.

(C) Mild nutritional stress does not trigger a change in reproductive strategy among yeast cultures.

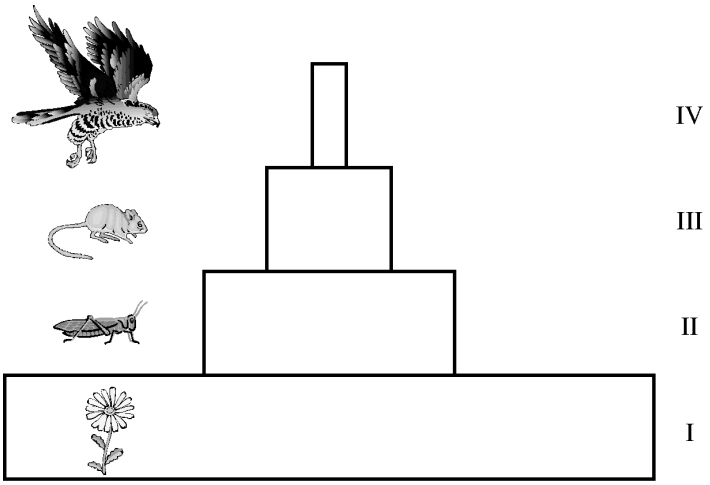
(D) Yeast respond to nutritional stress with sexual reproduction, producing new genetic combinations, or dormancy. ✓

8 Ecology MCQ

Answer D

Correct. Increasing stress leads to an increased number of colonies using sexual reproduction. Sexual reproduction could lead to new combinations of alleles that might allow the yeast to survive using a different energy source or produce a dormant form that could survive until the yeast cells were in a more favorable environment.

172.



Assume there are 50,000 joules (J) of energy available in trophic level II in the figure. According to the conventional model of energy flow in ecosystems, which of the following statements correctly describes the flow of energy in the system?

(A) Trophic level I generates a maximum of 50,000 J of energy.

(B) Trophic level I has approximately 5,000 J of available energy.

(C) Trophic level III has approximately 50 J of available energy.

(D) Trophic level IV has approximately 500 J of available energy. ✓

173. In the process of succession, which of the following is true of *K*-selected plant species?

(A) They keep their carrying capacity at a minimum level.

(B) They usually reproduce early in life.

(C) They are usually found in climax stages of succession. ✓

(D) They are the dominant species in early stages of succession.

(E) They have a short lifespan.

8 Ecology MCQ

174.

Level	Population
4	4
3	360
2	780
1	5,782

In a simple ecosystem, a census of the populations in four successive trophic levels was taken as shown above. If level 1 is composed of photosynthetic autotrophs, then the trophic level with 780 individuals will most likely represent

(A) primary consumers ✓

(B) secondary consumers

(C) tertiary consumers

(D) decomposers

(E) producers

An experiment to measure the rate of respiration in crickets and mice at 10° C and 25° C was performed using a respirometer, an apparatus that measures changes in gas volume. Respiration was measured in mL of O_2 consumed per gram of organism over several five-minute trials, and the following data were obtained.

Organism	Temperature (°C)	Average respiration (mL O_2/g/min)
Mouse	10	0.0518
Mouse	25	0.0321
Cricket	10	0.0013
Cricket	25	0.0038

175. According to the data, the crickets at 25° C have greater oxygen consumption per gram of tissue than do the crickets at 10° C. This trend in oxygen consumption is the opposite of that in the mice. The difference in trends in oxygen consumption among crickets and mice is due to their

(A) relative size

(B) mode of nutrition

(C) mode of internal temperature regulation ✓

(D) mode of ATP production

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8 Ecology MCQ

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(A) relative size

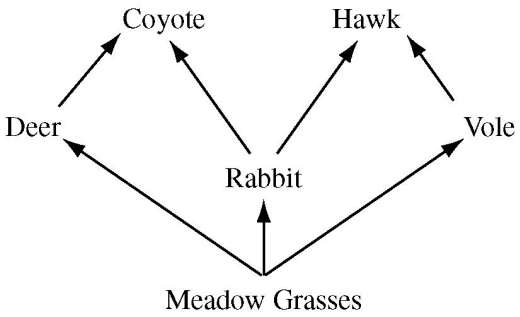
(B) mode of nutrition

(C) mode of internal temperature regulation ✓

(D) mode of ATP production

8 Ecology MCQ

177. The following is a food web for a meadow habitat that occupies 25.6km^2. The primary producers' biomass is uniformly distributed throughout the habitat and totals 1,500 kg/km^2



Developers have approved a project that will permanently reduce the primary producers' biomass by 50 percent and remove all rabbits and deer.

Which of the following is the most likely result at the completion of the project?

(A) The biomass of coyotes will be 6 kg, and the biomass of hawks will be 0.5 kg.

(B) The biomass of coyotes will be dramatically reduced. ✓

(C) The coyotes will switch prey preferences and outcompete the hawks.

(D) There will be 50 percent fewer voles and 90 percent fewer hawks.

178. All of the following statements about the pyramid of biomass are correct EXCEPT:

(A) Biomass is the total dry mass of the organisms present.

(B) The base of the pyramid generally represents primary consumers. ✓

(C) The amount of biomass at a particular level of the pyramid depends on the amount of energy available.

(D) Certain toxins tend to become concentrated at upper levels of the pyramid.

(E) Biomass pyramids tend to vary for different ecosystems.

179. Decades of seal hunting eliminated many fur seal breeding colonies. Since that time, fur seals have recolonized several islands in the Pacific Ocean. Scientists studying the populations of fur seals on three recently colonized islands observed different population growth rates among the three populations.

Which of the following best explains the different population growth rates on the recently colonized islands?

8 Ecology MCQ

(A) The islands differed in the number of predators of seals in the surrounding waters, resulting in lower growth rates on islands with fewer predators nearby.

(B) The islands differed in the amount of available prey, resulting in a higher population growth rate on the islands with fewer prey.

(C) The islands differed in the amount of competition for resources, resulting in higher population growth rates on islands with greater competition.

(D) The islands differed in the amount of available space for breeding, resulting in higher growth rates on islands with more available breeding space. ✓

Answer D

Correct. The islands are likely to differ in the amount of available space for breeding. More breeding space can support more breeding seals and offspring, so if other resources are also plentiful, it is expected that the seal population growth rate will be higher on those islands with more available breeding space.

180. An ecologist is studying a population of 70 penguins. The ecologist estimates that each year, 7 to 10 penguin chicks hatch and 4 to 5 of those chicks grow into viable adults.

Assuming that both the birth and death rates remain at their current maxima, what will the penguin population be after 20 years?

(A) 110

(B) 150

(C) 170 ✓

(D) 210

181. Which of the following best describes opportunistic, or *r*-strategy, organisms?

(A) They reach sexual maturity rapidly. ✓

(B) They reach their adult size slowly.

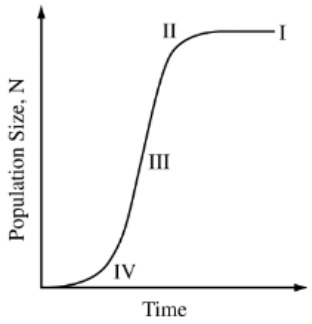
(C) They attain a large body size.

(D) They live for a long time.

(E) The size of their population remains fairly constant.

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182.



One model of a sustainable fisheries practice is for individual fish to be removed from a natural population at a rate equal to the highest possible growth rate of an ideal population. The graph above represents a population of bluefin tuna living along the Atlantic coast. At which labeled point in the graph is the population growth rate the highest?

(A) I

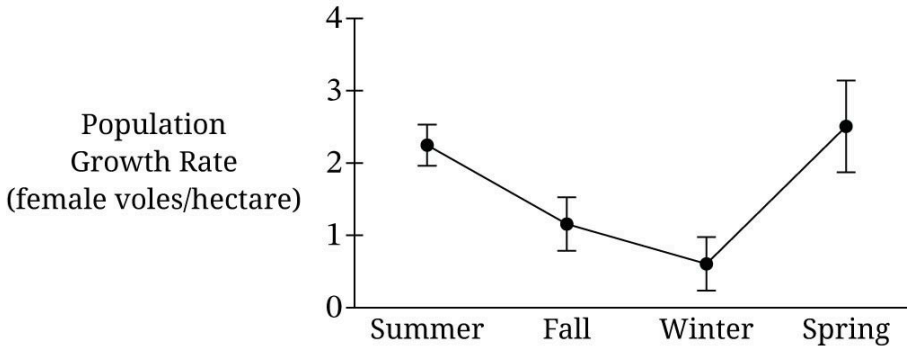
(B) II

(C) III ✓

(D) IV

183. Scientists studying small mammals called voles investigated how the growth rate of the female vole population changed during different seasons. Once a month, researchers captured voles, counted the females, and then released the voles back into their habitat. The data were used to estimate the total female population size and calculate the growth rate of the female population through each of the seasons (Figure 1).

Figure 1. Growth rate of female vole population across four seasons. The error bars represent $\pm SE_{\overline{x}}$



Which of the following statements about the growth rate of the female vole population is best supported by the data in Figure 1?

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(A) It is higher in the spring than it is in the summer.

(B) It decreases from summer to fall. ✓

(C) It decreases from fall to winter.

(D) It is three times greater in the spring than it is in the winter.

Answer B

Correct. The population growth rate declines from summer to fall, reflecting a downward trend during this seasonal transition.

184. A population of 250 birds inhabit the canopy of a tropical rain forest that has a carrying capacity of 400 birds.

What is the maximum population growth rate $(r_{\max})$ if the population grows to 283 in one year?

(A) 0.12

(B) 0.13

(C) 0.35 ✓

(D) 0.39

Answer C

Correct. This calculation uses the correct population size of $N = 250$ and includes consideration of the carrying capacity $(K = 400)$ as part of the logistical growth pattern, as described by $\frac{dN}{dt} = (283 - 250) = r_{\max} N\left(\frac{K-N}{K}\right) : 33 = r_{\max} \times 250 \times \left(\frac{400-250}{400}\right)$ and $r_{\max} = 0.35$.

185. Bison in Yellowstone National Park have a food-limited carrying capacity that determines how their population grows. If the population grows too large, some bison will either starve or migrate to search for more food. The park is able to support a maximum population of 4,500 bison. Park conservation officials must watch the population and prevent the bison from overpopulating, since they would migrate to nearby farms, causing agricultural damage and creating hazards to drivers. The bison begin to migrate when the population reaches 4,000 because of competition for food.

If the current population size of bison is 3,652 and the maximum growth rate of the population is 0.28, calculate the population size after one year and determine whether the park will need to take measures to control the population.

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(A) The population will be 3,845 after a year. The bison will not pose a problem until the following year, and no conservation action will need to take place. ✓

(B) The population will be 4,675 after a year. Some bison will need to be removed to prevent migration, since they will surpass the resource availability.

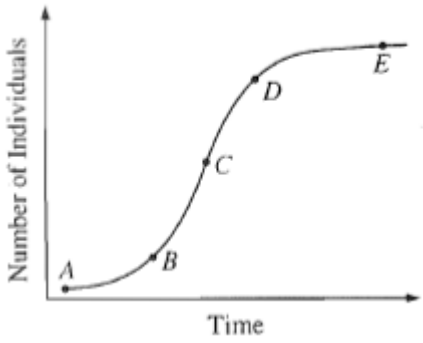
(C) The population will be 1,023 after a year, since the growth rate is too low. Conservation efforts will need to focus on increasing the bison population.

(D) The population will be 4,500 after a year. The bison cannot exceed the carrying capacity, so they will stop reproducing. Some bison will need to be removed.

Answer A

Correct. $\frac{dN}{dt} = r_{\max} N\left(\frac{K-N}{K}\right)$ should be used to determine the one-year change in population size: $0.28 \times 3,652\left(\frac{4,500-3,652}{4,500}\right) = 193$. Adding 193 to 3,652 projects a population of 3,845, so the population will not reach the point at which bison migration occurs. However, it will reach a level to initiate migration the following year.

186.



Which point on the curve in the diagram above best represents the carrying capacity of the environment for the population shown?

(A) *A*

(B) *B*

(C) *C*

(D) *D*

(E) *E* ✓

187. All of the following are density-dependent factors that limit animal populations EXCEPT

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(A) weather ✓
(B) predation
(C) birthrate
(D) food competition
(E) mortality

188. A fish population of 250 in a pond has a maximum annual rate of increase of 0.8.

If the carrying capacity of the pond is 1,500 for this species, what is the expected population size after one year?

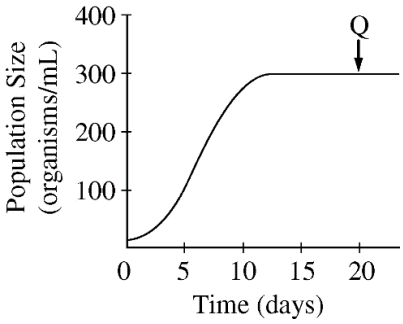
(A) 167
(B) 417 ✓
(C) 450
(D) 500

Answer B

Correct. Using the formula for logistic growth, $\frac{dN}{dt} = r_{\max}N\left(\frac{K-N}{K}\right)$, the increase in population is calculated: $8 \times 250\left(\frac{1{,}500-250}{1{,}500}\right) = 166.7$. Adding 166.7 to 250 (the current population) provides a predicted population of 417 after one year.

For following group of question first study the description of the diagram and then choose the one best answer to each question following it and fill in the corresponding oval on the answer sheet.

The graph below illustrates a logistic growth curve for a population of *Euglena*.



189. The population would be most likely to remain at the level shown at Q under which of the following conditions?

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(A) All predators are eliminated.

(B) The population size is well below its carrying capacity.

(C) *Euglena* is a primary producer.

(D) *Euglena* is a primary consumer.

(E) Environmental conditions do not fluctuate. ✓

190. The population density at point Q indicates that

(A) exponential population growth is occurring

(B) predation is beginning to occur

(C) the reproductive rate equals the death rate ✓

(D) the average age of individuals is decreasing

(E) the reproductive capacity for each individual is increasing

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191. The population of Japanese sika deer in central Japan was determined each year from 2005 to 2014. The sika deer population underwent logistic growth starting at 26,000 deer in 2005 and stabilized at 58,000 deer between 2012 and 2014. Based on these data, the $\mathrm{r}_{\max}$ for this population was determined to be 0.57. Central Japan contains a variety of habitats, including forests and grasslands. Sika deer benefit from the resources in grasslands more than forests; if deforestation occurred, producing more grasslands in the region, the carrying capacity for sika deer population would increase.

What would the population size of sika deer be one year after the carrying capacity increased to 70,000 as a result of deforestation? Assume $\mathrm{r}_{\max}$ does not change.

(A) The population would grow to 70,000 deer in a year, because they would occupy the entire environmental niche.

(B) The population would grow to 91,060 deer in a year, but then they would die off from a lack of resources.

(C) The population would grow to 63,667 deer in a year, and the growth rate would decrease the following year. ✓

(D) The population would remain at 58,000 deer, since it had already stabilized.

Answer C

Correct. $\frac{dN}{dt} = r_{\max} N\left(\frac{K-N}{K}\right)$ should be used to calculate the increase in the population size after a year: $0.57 \times 58,000 \times \left(\frac{70,000-58,000}{70,000}\right) = 5,667$. The new population size would be $5,667 + 58,000 = 63,667$. The growth rate would decrease the following year because the population would be closer to the new carrying capacity and $\left(\frac{K-N}{K}\right)$ would get smaller.

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192. Which of the following is an example of a density-independent limitation on a population?

(A) In larger population densities of rabbits, the percentage of rabbits infected with bacteria is greater.

(B) At greater concentrations of bacteria, nutrient supplies diminish and population growth ceases.

(C) At higher temperatures, more tomato plants in a population become infested with nematodes than at lower temperatures. ✓

(D) In denser stands of oak trees, greater percentages of trees are attacked by gypsy moths than in less dense stands.

(E) Once the population of grouse exceeds a certain number, the predation rate becomes very high.